F78AB

Actuarial and Financial Mathematics



Department of Actuarial Mathematics and Statistics

Heriot-Watt University

Contents

1 Introduction **7**
- 1.1 What Is Financial Mathematics? . . . 7
- 1.2 Financial Contracts . . . 7
  - 1.2.1 Examples of Financial Contracts . . . 7
  - 1.2.2 Timelines . . . 9
  - 1.2.3 Cashflows May Not Be Known In Advance . . . 12
- 1.3 Quantitative Questions Arising From Financial Contracts . . . 14

2 Interest and Present Value **17**
- 2.1 Time Value of Money . . . 17
- 2.2 Effective Rate of Interest . . . 18
- 2.3 More Than One Time Unit . . . 19
- 2.4 Compounding Over Any Number of Time Units . . . 21
- 2.5 Changing the Time Unit . . . 21
- 2.6 Accumulating and Discounting . . . 23
- 2.7 The Effective Rate of Discount . . . 25
- 2.8 The Value at Time s of a Cashflow at Time t . . . 26
- 2.9 Non-Constant Interest Rates . . . 26
- 2.10 General Accumulation and Discount Factors . . . 31

3 Cashflows and Annuities **33**
- 3.1 Accumulating and Discounting Cashflows . . . 33
- 3.2 Level Annuities-Certain . . . 37
- 3.3 Level Annuities-Due . . . 40
- 3.4 Some Important Relationships . . . 42
- 3.5 Perpetuities . . . 42
- 3.6 Deferred Annuities . . . 44
- 3.7 Increasing Annuities . . . 46
  - 3.7.1 Simple (Arithmetically) Increasing Annuities . . . 46
  - 3.7.2 Compound (Geometrically) Increasing Annuities . . . 48

4 Nominal Interest Rates **51**
- 4.1 Nominal Rates of Interest . . . 51
- 4.2 Nominal Rates of Discount . . . 52

5 Real Interest Rates **57**
5.1 The changing price of sugar . . . 57
5.2 Price inflation indices . . . 58
5.3 Retail Prices Index (RPI) . . . 59

6 Force of Interest and Continuously-paid Cashflows **61**
6.1 The Force of Interest . . . 61
6.2 Continuously-paid cashflows . . . 64

7 Loans and Loan Schedules **71**
7.1 Three Ways to Repay a Loan . . . 71
7.2 An Example . . . 72
7.3 The Principle of Equivalence . . . 73
7.4 The Mathematics of Loan Schedules . . . 76
7.5 Changing the Terms of a Loan . . . 77
7.6 Changing the Interest Rate . . . 79
7.7 Non-Level Repayments . . . 80

8 Yields **81**
8.1 An Example . . . 81
8.1.1 Use Suitable Computer Software . . . 83
8.1.2 Trial and Error, and Linear Interpolation . . . 83
8.2 Equations of Value . . . 84
8.3 Existence and Uniqueness of Yields . . . 86
8.3.1 Yields Do Not Always Exist . . . 86
8.3.2 Steffensen's Theorem . . . 88
8.4 Prices and Yields . . . 88

9 Measuring Fund Performance **91**
9.1 Fund Performance . . . 91
9.2 Money-Weighted Rate of Return (Yield) . . . 91
9.3 Linked Internal Rates of Return . . . 93
9.4 Time-Weighted Rate of Return . . . 96

10 Discounted Cashflows **99**
10.1 Yields Again . . . 99
10.2 Net Present Values . . . 101
10.3 The Discounted Payback Period . . . 104
10.3.1 Accumulated Profit . . . 105

11 Fixed Interest Securities **111**
11.1 Introduction and Terminology . . . 111
11.2 Bond Prices . . . 113
11.3 Bond Yields . . . 115
11.4 Income Tax . . . 115
11.5 Capital Gains Tax (CGT) . . . 116

11.6 Real yields . . . 119
11.7 Index-Linked Gilts (ILGs) . . . 120
11.7.1 Calculating the money coupon and redemption payments for ILGs . 121
11.7.2 Calculating the money yield on ILGs . . . 122
11.7.3 Calculating the real yield on ILGs . . . 125
11.7.4 Break-even inflation rate (BEIR) . . . 127

12 Duration, Convexity and Immunisation 129
12.1 Duration . . . 129
12.1.1 Use of duration . . . 130
12.2 Convexity . . . 135
12.3 immunisation . . . 137

13 Term Structure of Interest Rates 145
13.1 Yield Curves . . . 145
13.1.1 Explanations for the shape of the yield curve . . . 146
13.1.2 Expectations Theory . . . 146
13.1.3 Liquidity Preference Theory . . . 146
13.1.4 Market Segmentation Theory . . . 147
13.2 Zero-Coupon Bonds (ZCBs) and Spot Rates . . . 147
13.3 Forward Rates . . . 148
13.3.1 The relationship between spot rates and forward rates . . . 148
13.4 Continuous-time rates . . . 149

Chapter 1

Introduction

1.1 What Is Financial Mathematics?

Financial mathematics is a branch of applied mathematics dealing with financial contracts and financial markets. Examples of financial contracts include:

- bank accounts;
- loans;
- shares in companies;
- insurance contracts;

and there are many more. Within each category there are many different varieties: for example loans encompass personal loans, mortgages on houses, bonds issued by governments and corporations, and much besides. Insurance contracts can cover homes, cars, or the risks of life and death.
Some (but not all) financial contracts are traded on financial markets. The best known example is equity shares in quoted companies, that are traded on organised stock exchanges, such as the London Stock Exchange (LSE), Bursa Malaysia or the New York Stock Exchange (NYSE).

1.2 Financial Contracts

Every financial contract involves an *exchange of cashflows* between two parties, which can sometimes be identified as a buyer and a seller. Some examples follow.

1.2.1 Examples of Financial Contracts

Example 1.2.1 Savings Account
Suppose you open a savings account in a bank, by making a deposit of £1,000. The contract stipulates that at the end of one year, the account will be closed and the bank will

pay you £1,050. You and the bank have exchanged two sets of cashflows; one consisting of your deposit of £1,000, and the other consisting of the payment to you of £1,050 in one year's time.

Example 1.2.2 Mortgage
Suppose you obtain a loan (mortgage) of £100,000 from a bank to buy a house. You will repay the loan by making payments to the bank of £400 at the end of every month for the next 25 years. You and the bank have exchanged two sets of cashflows; one consisting of £100,000 now, and the other consisting of a stream of payments of £400 a month for 25 years.

Example 1.2.3 Fixed Interest Bond
Suppose on 4 September 2025 you purchase a £10,000 worth of a fixed interest investment that pays 4% interest per annum. The interest is paid annually, beginning 4 September 2026 until the bond matures at par on 4 September 2035.

Example 1.2.4 Share purchase
Suppose you buy some shares in a quoted company on the stock exchange in London. For example, you might buy some shares in British Airways (BA) which cost you, at their current market price, £10,000. In return, you have the right to receive all the future dividends alloted to those shares, declared in future, until you decide to sell the shares, at which point you can sell them on the stock market for their then-current market price. You and BA have exchanged two sets of cashflows; one consisting of the £10,000 purchase price now, and the other consisting of the future dividends and then the price you obtain when you sell the shares.

Example 1.2.5 Life Insurance
You decide to insure your life, in the following way. You agree to pay premiums of £100 each month to a life insurance company. In return, the life insurance company agrees to pay you (or your estate) £100,000 when you die, no matter whether you die next year or live to be a hundred. You and the life insurance company have exchanged two sets of cashflows; one consisting of the payments of £100 each month until you die, and the other consisting of the payment of £100,000 to you when you die.

Example 1.2.6 House Insurance
If you insure your house against destruction or damage by fire during the next year, you are the buyer of the insurance contract, and the insurance company is the seller. At outset you pay a premium of £400 to the insurer, who who may or may not pay anything back, depending on whether or not your house suffers a fire, but the insurer is *liable* to pay you in the event of a fire during the next year.

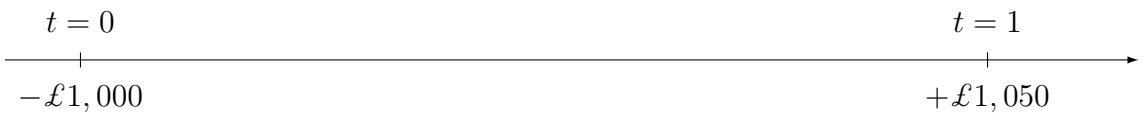
These examples display many different features that are characteristic of different kinds of financial contracts. Before discussing these, we consider a very useful way to illustrate financial contracts called a timeline.

1.2.2 Timelines

A timeline is just a graphic showing a line representing the passage of time, on which are marked the times and amounts of all the cashflows involved in a financial contract. It is a simple idea but it can be very effective in keeping track of contracts with more complicated cashflows. For **example 1.2.1** above, a timeline might look like this:



In this timeline, we have chosen, quite arbitrarily, to indicate payments above the line and time units below the line. We could have done it the other way round as follows:



and it makes no difference; it is just a rearrangement of the same information. What is significant is that signs have been attached to the cashflows. The deposit of £1,000 has a negative sign, which we take to mean that it is outgo. The repayment of £1,050 has a positive sign, which means that it is income. Therefore, this timeline illustrates the contract from your point of view. From the bank's point of view, of course, the signs are reversed (your outgo is the bank's income and *vice versa*). So from the bank's point of view, the timeline is:



Every timeline illustrates a financial contract from the point of view of one of the parties to the contract. The point of view of the other party is illustrated simply by reversing the signs of all the cashflows. This is true of *all* financial contracts that we will deal with. In general, it does not matter which party's point of view we adopt.

Something else that is important is the timescale we have chosen. The initial deposit is shown as being made at time $t = 0$, and the repayment at time $t = 1$. Since the contract is in force for exactly one year, clearly the time unit we have chosen for our timelines is the year. It need not be so. For example, suppose the unit of time used for the bank's management accounts is a month. Then a more natural timeline for this contract, from the bank's point of view, might be as follows:



where, clearly, the time unit is now a month. A careful choice of time units will be important later on. It is good practice to specify the time units on the timeline.
An alternative presentation, conveying exactly the same information, may specify the calendar dates of the cashflows. For example, if the deposit is made on 10 January 2025, and the repayment is on 10 January 2026, an equally good timeline, from your point of view, might be the following:



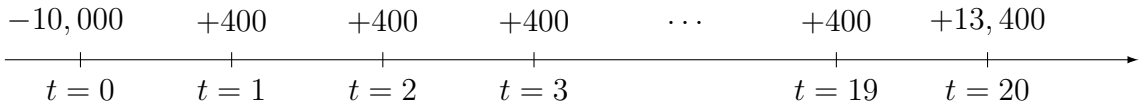
which, incidentally, illustrates the European manner of writing dates, as *dd/mm/yy*, that we shall use throughout, rather than the American convention of *mm/dd/yy*.

For **example 1.2.2** above, we note that your payments to the bank are monthly, so a natural time unit to choose is the month. Then a timeline, from your point of view, might look like this:



where the ellipsis (...) represents the repetitive payments between the beginning and end of the contract, of which there would be far too many to illustrate. Note that we have dropped the pounds signs (£) which we often do in timelines if a financial contract involves just one currency.

Turning to **example 1.2.4**, we discover a fundamental difference between it and the two examples above. Both the bank deposit and the mortgage specified in advance the timings and the amounts of all the cashflows. With the shares in BA, you know at outset how much you paid for them (£10,000) but you do *not* know at outset what dividends and sale price you will receive in the future. These depend on how profitable BA is in the future and how well its share price performs on the stock exchange. These can only be known after the event. So, suppose you hold on to the shares for ten years, and that a dividend of £400 is paid at the end of every six months, and that after ten years you sell the shares for £13,000. If you choose half-year as the time unit, because that is how often the dividends are paid, then from your point of view the timeline would be as follows.



Consider now **example 1.2.5**, we discover another source of uncertainty — you do not know when you will die (we hope). Like the financial success of BA, this is something that can be known only after the event. Let us suppose that you take out the insurance policy on your 30th birthday, and you live to your 85th birthday. Then, adopting the month as time unit, because that is how often you pay premiums, the timeline from the life insurance company's point of view (for a change) is the following.

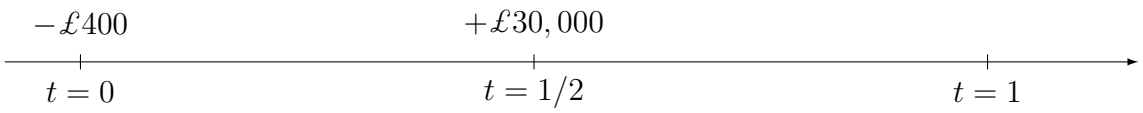


Note that you pay premiums each month *including* at outset, which is conventional with life insurance contracts, and we assume that you not pay the premium on your 85th birthday, when you die.

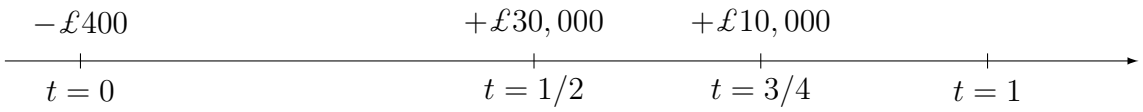
Finally, consider **example 1.2.6**, insuring your house against fire. There are two cases. First, your house does not suffer any fire during the year, in which case (from your point of view and with time unit one year) the timeline is as follows,



and only you actually make a payment. Suppose, however, that your house suffers a fire six months from outset, which causes damage worth £30,000. Then the insurance company will make a payment and the corresponding timeline will be the following.



In fact, more could happen. Most house insurance contracts will cover you for the whole of the year, so if you were really unlucky and suffered a *second* fire during the year, the insurance company will pay out for that too. For example, if your house suffers another fire nine months from outset, which causes damage worth £10,000, the timeline will be the following.



1.2.3 Cashflows May Not Be Known In Advance

As we have developed the timelines for the examples above, we have encountered the properties of cashflows that characterise a financial contract, namely:

- the direction of a cashflow: which party pays it and which party receives it;
- the timing of a cashflow;
- the amount of a cashflow; and
- the number of cashflows.

Table 1.1: Properties of cashflows defining the financial contracts in Examples 1 to 5: known in advance or not?

	Direction	Timing	Amount	Number
example 1.2.1	Yes	Yes	Yes	Yes
example 1.2.2	Yes	Yes	Yes	Yes
example 1.2.3	Yes	Yes	Yes	Yes
example 1.2.4	Yes	No	No	No
example 1.2.5	Yes	No	Yes	No
example 1.2.6	Yes	No	No	No

A key question is which of these are known in advance, and which can be known only after the event. For our examples, we summarise these properties in Table 1.1.

Note that there do exist financial contracts where the direction of a cashflow is not known at outset; that is, future events determine which party pays the other. However, we did not include any of these among our examples.
The reasons for features of cashflows being unknown are various.

- In the case of the equity shares, the amounts of future dividends and sales proceeds depended on the future performance of the company, and you would not necessarily know in advance when you might sell the shares.
- In the case of the life insurance contract, the time of your death is not known in advance.
- In the case of the house insurance contract, the number of fires your house will suffer during the year is unknown, and for each such event its timing is unknown, and the monetary amount of the damage it will cause is unknown.

In addition, even when the amount, timing and so on of some cashflows appear to be known in advance, because they are specified in the contract, in reality some uncertainty is always present.

- In example 1.2.1, the bank could go bust before it can repay you. This is an example of *credit risk* or *counterparty risk*.
- In example 1.2.2, you could decide to repay the mortgage early, or you might die before completing the repayments.
- In example 1.2.5, you might decide to cancel the life insurance policy, for example if you could no longer afford the premiums.
- In example 1.2.6, the insurance company might suffer catastrophic losses elsewhere and be unable to meet your claims (another example of credit risk).

To deal with any of the above sources of uncertainty, we need the mathematical tools to describe uncertainty, namely the theory of probability. In these notes, we shall deal only with cashflows whose existence, timing and amount are assumed to be known with certainty, as in example 1.2.1 and example 1.2.2 above. Moreover, we shall ignore credit and counterparty risk, mortality, and the foibles of human behaviour, that in reality inject a degree of uncertainty into cashflows that are otherwise completely determined by the terms of a financial contract. We shall assume that we operate in a genuinely risk-free world, and we shall not have any need for probability theory. There is plenty of time for that later!

1.3 Quantitative Questions Arising From Financial Contracts

The examples above mostly contained some numbers. Where did they come from?

- In example 1.2.1 (the bank deposit account), why would the bank return £1,050 at the end of the year, for a deposit of £1,000 at the start of the year? What would happen if you wanted to withdraw your money half-way through the year? What would happen if you wanted to leave it on deposit for another year?
- In example 1.2.2 (the mortgage), why did you have to repay £400 each month for 25 years? Why not £390 or £410? What happens if you want to repay the mortgage early?
- In example 1.2.3 (bond), why does the bond pay once a year? And why at the end and not beginning of the year? Could the bond pay dividends twice (or more times) a year?
- In example 1.2.4 (the equity shares in BA) Suppose you had the choice between buying these shares, and investing the same amount elsewhere, for example in bonds issued by the UK Government. How do you decide which is the better investment?
- In example 1.2.5 (the life insurance contract) why was the monthly premium £100, and not £95 or £105? What happens if you live to age 65 and then decide you want to cancel the contract — should you get any money back? From the insurance company's point of view, how profitable is this contract likely to be?
- In example 1.2.6 (the house insurance contract) why should the premium be £400, not £300 or £500? From the insurance company's point of view, how profitable is this contract likely to be?

These are all quantitative questions and we need to formulate them mathematically, in a sensible way, in order to answer them. That is what financial and actuarial mathematics is about.
The questions above are mostly posed from the customer's point of view — why am I paying a particular price for this contract? Except for example 1.2.4, on the other side of

the contract is a financial institution; a bank or insurance company. Any such institution has to publish a balance sheet at least annually, to show that its assets are greater in value than its liabilities. Its liabilities are precisely all the contracts it has entered into with its customers, such as you. By entering into such contracts, it has assumed a liability to pay you in the future, under stated conditions. A very important question in all these cases is: what value should the financial institution place upon the contract in its balance sheet, to show whether or not it is solvent? This is called the *valuation* of financial contracts and is at the very heart of the proper management of financial institutions[1]. Two examples will suffice.

- In example 1.2.2, after the mortgage has been in force for twelve years, what value should the bank place upon the repayments it expects to receive from you in the future? Since this represents future income, the value will appear on the balance sheet as an asset. It will have to be calculated mathematically.
- In example 1.2.5, should you live to age 60, what value should the life insurance company place on its obligation to pay you £100,000 whenever you die in the future? Since this is outgo, it will appear on the balance sheet as a liability. It too will have to be calculated mathematically.

Complete answers to all these questions will take us far beyond these notes. For the moment, we will consider simple settings from which all risk and uncertainty is excluded. That is, we will study *deterministic* problems and leave *stochastic* problems for the future.

[1]One of the main causes of the *credit crunch* of 2007–2008 was that banks had bought and sold huge quantities of extremely complex financial contracts called collateralized debt obligations (CDOs) which we will not describe here. When it became apparent that these were running into difficulties it became impossible for the banks to value these either as assets (if they had bought them) or liabilities (if they had sold them). The banks were basically incapable of saying whether or not they were solvent. In these circumstances, no-one was willing to lend to banks, which is to say banks were unable to obtain credit, hence the 'credit crunch'.

Chapter 2

Interest and Present Value

2.1 Time Value of Money

At a given point in time, cash has a monetary value, but also has a time value. For example, it is clear that you would rather have £1.10 now than £1 now. But, is it also clear that you would be better off with £1.10 in 2 years' time than £1 now?
The point is that receiving and holding £1 now is worth more than receiving £1 at some time in the future. Why is this? There are three reasons:

1. **Deferred Use:** you do not have the use of the £1 until you receive it, someone else does.

2. **Risk:** the £1 you have been promised might never be paid (the company might go bankrupt, the borrower might die, etc).

3. **Inflation:** prices might rise.

Therefore, if you deposit or invest your money now, you expect not only your money back, but also a reward for letting someone else have the use of the money for a period of time.

Definition 2.1 The amount deposited or invested is called **capital** (or **principal**).

Definition 2.2 The reward received for investing or depositing it is called **interest**.

Notation: p.a. or per annum is used in these notes to denote **per year**.

Example 2.1.1 You deposit £1,000 in a bank account for one year. The bank pays interest at 3% p.a. (per annum, per year). In one year's time you will receive £1,030, which consists of the return of your capital (£1,000) and interest ($£30 = 0.03 \times £1,000$).

Example 2.1.2 You deposit £1,000 in a bank account for one year. In one year's time you will receive £1,050, which consists of the return of your capital (£1,000) and interest ($£50 = 0.05 \times £1,000$). You deduce that that bank pays interest at the rate of 5% p.a.

2.2 Effective Rate of Interest

The rate at which interest is received depends on a choice of time unit. The year is the most common time unit, but the week, month, quarter-year or half-year are also often used.

Definition 2.3 *The effective rate of interest per time unit is the amount of interest received at the end of one time unit per £1 invested at the start of that time unit.*

Notation: The effective rate of interest per time unit is most often denoted i, although when more than one rate of interest is involved, other letters such as j may be used, or subscripted symbols such as i_1 and i_2.

Comment: Note that the effective *rate* of interest is *not* a percentage, although it is usually *expressed* as a percentage. In the first example above, the effective rate of interest p.a. (the time unit is a year) was 0.03, although expressed as 3%. The point is that '3%' is not a valid quantity in an arithmetical expression: $3\% \times 1,000$ does not make sense, while $0.03 \times 1,000$ makes perfect sense. This is a suitable point at which to remind the reader that mathematics is a language in which *absolutely precise* statements can be expressed and anything that falls short of *absolute precision* has no place.

Comment: The time unit is an *essential* part of the definition of effective interest.

The timeline for effective interest of i per time unit, over one time unit, looks like this.



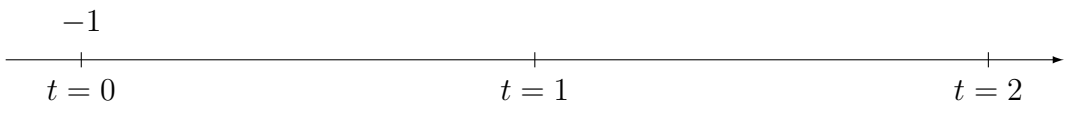
Example 2.2.1 Invest £100 for 6 months at 3.75% per half-year effective. Then the natural time unit is the half-year, $i = 0.0375$ per half-year and the investment is worth £103.75 after half a year (= 6 months).



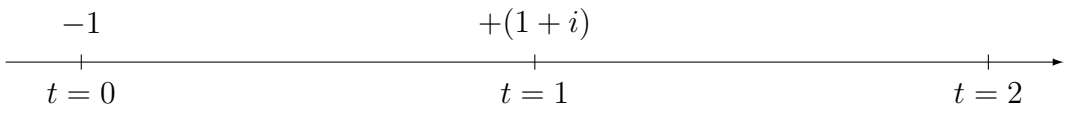
2.3 More Than One Time Unit

Suppose the effective rate of interest is i per time unit. What happens if we deposit a sum of money for two time units? We reason as follows:

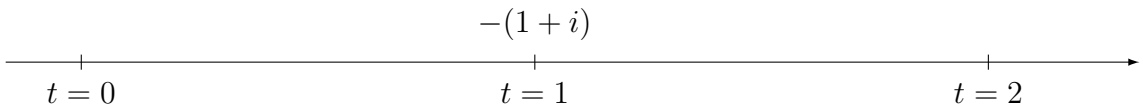
- Invest £1 at outset ($t = 0$).



- At the end of one time unit ($t = 1$) we have $£(1 + i)$.

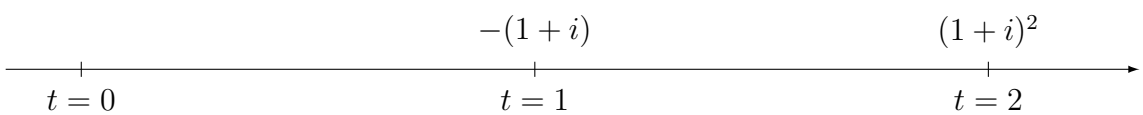


- We withdraw our money from the bank (we suppose we are free to do so) and reinvest it as fresh capital at $t = 1$.



- At the end of the second time unit ($t = 2$) we have:

$$(1 + i) + i(1 + i) = (1 + i)(1 + i) = (1 + i)^2.$$



Clearly, we can repeat the process over three time units: at $t = 2$, reinvest our $(1 + i)^2$ as fresh capital and at $t = 3$ we have:

$$(1 + i)^2 + i(1 + i)^2 = (1 + i)(1 + i)^2 = (1 + i)^3.$$

If we reinvest our money as fresh capital again at $t = 3$ then at $t = 4$ we have:

$$(1 + i)^3 + i(1 + i)^3 = (1 + i)(1 + i)^3 = (1 + i)^4$$

and so on. It is easy to see (and to prove inductively) that £1 invested at $t = 0$ will grow to $(1 + i)^n$ at time $t = n$, where n is any positive integer.

Example 2.3.1

Invest £50 for 7 years at 4% p.a. effective. $50(1.04)^7 = 65.80$.

Example 2.3.2

Invest £100 for 9 months at 2% per quarter year effective. $100(1.02)^3 = 106.12$.

Example 2.3.3

Invest £80 for 1 year at 6.25% per half year effective. $80(1.0625)^2 = 90.31$.

Remarks:

1. This is a model of *compound interest*, where interest is earned on interest previously earned, not just on the original capital (or principal).

2. The idea of compound interest above depends on our being able to withdraw our funds from the account at any time, and reinvest them as fresh capital. In practice, this is inconvenient and unnecessary, as virtually all bank accounts and the like carry compound interest anyway.

3. If interest is earned only on the original principal even over more than one time unit, it is called *simple interest* and after n time periods we have $(1 + in)$ rather than $(1 + i)^n$. This is sometimes met in special circumstances, but it is uncommon.

4. If $i > 0$, as is usually the case, then $(1 + i)^n > (1 + in)$ for $n > 0$, so compound interest is always preferable to simple interest.

Unless otherwise stated, you should always assume that i is an effective rate of *compound* interest.

2.4 Compounding Over Any Number of Time Units

Suppose you invest £1 today at an effective rate of interest of $i = 5\%$ p.a.. How much will you have after 1 month? Or 7 months? Or 123 months?

In fact the result of Section 2.3 extends to any real number $t \geq 0$. That is, if we invest £1 for t time units ($t \geq 0$) at an effective rate of interest i per time unit, at time t we get $£(1+i)^t$.

Example 2.4.1

Invest £2,000 for 4 months at 10.5% p.a. effective. $2,000(1.105)^{4/12} = 2.067.68$.

Example 2.4.2

Invest £2,500 for 1 month at 2% per quarter year effective. $2,500(1.02)^{1/3} = 2,516.56$.

Example 2.4.3

Invest £100 for 3.6 years at 4.1% per half year effective. $100(1.041)^{2\times 3.6} = 133.55$.

2.5 Changing the Time Unit

It is often convenient to change from one time unit to another. For example: The effective rate of interest may be defined per annum, but the financial contract we are dealing with has cashflows every month. It would be more convenient to work with an effective rate of interest per month (we will see this later on). How would we find the equivalent effective rate of interest per month?

Suppose we have two time units, denoted t_i years and t_j years. For example, $t_i = 1$ and $t_j = 1/12$ means that our two time units are the year and the month. Let the effective rate of interest per t_i years be denoted i, and let the effective rate of interest per t_j years be denoted j. Suppose we invest £1 for one year.

- Using time unit t_i years, we have $£(1+i)^{1/t_i}$ after 1 year.
- Using time unit t_j years, we have $£(1+j)^{1/t_j}$ after 1 year.
- The two effective rates of interest are defined to be equivalent if we have the same amount of money after one year; that is, if:

$$(1+i)^{1/t_i} = (1+j)^{1/t_j}.$$

- Thus, if we start with effective interest of i per t_i years, we can switch to time unit t_j years by using effective interest rate:

$$j = (1+i)^{t_j/t_i} - 1$$

per t_j years.

- Or, if we start with effective interest of j per t_j years, we can switch to time unit t_i years by using effective interest rate:

$$i = (1+j)^{t_i/t_j} - 1$$

  per t_i years.

- It is easily seen that if two effective rates of interest are equivalent in this sense, giving the same amount after 1 year, then they also give the same amount after any time $t > 0$ (can you prove this?).

Example 2.5.1 What effective rate of interest per half year is equivalent to 7% p.a. effective?

Here, t_i is 1 year and t_j is 1/2 year.

$1.07^{(1/2)/1} = 1.07^{1/2} = 1.03440804 \Rightarrow 0.034408$ per half year to 6 decimal places.

Example 2.5.2 What effective rate of interest per quarter year is equivalent to 15% per two years effective?

Here, t_i is 2 years and t_j is 1/4 year.

$1.15^{(1/4)/2} = 1.15^{(1/8)} = 1.01762374 \Rightarrow 0.017624$ per quarter year to 6 decimal places.

Example 2.5.3 What effective rate of interest per 1.7 years is equivalent to 1.3% per month effective?

Here, t_i is 1/12 years and t_j is 1.7 year.

$1.013^{1.7/(1/12)} = 1.301465576 \Rightarrow 0.301466$ per 1.7 years to 6 decimal places.

Comment on Rounding: Why did we round the results of the calculations above to six decimal places? In practice, the number of decimal places that is appropriate depends on the magnitude of the monetary amounts involved. Suppose, for example, that our problem is to invest £1 for one year. In the first example above, we find that, rounded to pence, the answers are equal:

$$£1 \times 1.034408^2 = £1.06999991 = £1.07 \text{ to 2 d.p.}$$

We can see that six decimal places will suffice as long as we have to invest amounts of not more than about £100,000:

$$£100,000 \times 1.034408^2 = £106,999.991 = £107,000.00 \text{ to 2 d.p..}$$

However, if we had to invest larger amounts, say £100,000,000, even rounding to six decimal places would introduce non-trivial accounting errors:

$$£100,000,000 \times 1.034408^2 = £106,999,991.00 \text{ to 2 d.p.}$$

a difference of £9 from the correct answer of £107,000,000.00.

Hence, there is no absolutely correct approach to rounding interest rates when converting from one time unit to another. *Purely for the purposes of these notes*, we adopt as a *convention* that we will round to not less than six decimal places, to show awareness of the issue. Monetary amounts, of course, cannot be shown to more than two decimal places.

In most cases now such calculations will be done on a spreadsheet, which will give answers accurate enough that rounding errors such as the above are not an issue (at least, not a mathematical issue). However, this introduces a different problem, of being able to audit and validate the workings and outputs of the spreadsheet. If the spreadsheet is rounding calculations (as it must) in ways that the user is unaware of, it is impossible to be absolutely certain that the spreadsheet is doing what the user intends. For this reason, calculations in Excel financial spreadsheets are often rounded explicitly using the `ROUND()` function. For example, `ROUND( A4 * B4 , 2 )` will multiply the contents of cells A4 and B4 and round the answer to two decimal places.

2.6 Accumulating and Discounting

We say that money *accumulates* with interest. The amount to which capital accumulates with the addition of interest is called its *accumulation* or *accumulated value*

Definition 2.4 If i is the effective rate of interest per time unit, we call $(1 + i)$ the *accumulation factor* per time unit. Also, we call $(1 + i)^t$ the accumulation factor over t time units.

Accumulation is retrospective in nature; it tells us how much we have at the end of some time period.

Often, what we want to know is the converse i.e. how much we need at the start of the period in order to meet some target at the end of the period. For example, we might need to repay a loan in 20 years' time. How much should we invest now, so that in 20 years, with the addition of interest, we have exactly the amount needed to repay the loan?

Example 2.6.1 We need £1,000 in one year's time. The bank offers interest at an effective rate of 6% p.a. How much should we invest now to meet this liability?

Answer: We need $£1,000/1.06$, since:

$$(1,000/1.06) \times 1.06 = 1,000.$$

Terminology:

- We say that £1,000/1.06 is the *discounted value* of £1,000 over one year at 6% p.a. effective.
- We also say that £1,000/1.06 is the *present value* of £1,000 over one year at 6% p.a. effective. We often abbreviate 'present value' to PV or pv.
- When we calculate a present value of a future payment like this, we say we are *discounting* it.
- Discounting and accumulation are inverse operations.

Definition 2.5 The discount factor per time unit, at rate of interest i per time unit effective is denoted v and defined as:

$$v = (1+i)^{-1}.$$

Example 2.6.2 How much should we invest now to meet a liability of £2,000 in 2 years' time if the effective rate of interest is 6% p.a?

Answer: We need X such that $X(1.06)^2 = 2,000$ so $X = 2,000(1.06)^{-2} = £1{,}779.99$.

We say that £1,779.99 is the PV of £2,000 in 2 years or the discounted value of £2,000 in 2 years. Here, $v = 1/1.06 = 0.943396$ to 6 decimal places.

Example 2.6.3 What is the PV of £10,000 in 5 years at an effective rate of interest of 3% per half-year?

Answer: $£10,000/1.03^{10} = £7,440.94$. Here, $v = 1/1.03 = 0.970874$ (to 6 d.p.) per half year.

Comments:

- If the effective rate of interest is i per time unit, then the PV at time 0 of £1 at time $t > 0$ is $v^t = (1+i)^{-t}$.
- We call v^t the *discount factor* from time t to time 0.
- It should be clear what we mean when we say that accumulating and discounting are inverse operations. If we accumulate and then discount, or discount and then accumulate, over the same time period, we get back to where we started.
- Because discount factors, like accumulation factors, may be applied to arbitrarily small or large amounts of money, rounding affects them509k in the same way. When we have to round a discount factor, we will adopt the same convention of at least six decimal places.

Example 2.6.4 Let $v = 0.94$ be the discount factor per half-year.

(i) What is the effective rate of interest per half-year?

Answer: $v = (1+i)^{-1} \Rightarrow i = v^{-1} - 1 = 0.063830$ to 6 d.p..

(ii) What is the equivalent discount factor per year?

Answer: The equivalent effective rate of interest p.a., call it j, is given by $(1+j) = (1+i)^2$, so the equivalent discount factor p.a., call it w, is $w = (1+j)^{-1} = (1+i)^{-2} = v^2 = 0.883600$ (to 6 d.p.).

2.7 The Effective Rate of Discount

Definition 2.6 Let i be the effective rate of interest per time unit, so that $v = (1+i)^{-1}$ is the corresponding discount factor. The effective rate of discount per time unit, denoted d, is defined as:

$$d = 1 - v.$$

Hence, for each of the operations of accumulating and discounting, we have an *effective rate* and a corresponding *factor*.

Operation	Rate	Factor
Accumulation	i	$1+i$
Discounting	d	$1-d =$ v

Comments:

- It is easy to remember which way the signs work: accumulation makes money grow so the accumulation factor must be greater than one; discounting makes money shrink so the discount factor must be less than one.
- The rounding convention of at least six decimal places applies to all the quantities in this table.

Useful Result: We will very often use the fact that $d = i/(1+i)$, which we show as follows:

$$d = 1 - v = 1 - \frac{1}{1+i} = \frac{1+i-1}{1+i} = \frac{i}{1+i}.$$

Interpretation of Useful Result: Imagine a rather odd bank account, where interest was added to the account at the start of the time unit, and the account holder was then forbidden to withdraw their funds until the end of the time unit. What amount of interest, paid in advance in this way, would be equivalent to effective interest of i per time unit? The answer is d, because we note that only the interest added at the start of the time unit accumulates (or we would be adding two lots of interest to the capital) and:

$$1 + d(1+i) = 1 + \left(\frac{i}{1+i}\right)(1+i) = 1 + i.$$

Example 2.7.1 Given $i = 0.04$ effective per time unit, find v and d.

Answers: $v = 1.04^{-1} = 0.961538$, and $d = 1 - v = 0.038462$, both to six decimal places.

2.8 The Value at Time s of a Cashflow at Time t

So far, we have looked at accumulating a cashflow from time 0 to time $t > 0$, or discounting a cashflow due at time $t > 0$ back to time 0. More generally, given a cashflow at time t, we will want to find its value at *any* time s.

Suppose the effective rate of interest is i per time unit, and we have a cashflow X at time t. Then we have two cases to consider:

- Case 1: $s \geq t$. Then we must accumulate the cashflow at time t up to time s, through a period of length $s-t$, so the value of the cashflow at time s is $X(1+i)^{s-t}$. Note that we have chosen, arbitrarily, to include the possibility $t = s$ in this case.
- Case 2: $s < t$. Then we must discount the cashflow at time t back to time s, through a period of length $t - s$, so the value of the cashflow at time s is $Xv^{(t-s)}$.

In either case, we call the result the *value* at time s of the cashflow X payable at time t.

Some authors dispense with the notational distinction between discounting and accumulating by just defining either $(1+i)$ or v and then allowing negative exponents to be used. Then, for example, the value at time s of the cashflow X payable at time t can be written as either $(1+i)^{s-t}$ or v^{t-s} regardless of whether $s \geq t$ or $s < t$. This is sometimes helpful in simplifying expressions.

Example 2.8.1 Let the effective rate of interest be 5% per half year.

(i) What is the value at time $s = 3$ of a cashflow of £1,000 payable at time $t = -2$?

Answer: $1,000(1.05)^{10} = £1,628.89$ (rounded to pence).

(ii) What is the value at time $s = 4$ of a cashflow of £500 due at time $t = 7.75$?

Answer: $500(1.05)^{-7.5} = £346.78$ (rounded to pence).

2.9 Non-Constant Interest Rates

In reality, interest rates may not remain constant over long periods of time. A well-known example is the Bank rate in the UK, which is the interest rate that the Bank of England charges for holding overnight deposits of cash from commercial banks. This changes from time to time, and in periods of financial turmoil it can change rapidly. Therefore, we have to be able to accumulate and discount cashflows through periods in which the rate of interest is not constant.

In theory the rate of interest could be almost any function of time, and it could change continuously. For example, $i(t) = (2 + sin(t))/100$ would define a rate of interest that forever cycled between its upper bound $i(t) = 0.03$ and its lower bound $i(t) = 0.01$. This example is not particularly realistic, but there are models in which the rate of interest may vary continuously which are of the greatest importance in financial mathematics. Such models lie beyond these notes.

We will restrict ourselves to models of interest rates which have the following properties.

- **Property 1:** Our accumulating and discounting all takes place within the finite time interval $[T_0, T_n]$.
- **Property 2:** As a function of time, the effective rate of interest is piecewise constant. That is, there is a finite series of distinct times $T_0 = t_0 < t_1 < \ldots < t_n = T_n$ and the effective rate of interest, now denoted $i(t)$, changes at the interior points $t_1, \ldots, t_{n-1}$. In between these times, $i(t)$ stays constant. For $t_{r-1} \leq t < t_r$ we use the notation $i(t) = i_r$ $(r = 1, \ldots, n)$.
- **Property 3:** The set of times $t_0 < t_1 < \ldots < t_n$, and the set of interest rates $i_1, \ldots, i_n$, are known in advance.

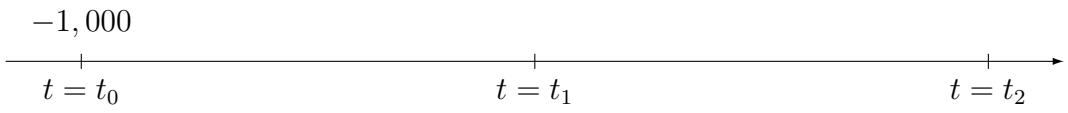
Property 3 is needed to ensure that we can carry out calculations involving future interest rates without introducing uncertainty and, hence, requiring the use of probability theory.

To accumulate a cashflow through a period with interest rates changing as above, we turn again to the idea that we may, at any time, withdraw our funds and reinvest them as fresh capital. Whenever we meet a time t_r, when the effective interest rate changes from i_r to i_{r+1}, we do exactly that.

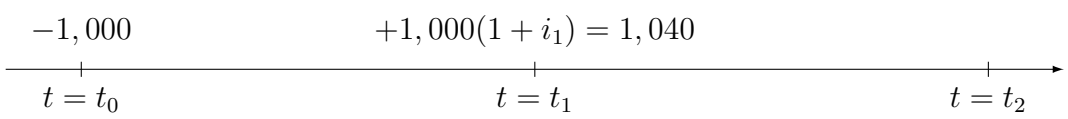
Example 2.9.1 Let the effective rate of interest be 4% p.a. during 2023 and 6% p.a. during 2024. What is the accumulation on 31/12/24 of £1,000 deposited in a bank account on 1/1/23?

Answer: In this case $T_0 = t_0 = 1/1/23$, $t_1 = 1/1/24$ and $t_2 = T_2 = 1/1/25$ (or 31/12/24 if we wish, it does not matter). Also, $i_1 = 0.04$ and $i_2 = 0.06$. We now repeat the reasoning we used in accumulating over two time units.

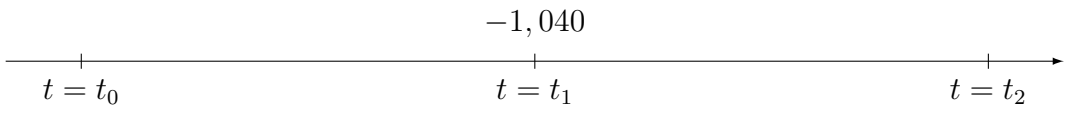
- Invest £1,000 at outset $(t = 0)$.



- At 1/1/24 (time t_1) we have $£1,000(1 + i_1) = £1,040$.

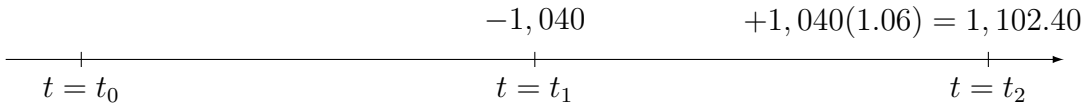


- We withdraw our money from the bank (we suppose we are free to do so) and reinvest it as fresh capital at t_1.



- At the end of 2024 (time t_2) we have:

$$£1,040(1+i_2) = 1,040(1.06) = £1,102.40.$$



This process is carried out as needed, over any number of periods in each of which the effective rate of interest is constant.

Example 2.9.2 The effective rate of interest will be 5% p.a. during 2023, 5.5% p.a. during 2024 and 6% p.a. during 2025. What will be the accumulation of:

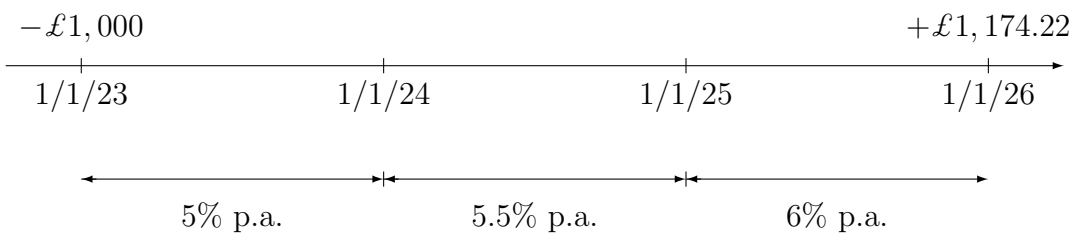
(i) £1,000 invested on 1/1/23 for three years?

(ii) £1,000 invested on 1/7/23 for two years?

(iii) £1,000 invested on 1/10/24 for one year?

Answers:

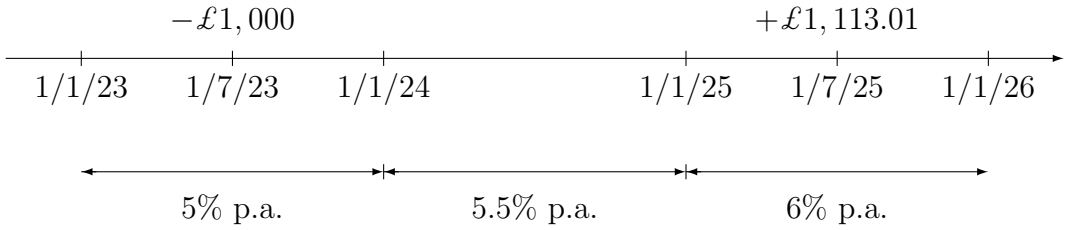
(i) The timeline for this financial contract, with interest rates also indicated, is below:



The accumulation is one full year at 5% p.a., one full year at 5.5% p.a. and one full year at 6% p.a., giving:

$$1,000 \times (1.05)^1(1.055)^1(1.06)^1 = 1,174.22 \text{ to 2 d.p.}$$

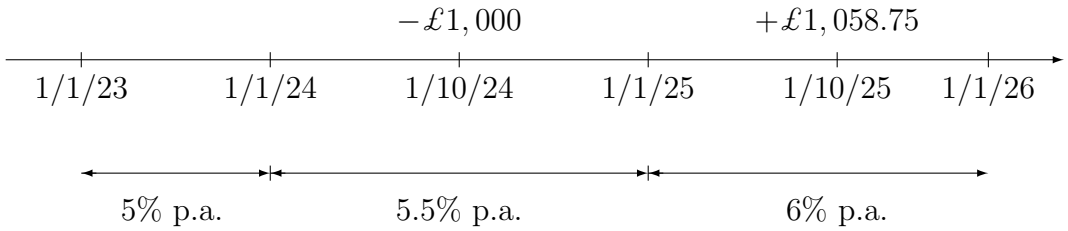
(ii) The timeline for this financial contract, with interest rates also indicated, is below:



The accumulation is half a year at 5% p.a., one full year at 5.5% p.a. and half a year at 6% p.a., giving:

$$1,000 \times (1.05)^{1/2}(1.055)^1(1.06)^{1/2} = 1,113.01 \text{ to 2 d.p.}$$

(iii) The timeline for this financial contract, with interest rates also indicated, is below:



The accumulation is a quarter year at 5.5% p.a. and three-quarters of a year at 6% p.a., giving:

$$1,000 \times (1.055)^{1/4}(1.06)^{3/4} = 1,058.75 \text{ to 2 d.p.}$$

In general, suppose that an amount X is invested between time T_0 and T_n, with effective interest being i_r between times t_{r-1} and t_r, as above. Then the accumulated value of X at time T_n is:

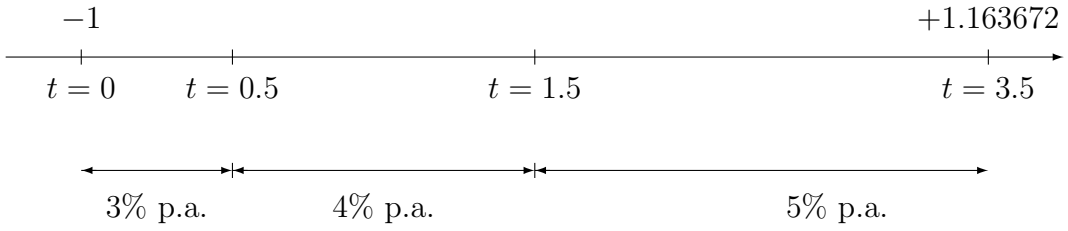
$$X(1+i_1)^{t_1-t_0}(1+i_2)^{t_2-t_1}\dots(1+i_n)^{t_n-t_{n-1}} = X\prod_{r=1}^{n}(1+i_r)^{t_r-t_{r-1}}.$$

Alternatively, suppose we require the PV at time T_0 of X due at time T_n. Then this PV is:

$$X(1+i_1)^{t_0-t_1}(1+i_2)^{t_1-t_2}\dots(1+i_n)^{t_{n-1}-t_n} = X\prod_{r=1}^{n}(1+i_r)^{t_{r-1}-t_r}.$$

Example 2.9.3 Starting now, the effective rate of interest will be 3% p.a. for 6 months, 4% p.a. for the following one year, and 5% p.a. for the following two years. What is the accumulation factor over the period [0,3.5] (measuring time in years)?

Answer: The timeline, with interest rates also indicated, is as follows.



The accumulation is for half a year at 3%, one full year at 4% and two full years at 5%, so the accumulation factor (to six decimal places) is:

$$(1.03)^{0.5}(1.04)^{1}(1.05)^{2} = 1.163672.$$

Example 2.9.4 Starting now, the effective rate of interest will be 2% per half-year for 1 year, 1.5% per quarter year for 6 months, and 8% p.a. for 18 months. What is the PV now of £1,000 due in 3 years' time?

Answer: It will be convenient to choose one time unit and convert all effective rates to that time unit. In this case, we will choose the quarter year (3 months) and define:

$$\begin{aligned} i_1 &= 0.009950 \text{ per quarter year (equivalent to 2\% per half year)} \\ i_2 &= 0.015 \text{ per quarter year} \\ i_3 &= 0.019427 \text{ per quarter year (equivalent to 8\% p.a.)} \end{aligned}$$

In units of a quarter year, $t_0 = 0, t_1 = 4, t_2 = 6$ and $t_3 = 12$. Therefore the PV is:

$$1,000\prod_{r=1}^{3}(1+i_r)^{t_{r-1}-t_r} = 1,000(1.009950)^{-4}(1.015)^{-2}(1.019427)^{-6} = £831.25.$$

2.10 General Accumulation and Discount Factors

It is sometimes convenient to work with a more general and abstract model of accumulation and discounting, since there is no limit to the complexity of the pattern of interest rates once we allow them to be other than constant. We can abstract all the features of compound interest we have observed in the foregoing with a definition and two rules.

Definition 2.7 For arbitrary times t and s, with $t \leq s$, define the following:

$$\begin{aligned} V(t,s) &= \text{the value at time } t \text{ of 1 payable at time } s. \\ V(s,t) &= V(t,s)^{-1} = \text{ the accumulation at time } s \text{ of 1 payable at time } t \end{aligned}$$

Rules: Assume that the quantities just defined satisfy the following rules:

1. $V(t,s) > 0$ for any t and s
2. $V(t,s) = V(t,r)V(r,s)$ for any t, s and r.

The first rule excludes examples that would not make sense in the real world. The second rule ensures that interest is of the compound type.

Any set of factors satisfying these rules gives a well-defined model of compound interest We have seen two, namely constant effective interest and piecewise constant effective interest, but there are many more.

As with timelines, the arguments t and s of these general factors could be algebraic symbols as above, or calendar dates as in $V(1/1/23, 1/1/24)$, depending on the setting; and the time unit can be anything.

Result: Show that $V(t,t) = 1$ for all t.

Proof: Put $r = s = t$ in the second rule. Then:

$$V(t,t) = V(t,t)V(t,t) = [V(t,t)]^2$$

which has solutions 0 and 1, but by the first rule $V(t,t) > 0$, hence $V(t,t) = 1$.

Chapter 3

Cashflows and Annuities

3.1 Accumulating and Discounting Cashflows

A series of *certain* cashflows is determined by:

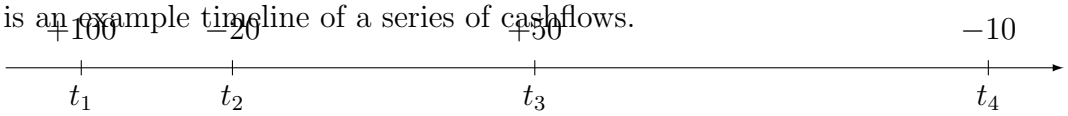
- the times at which they will be paid, which we denote $t_1, t_2, \ldots, t_n$; and
- their amounts. We denote the amount of the cashflow payable at time t_r by c_r or $c(t_r)$ $(r = 1, \ldots, n)$.

The most basic question is: at any time t, what is the value of these cashflows?

Comments:

- The number of cashflows is implicitly defined by the number of times at which cashflows are payable, namely n above.
- The direction of a cashflow is determined by its sign; the c_r may be positive or negative.

Here is an example timeline of a series of cashflows.



An obvious summary figure, of a series of cashflows, is their sum:

$$\text{Total Cashflows} = \sum_{r=1}^{n} c_r.$$

For some purposes, for example drawing up a company's accounts over a short period such as a month, a quarter or a year, this is adequate and useful. However, when we

have investment or actuarial applications in mind, it is not adequate, because it ignores the time value of money, represented by accumulating or discounting payments made at different times. When we allow for the time value of money, cashflows made at different times are different in kind, and adding them together would be like adding apples and oranges. What we can add together is the *value* of different cashflows, taken at the same point of time.

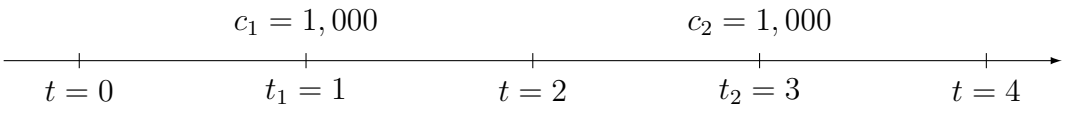
Principle: Two or mare cashflows paid at different times may legitimately be compared or combined if and only if we have allowed for the time value of money by finding their accumulations or present values at the same point of time.

Definition 3.1 The *value* at time t of the series of cashflows defined above is defined as:

$$\sum_{r=1}^{n} c_r V(t, t_r).$$

In words, any cashflows at times up to and including time t (that is, with $t_r \leq t$) are accumulated to time t, and any cashflows after time t (that is, with $t \leq t_r$) are discounted to time t, and the resulting accumulations and PVs are summed.

If we seek the value of the cashflows at time $t \geq t_n$, the result is their accumulation to time t. If we seek the value of the cashflows at time $t \leq t_1$, the result is their present value at time t. If we seek the value of the cashflows at time t where $t_1 < t < t_n$, the result is mixed; it is the accumulation of past cashflows and the present value of future cashflows. Consider the following very simple example with two cashflows.



Example 3.1.1 Suppose the effective rate of interest is 4% p.a. throughout.

- The value of these cashflows at $t = 0$ is:

$$\begin{aligned} \sum_{r=1}^{n} c_r V(t, t_r) &= \sum_{r=1}^{2} c_r V(0, t_r) \\ &= 1,000(1.04)^{-1} + 1,000(1.04)^{-3} \\ &= 1,850.53 \qquad \text{to 2 d.p.} \end{aligned}$$

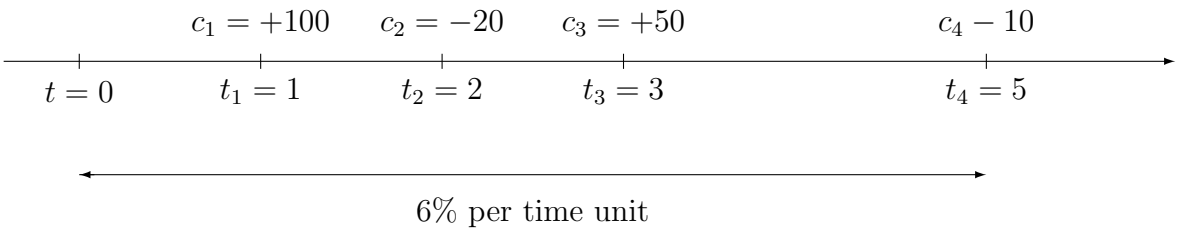
- The value of these cashflows at $t = 4$ is:

$$\begin{aligned}\sum_{r=1}^{n} c_r V(t, t_r) &= \sum_{r=1}^{2} c_r V(3, t_r) \\ &= 1,000(1.04)^3 + 1,000(1.04)^1 \\ &= 2,164.86 \qquad \text{to 2 d.p.}\end{aligned}$$

- The value of these cashflows at $t = 2$ is:

$$\begin{aligned}\sum_{r=1}^{n} c_r V(t, t_r) &= c_1 V(1.5, t_1) + c_2 V(1.5, t_2) \\ &= 1,000(1.04)^1 + 1,000(1.04)^{-1} \\ &= 2,001.54 \qquad \text{to 2 d.p.}\end{aligned}$$

Example 3.1.2 Suppose the effective rate of interest is 6% per time unit, and the cashflows are defined by the following timeline (we may often define cashflows in this way rather than writing them down).



- The accumulation at time $t = 6$ of these cashflows is:

$$\begin{aligned}\text{Accumulation} &= \sum_{r=1}^{n} c_r V(t, t_r) \\ &= \sum_{r=1}^{4} c_r V(6, t_r) \\ &= 100(1.06)^5 - 20(1.06)^4 + 50(1.06)^3 - 10(1.06)^1 \\ &= 157.5238 \to 157.52 \qquad \text{to 2 d.p.}\end{aligned}$$

- The present value at time $t = 0$ of these cashflows is:

$$\begin{aligned}
\text{Present Value} &= \sum_{r=1}^{n} c_r V(t, t_r) \\
&= \sum_{r=1}^{4} c_r V(0, t_r) \\
&= 100(1.06)^{-1} - 20(1.06)^{-2} + 50(1.06)^{-3} - 10(1.06)^{-5} \\
&= 111.0481 \to 111.05 \qquad \text{to 2 d.p.}
\end{aligned}$$

- The value at time $t = 3$ of these cashflows is:

$$\begin{aligned}
\text{Value} &= \sum_{r=1}^{n} c_r V(t, t_r) \\
&= \sum_{r=1}^{4} c_r V(3, t_r) \\
&= 100(1.06)^{2} - 20(1.06)^{1} + 50(1.06)^{0} - 10(1.06)^{-2} \\
&= 132.2600 \to 132.26 \qquad \text{to 2 d.p.}
\end{aligned}$$

Observation: The observant reader will notice straightaway that:

$$\begin{aligned}
111.0481 &= 157.5238(1.06)^{-6} \\
132.2600 &= 157.5238(1.06)^{-3} = 111.0481(1.06)^{3}.
\end{aligned}$$

It seems that once we have calculated the value of a series of cashflows at any one time t, we can find its value at any other time s, say, just by applying the appropriate factor $V(s, t)$. In fact this is correct. Given an arbitrary series of cashflows as defined at the start of this chapter, its value at time t is, by definition:

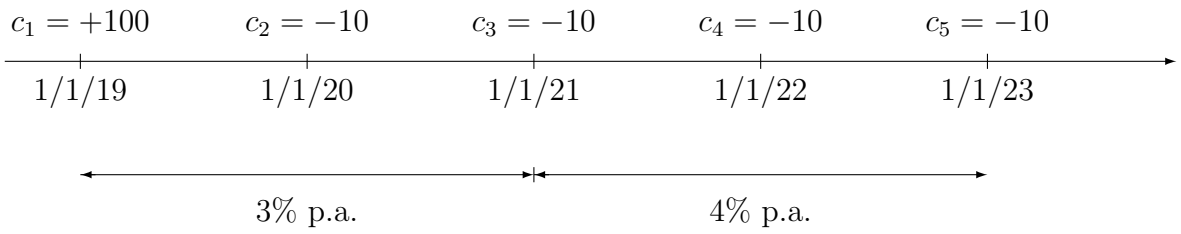
$$\sum_{r=1}^{n} c_r V(t, t_r).$$

and its value at time s is:

$$\begin{aligned}
\sum_{r=1}^{n} c_r V(s, t_r) &= \sum_{r=1}^{n} c_r V(s, t) V(t, t_r) \qquad \text{(Second Rule)} \\
&= V(s, t) \sum_{r=1}^{n} c_r V(t, t_r) \\
&= V(s, t) \times \text{value of cashflows at time } t.
\end{aligned}$$

One way to view this is that, once we have calculated the value of a series of cashflows at some time t, that value is equivalent to a lump sum or single cashflow, equal to the calculated value and payable at time t.

Example 3.1.3 Suppose the effective rate of interest is 3% p.a. during 2019 and 2020, and 4% p.a. thereafter. The cashflows are as in the timeline below.



What are the values of these cashflows on 1/1/24 (an accumulation), on 1/1/19 (a present value), and on 1/7/21 (a mixture)?

- The accumulation on 1/1/24 is $\sum_{r=1}^{n} c_r V(1/1/24, t_r)$ which is:

$$\begin{aligned} &= c_1 V(1/1/24, 1/1/19) + c_2 V(1/1/14, 1/1/20) + \\ &\quad + c_3 V(1/1/24, 1/1/21) + c_4 V(1/1/24, 1/1/22) + c_5 V(1/1/24, 1/1/23) \\ &= 100(1.03)^2(1.04)^3 - 10(1.03)^1(1.04)^3 - 10(1.04)^3 - 10(1.04)^2 - 10(1.04)^1 \\ &= 75.28608 \to 75.29 \text{ to 2 d.p..} \end{aligned}$$

- The present value at 1/1/19 is then:

$$\begin{aligned} \text{Value at } 1/1/24 \times V(1/1/19, 1/1/24) &= 75.28608(1.03)^{-2}(1.04)^{-3} \\ &= 63.08705 \to 63.09 \text{ to 2 d.p..} \end{aligned}$$

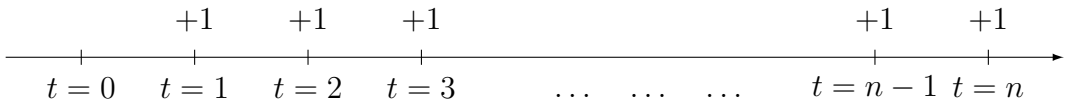
- The value on 1/7/21 is then:

$$\begin{aligned} \text{Value at } 1/1/19 \times V(1/7/21, 1/1/19) &= 63.08705(1.03)^2(1.04)^{1/2} \\ &= 68.25451 \to 68.25 \text{ to 2 d.p..} \end{aligned}$$

3.2 Level Annuities-Certain

An *annuity* is a regular series of cashflows, usually purchased from, or payable to, a financial institution. A pension is an example. When the cashflows are certain, we call it an *annuity-certain*. Simple annuities such as a constant amount of money per time unit for a prespecified number n time units appear very often in financial problems. Consequently, they have been codified, and the most useful present values and accumulations of these annuities are part of the standard *international actuarial notation*.

The simplest level annuity-certain has cashflows of 1 at the *end* of each of the next n time units. We say this annuity is payable once per time unit *in arrears* for n time units (for n a positive integer). In actuarial practice, payments are made 'in arrears' if they are made at the end of the relevant time units. The timeline of such an annuity, from the point of view of the party receiving the cashflows, is as follows:



Notation: Let i be the effective rate of interest per time unit, assumed constant. Then the following symbols are standard in actuarial practice.

$$\begin{aligned} s^{i}_{\overline{n}|} &= \text{the accumulation at time } n \text{ of this annuity} \\ a^{i}_{\overline{n}|} &= \text{the present value at time 0 of this annuity.} \end{aligned}$$

Comments:

- The number of time units for which the annuity is payable is called the *term* of the annuity.
- When the rate of interest i is clear from the context and there is no risk of ambiguity, it is often omitted and the symbols $s_{\overline{n}|}$ and $a_{\overline{n}|}$ are used.
- If the rate of interest is not constant there is no standard actuarial notation for annuity accumulations and present values and the use of these symbols is then *not correct*.

Very Important Result:

(i) $s^{i}_{\overline{n}|} = \dfrac{(1+i)^n - 1}{i}$

(ii) $a^{i}_{\overline{n}|} = \dfrac{1 - v^n}{i}$.

Proof of (i):

$$\begin{aligned} s^{i}_{\overline{n}|} &= \sum_{r=1}^{n} c_r V(n, t_r) \\ &= (1+i)^{n-1} + (1+i)^{n-2} + \ldots + (1+i)^1 + (1+i)^0. \end{aligned}$$

Therefore;

$$(1+i)s^{i}_{\overline{n}|} = (1+i)^n + (1+i)^{n-1} + \ldots + (1+i)^2 + (1+i)^1.$$

Therefore;

$$\begin{aligned} i s^{i}_{\overline{n}|} &= (1+i)s^{i}_{\overline{n}|} - s^{i}_{\overline{n}|} \\ &= (1+i)^n + (1+i)^{n-1} + \ldots + (1+i)^2 + (1+i)^1 \\ &\quad - (1+i)^{n-1} - (1+i)^{n-2} - \ldots - (1+i)^1 - (1+i)^0 \\ &= (1+i)^n - 1. \end{aligned}$$

So on dividing by i:

$$s^{i}_{\overline{n}|} = \frac{(1+i)^n - 1}{i}.$$

Comment: The reader may recognize that this is just summing a geometric series, a standard result in basic algebra.

Proof of (ii): $a^{i}_{\overline{n}|} = V(0,n)s^{i}_{\overline{n}|} = (1+i)^{-n}s^{i}_{\overline{n}|} = \dfrac{1-v^n}{i}$.

Example 3.2.1 Let the rate of interest be 12% p.a. effective. Write down the standard symbol for, and evaluate, the following.

(i) The accumulation at time 20 years of £1,000 per annum payable yearly in arrears from now, for twenty years.

Answer: $1,000\, s^{0.12}_{\overline{20}|} = 1,000\, \dfrac{1.12^{20} - 1}{0.12} = £72,052.44.$ (to 2 d.p.).

(ii) The present value now of £7,000 payable yearly in arrears for the next 15 years.

Answer: $7,000\, a^{0.12}_{\overline{15}|} = 7,000\, \dfrac{1 - 1.12^{-15}}{0.12} = £47,676.05$ (to 2 d.p.).

(iii) The present value now of £500 payable at the end of each of the next 14 half-years.

Answer: It is easiest to change time unit to the half year, and use effective interest of:

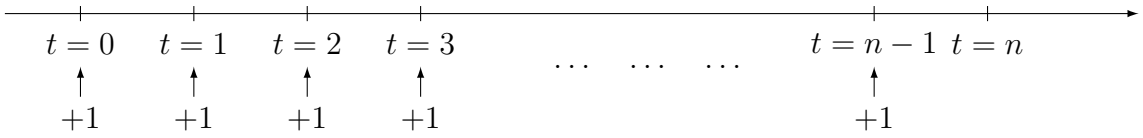
$$1.12^{1/2} - 1 = 0.058301 \qquad (6 \text{ d.p.})$$

per half year. Then the annuity present value is:

$500\, a^{0.058301}_{\overline{14}|} = 500\, \dfrac{1 - 1.058301^{-14}}{0.058301} = £4,696.78$ (to 2 d.p.).

3.3 Level Annuities-Due

An *annuity-due* is an annuity where the payments are made at the *start* of each time unit, rather than at the end of each time unit. We also say that the payments are *in advance* rather than in arrears. The simplest level annuity-due pays a constant amount of 1 per time unit for a total of n time units, which again is called the *term* of the annuity. The timeline for this annuity is as follows.



Comment: Note that there is a payment at time 0 but no payment at time n.

Notation: Let i be the effective rate of interest, assumed constant. Then the following symbols are standard in actuarial practice.

$$\begin{aligned} \ddot{s}^{i}_{\overline{n}|} &= \text{the accumulation at time } n \text{ of this annuity} \\ \ddot{a}^{i}_{\overline{n}|} &= \text{the present value at time 0 of this annuity.} \end{aligned}$$

Comments:

- In speech, we say 's-due-n' and 'a-due-n'.
- When the rate of interest i is clear from the context and there is no risk of ambiguity, it is often omitted and the symbols $\ddot{s}_{\overline{n}|}$ and $\ddot{a}_{\overline{n}|}$ are used.
- If the rate of interest is not constant there is no standard actuarial notation for annuity accumulations and present values and the use of these symbols is then *not correct*.

Very Important Result:

(i) $\ddot{s}^{i}_{\overline{n}|} = \dfrac{(1+i)^n - 1}{d}$

(ii) $\ddot{a}^{i}_{\overline{n}|} = \dfrac{1 - v^n}{d}$.

Proof of (i):

$$\begin{aligned} \ddot{s}^{i}_{\overline{n}|} &= \sum_{r=1}^{n} c_r V(n, t_r) \\ &= (1+i)^n + (1+i)^{n-1} + \ldots + (1+i)^2 + (1+i)^1. \end{aligned}$$

Therefore;

$$(1+i)\ddot{s}^{i}_{\overline{n}|} = (1+i)^{n+1} + (1+i)^{n} + \ldots + (1+i)^{3} + (1+i)^{2}.$$

Therefore;

$$\begin{aligned} i\ddot{s}^{i}_{\overline{n}|} &= (1+i)\ddot{s}^{i}_{\overline{n}|} - \ddot{s}^{i}_{\overline{n}|} \\ &= (1+i)^{n+1} + (1+i)^{n} + \ldots + (1+i)^{3} + (1+i)^{2} \\ &\quad - (1+i)^{n} - (1+i)^{n-1} - \ldots - (1+i)^{2} - (1+i)^{1} \\ &= (1+i)^{n+1} - (1+i)^{1} \\ &= (1+i)((1+i)^{n} - 1). \end{aligned}$$

So on dividing by i, and using $d = i/(1+i)$, we get:

$$\ddot{s}^{i}_{\overline{n}|} = \frac{(1+i)^{n} - 1}{d}.$$

Proof of (ii): $\ddot{a}^{i}_{\overline{n}|} = V(0,n)\ddot{s}^{i}_{\overline{n}|} = (1+i)^{-n}\ddot{s}^{i}_{\overline{n}|} = \dfrac{1-v^{n}}{d}$.

Example 3.3.1 Let the rate of interest be 12% p.a. effective.

(i) Find $\ddot{s}^{0.12}_{\overline{20}|}$.

Answer: $\ddot{s}^{0.12}_{\overline{20}|} = (1.12)\dfrac{1.12^{20} - 1}{0.12} = 80.698735$ (to 6 d.p.).

(ii) Find $\ddot{a}^{0.12}_{\overline{15}|}$.

Answer: $\ddot{a}^{0.12}_{\overline{15}|} = (1.12)\dfrac{1 - 1.12^{-15}}{0.12} = 7.628168$ (to 6 d.p.).

(iii) Find the present value now of £500 payable at the start of each of the next 14 half-years.

Answer: It is easiest to change time unit to the half year, and use effective interest of 0.058301 per half year (see last example). Then the annuity present value is:

$$500\,\ddot{a}^{0.058301}_{\overline{14}|} = 500\,(1.058301)\frac{1 - 1.058301^{-14}}{0.058301} = £4,970.60 \text{ (to 2 d.p.)}.$$

Comments:

- As long as interest is positive ($i > 0$) a cashflow received sooner is worth more than a cashflow received later. Under an annuity-due, each cashflow is received at the start of a time unit instead of at the end of a time unit, so we intuitively expect that $\ddot{s}^{i}_{\overline{n}|} > s^{i}_{\overline{n}|}$ and $\ddot{a}^{i}_{\overline{n}|} > a^{i}_{\overline{n}|}$. This is indeed the case, and is easily shown from the identities in the next section.

- An annuity with cashflows of 1 per time unit is obviously meant to be applied to level cashflows of any amount, so the question of truncation error when applying quantities like $\ddot{a}^i_{\overline{n}|}$ to very large cashflows arises, just as it did for interest and discount rates. In the latter case we adopted the convention of using at least six decimal places of accuracy. For annuities we will be satisfied with three decimal places of accuracy. Thus, the answers to (i) and (ii) above could have been rounded to 80.699 and 7.628, respectively[1].

3.4 Some Important Relationships

The following relationships are straightforward to prove algebraically.

(i) $\ddot{s}^i_{\overline{n}|} = (1+i)s^i_{\overline{n}|}$.

(ii) $\ddot{a}^i_{\overline{n}|} = (1+i)a^i_{\overline{n}|}$.

(iii) $s^i_{\overline{n}|} = 1 + \ddot{s}^i_{\overline{n-1}|}$.

(iv) $\ddot{a}^i_{\overline{n}|} = 1 + a^i_{\overline{n-1}|}$.

The first two of these give us a proof of the intuitive statement given above, that if $i > 0$ we should expect that $\ddot{s}^i_{\overline{n}|} > s^i_{\overline{n}|}$ and $\ddot{a}^i_{\overline{n}|} > a^i_{\overline{n}|}$.

Exercise: Prove these results.

3.5 Perpetuities

A *perpetuity* is an annuity that never ceases payment; it has an infinite term. Examples do exist, for example certain UK Government securities called Consols.

As usual, we consider the simplest examples, perpetuities with level payments of 1 per time unit. These can be payable at the start or the end of the time unit (the former may be called a 'perpetuity-due') and their present values have symbols in the international actuarial notation.

Notation: Let the effective rate of interest per time unit be i, a constant. Then define:

$a^i_{\overline{\infty}|}$ = the present value at time 0 of a perpetuity of 1 per time unit payable in arrears

$\ddot{a}^i_{\overline{\infty}|}$ = the present value at time 0 of a perpetuity of 1 per time unit payable in advance.

We have not defined accumulations of perpetuities, for the good reason that their term never ends; there are always payments in the future.

[1] The reason for choosing six decimal places for interest rates and three decimal places for annuity values is that this is the convention used in the *Formulae and Tables for Actuarial Examinations* published by the Institute and Faculty of Actuaries. Remember that it is just a convention.

Result: If $i > 0$ then:

(i) $a^{i}_{\overline{\infty}|} = \dfrac{1}{i}$.

(ii) $\ddot{a}^{i}_{\overline{\infty}|} = \dfrac{1}{d}$.

Proof of (i): Clearly, $a^{i}_{\overline{\infty}|}$ is the sum of an infinite series, which by definition (if it converges) is the limit of the series of its partial sums. The partial sums are just the present values of annuities with terms 1, 2, 3, ... time units. So we have:

$$\begin{aligned} a^{i}_{\overline{\infty}|} &= \lim_{n\to\infty} a^{i}_{\overline{n}|} \\ &= \lim_{n\to\infty} \frac{1-v^n}{i} \\ &= \frac{1}{i} \qquad \text{since } i > 0 \Rightarrow \lim_{n\to\infty} v^n = 0. \end{aligned}$$

(ii) is proved similarly.

Comment: A common student error is to 'prove' these results by writing, for example:

$$\lim_{n\to\infty} v^n = v^{\infty} = 0.$$

This is incorrect mathematics.

Example 3.5.1 Let the effective rate of interest be 5% per time unit.

(i) Find $a^{0.05}_{\overline{\infty}|}$.

Answer: $a^{0.05}_{\overline{\infty}|} = 1/0.05 = 20.000$.

Note that we write 20.000, not 20. This is because of our convention about displaying a minimum number of decimal places. We do not drop the final decimal places even if they are zeros.

(ii) Find $\ddot{a}^{0.05}_{\overline{\infty}|}$.

Answer: $\ddot{a}^{0.05}_{\overline{\infty}|} = 1.05/0.05 = 21.000$.

(iii) Find the value at time $t = 10$ of a perpetuity of £100 per time unit, payable annually in arrears and commencing now.

Although we cannot find the accumulation of a perpetuity, we may certainly calculate its value at any finite time in the future. Also, note that when we say the annuity commences now, that does *not* mean that the first payment is made now. Because payment is in arrears, the first payment will be received at the end of the first time unit.

At $t = 10$, we have past cashflows of 1 per time unit, in arrears, for the last ten time units, while in the future we have a perpetuity of 1 per time unit payable in arrears. Hence the value we want is:

$$100(s_{\overline{10}|}^{0.05} + a_{\overline{\infty}|}^{0.05}) = 100(12.578 + 20.000) = £3,257.80 \text{ to 2 d.p.}$$

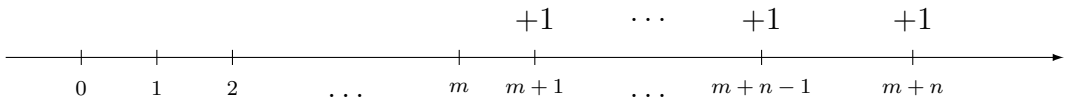
Note that (ii) could have been deduced from (i) using result (iv) in Section 3.4, because $\lim_{n\to\infty} a_{\overline{n-1}|} = \lim_{n\to\infty} a_{\overline{n}|}$.

3.6 Deferred Annuities

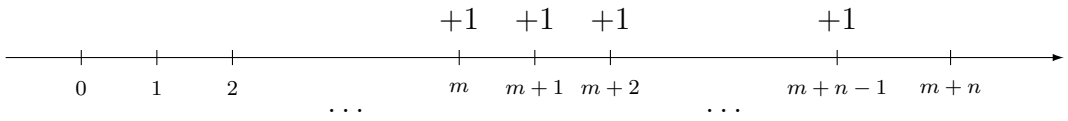
Sometimes an annuity does not start immediately[2], but is deferred for a period of time. For example, an employer might promise to pay an annuity to an employee when he/she retires, in other words, a pension. Pensions are the most important example of deferred annuities.

Notation: International actuarial notation indicates an annuity deferred for m time units by a prefix $m|$. So, for example:

- The symbol ${}_{m|}a_{\overline{n}|}^{i}$ denotes the present value at time 0 of an annuity of 1 per time unit payable in arrears for n time units, commencing after m time units, where the effective rate of interest is a constant i per time unit. The first payment is at time $m + 1$, because payment is in arrears. In speech we say 'm-deferred-a-n'. The timeline fot this annuity is as follows.



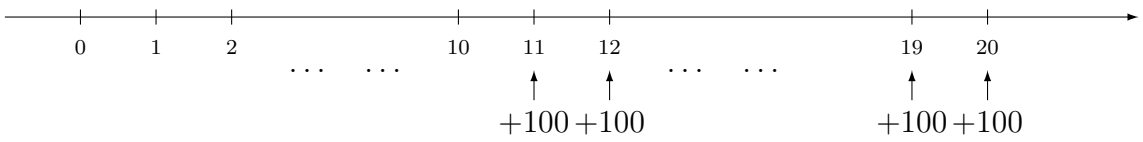
- The symbol ${}_{m|}\ddot{a}_{\overline{n}|}^{i}$ denotes the present value, at time 0, of an annuity of 1 per time unit payable in advance for n time units, commencing after m time units, where the effective rate of interest is a constant i per time unit. The first payment is at time m, because payment is in advance, and there is no payment at time $m + n$. In speech we say 'm-deferred-a-due-n'. A representative timeline for these cashflows is as follows.



[2]An annuity that starts immediately is called an immediate annuity, to distinguish it from a deferred annuity. However, an immediate annuity does not necessarily commence with the first payment being made immediately. It does if it is payable in advance (an annuity-due) but if it is payable in arrears the first payment is at the end of the first time unit after the annuity commences.

Example 3.6.1 Find the present value at outset of a level annuity of £100 per year, payable annually in arrears for ten years, deferred for ten years, if the effective rate of interest is 10% p.a..

Answer: We first illustrate the timeline for this annuity.



Then we have:

$$\begin{aligned} {}_{10}|a^{i}_{\overline{10}|} &= \sum_{r=1}^{10} c_r V(0,t_r) \\ &= \sum_{r=1}^{10} 100\, v^{10+r} \\ &= 100\, v^{10} \sum_{r=1}^{10} v^r \\ &= 100\, v^{10} a^{i}_{\overline{10}|} \end{aligned}$$

which, evaluated at $i = 0.1$ is £236.90 (to 2 d.p.). Note that this is an example of finding the value at $t = 0$ by first finding the value at $t = 10$, and then discounting back to $t = 0$.

Comment: We could define symbols such as ${}_{m}|s^{i}_{\overline{n}|}$ for the accumulation of deferred annuity payments at the end of the payment term, but a glance at the timelines above should be enough to realize that the 'deferred' part of the notation would be superfluous. Once the annuity is in payment, the deferred period is irrelevent, and clearly ${}_{m}|s^{i}_{\overline{n}|} = s^{i}_{\overline{n}|}$ and so on.

Example 3.6.2 Find the present value at outset of a level annuity of £100 per year, payable annually in arrears for ten years, deferred for ten years, if the effective rate of interest is 10% p.a. during the first ten years and 8% p.a. during the last ten years.

Answer: First note that we can no longer use the symbol ${}_{m}|a^{i}_{\overline{n}|}$ because there is not now a single unambiguous interest rate i. We have:

$$\begin{aligned} \text{PV} &= \sum_{r=1}^{10} c_r V(0,t_r) \\ &= \sum_{r=1}^{10} 100\, V(0,10)\, V(10,t_r) \\ &= 100\, V(0,10) \sum_{r=1}^{10} V(10,t_r) \\ &= 100\,(1.10)^{-10} a_{\overline{10}|}^{0.08} \end{aligned}$$

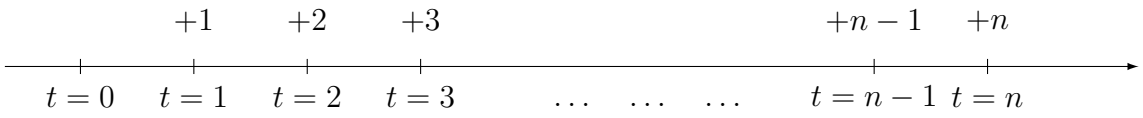
which evaluates to £258.70 (to 2 d.p.).

3.7 Increasing Annuities

We find it useful sometimes to work with annuities that increase at a regular rate.

3.7.1 Simple (Arithmetically) Increasing Annuities

The simplest form of increasing annuity is one that increases by the same amount each time unit, and the simplest such annuity is that which has first payment 1 and then increases by one per time step. Its timelines are illustrated below, first when payable in arrears:



and second when payable in advance.



Notation: Let i be the effective rate of interest, assumed constant. Then the following symbols are standard in actuarial practice.

$$\begin{aligned} (Is)_{\overline{n}|}^{i} &= \text{the accumulation at time } n \text{ of this annuity paid in arrears.} \\ (Ia)_{\overline{n}|}^{i} &= \text{the present value at time 0 of this annuity paid in arrears.} \\ (I\ddot{s})_{\overline{n}|}^{i} &= \text{the accumulation at time } n \text{ of this annuity paid in advance.} \\ (I\ddot{a})_{\overline{n}|}^{i} &= \text{the present value at time 0 of this annuity paid in advance.} \end{aligned}$$

Very Important Result:

(i) $(Is)^{i}_{\overline{n}|} = \dfrac{\ddot{s}^{i}_{\overline{n}|} - n}{i}$

(ii) $(Ia)^{i}_{\overline{n}|} = \dfrac{\ddot{a}^{i}_{\overline{n}|} - nv^{n}}{i}$

(iii) $(I\ddot{s})^{i}_{\overline{n}|} = \dfrac{\ddot{s}^{i}_{\overline{n}|} - n}{d}$

(iv) $(I\ddot{a})^{i}_{\overline{n}|} = \dfrac{\ddot{a}^{i}_{\overline{n}|} - nv^{n}}{d}$.

Proof of (i):

$$\begin{aligned}(Is)^{i}_{\overline{n}|} &= \sum_{r=1}^{n} c_r V(n, t_r) \\ &= (1+i)^{n-1} + 2(1+i)^{n-2} + \ldots + (n-1)(1+i)^{1} + n(1+i)^{0}.\end{aligned}$$

Therefore:

$$(1+i)(Is)^{i}_{\overline{n}|} = (1+i)^{n} + 2(1+i)^{n-1} + \ldots + (n-1)(1+i)^{2} + n(1+i)^{1}.$$

Therefore:

$$\begin{aligned}i(Is)^{i}_{\overline{n}|} &= (1+i)(Is)^{i}_{\overline{n}|} - (Is)^{i}_{\overline{n}|} \\ &= (1+i)^{n} + 2(1+i)^{n-1} + \ldots + (n-1)(1+i)^{2} + n(1+i)^{1} \\ &\quad - (1+i)^{n-1} - 2(1+i)^{n-2} - \ldots - (n-1)(1+i)^{1} - n(1+i)^{0} \\ &= (1+i)^{n} + (1+i)^{n-1} + \ldots + (1+i)^{2} + (1+i)^{1} - n \\ &= \ddot{s}^{i}_{\overline{n}|} - n.\end{aligned}$$

So on dividing by i, we get:

$$(Is)^{i}_{\overline{n}|} = \frac{\ddot{s}^{i}_{\overline{n}|} - n}{i}.$$

The proofs of (ii) to (iv) are left as an exercise.

Example 3.7.1 Let the effective rate of interest be a constant 5% p.a.. Find the present value at outset of an annuity payable annually in advance, for ten years, whose first payment is £1 and whose payments increase by £1 p.a..

Answer: We want $(I\ddot{a})^{0.05}_{\overline{10}|}$. First we evaluate:

$$1.05^{-10} = 0.613913 \qquad \text{and} \qquad \ddot{a}^{0.05}_{\overline{10}|} = 8.108$$

and then:

$$(I\ddot{a})^{0.05}_{\overline{10}|} = (1.05)\left(\frac{\ddot{a}^{0.05}_{\overline{10}|} - 10(1.05)^{-10}}{0.05}\right) = 41.346$$

where we have used the identity $d = i/(1+i)$.

Level annuities and the basic arithmetically increasing annuities defined can be used as building blocks to construct any annuity with arithmetically increasing or decreasing payments. In the next two examples, let the effective rate of interest be a constant i p.a..

Example 3.7.2 Write down, in terms of the standard annuity symbols defined in this chapter, an expression for the present value of an annuity payable annually in arrears for 20 years, whose first payment is £10,000 and whose payments increase by £1,000 each year.

Answer: $9,000\, a^{i}_{\overline{20}|} + 1,000\, (Ia)^{i}_{\overline{20}|}$.

Make sure you understand why the level annuity is of amount £9,000 p.a. and not £10,000 p.a..

Example 3.7.3 Write down, in terms of the standard annuity symbols defined in this chapter, an expression for the present value of an annuity payable annually in advance for 15 years, whose first payment is £30,000 and whose payments decrease by £2,000 each year.

Answer: $32,000\, \ddot{a}^{i}_{\overline{15}|} - 2,000\, (I\ddot{a})^{i}_{\overline{15}|}$.

Make sure you understand why the level annuity is of amount £32,000 p.a. and not £30,000 p.a..

3.7.2 Compound (Geometrically) Increasing Annuities

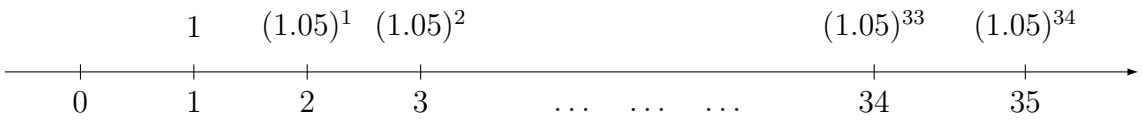
Another common form of increasing annuity is one which increases by a fixed multiple or factor each time unit. An example might be a pension which increases by 5% each year to allow for possible inflation.

Such annuities are not included in the international actuarial notation, so they just have to be described in words. There are also no standard formulae for dealing with them, so expressions for values have to be written down from first principles. Fortunately, the actions of applying compound interest, and increasing (or decreasing) payments by a fixed multiple each year, are basically the same, so we can usually reduce a present value or accumulation to a familiar looking form.

Example 3.7.4 Suppose a person now age 65 has a pension payable annually in arrears, whose first payment is £1 (so will be paid on their 66th birthday). The last payment will be made on their 100th birthday. To combat inflation, payments will grow by 5% each year; that is, each payment will be 5% greater than the previous payment. If the effective

rate of interest is a constant 7% p.a. throughout, what is the present value of this pension now?

Answer: First we draw up a timeline for the pension payments, noting that the annuity has a term of 35 years. We suppress the signs of the cashflows just for a less cluttered appearance.



Now we just have to proceed from first principles.

$$\begin{aligned} \text{PV} &= \sum_{r=1}^{35} c_r \, V(0, t_r) \\ &= \sum_{r=1}^{35} (1.05)^{r-1} (1.07)^{-r} \\ &= (1.05)^{-1} \sum_{r=1}^{35} (1.05)^{r} (1.07)^{-r}. \end{aligned}$$

Now notice that the terms in the sum form a geometric series, exactly like we are used to dealing with for level annuities. If we define an 'artificial' interest rate as follows:

$$1 + j = \frac{1.07}{1.05} \qquad \Rightarrow j = \frac{1.07}{1,05} - 1 = 0.019048 \quad (6 \text{ d.p.})$$

then the present value is:

$$\begin{aligned} \text{PV} &= (1.05)^{-1} \sum_{r=1}^{35} (1 + j)^{-r} \\ &= (1.05)^{-1} \, a_{\overline{35}|}^{j}. \end{aligned}$$

Thus, the calculation reduces to the evaluation of a *level* annuity, but at the artificial interest rate j. We find the present value to be $24.168 \rightarrow$ £24.17 (rounded to the nearest pence).

Comments:

- All problems involving compound increasing annuities can be resolved in this way, by defining the 'interest rate' j and then trying to extract the right common factors from the payments so we are left with the present value of a *level* annuity evaluated at rate j. In the example above, the right common factor was $(1.05)^{-1}$. The details will depend on the circumstances *and must be worked out from first principles each time.*

- It is instructive to consider the answer just given. For every £1 of income in retirement, with some protection against inflation and living to 100, but assuming you can invest your retirement fund to earn 7% p.a., you need to have saved a bit more than £24 by the time you retire. So, if you had just finished a well-paid career and you wanted to maintain that standard of living in retirement, let us say a minimum pension of £30,000 p.a., you will need to have saved nearly £0.75 million in today's prices. In reality it is a bit more complicated than that — see Life Insurance Mathematics — but the message remains true: good pensions are very, very expensive.

Chapter 4

Nominal Interest Rates

Throughout this section the time unit will be one year.

4.1 Nominal Rates of Interest

Say you put £100 into a savings account at a bank for a year. Which effective interest rate would you rather have

- 6% per annum?
- 2.9563% per half-year?

Consider a rate of interest i per annum.

- **Effective** interest means that interest of £i is paid at the year end on £1 invested at the start of the year.
- **Nominal** interest of $i^{(m)}$ per annum compounding m times a year means that £$\frac{i^{(m)}}{m}$ is paid at the end of each fraction $\frac{1}{m}$ of a year on £1 invested at the start of the year.

Example 4.1.1 The following are equivalent:

Effective Rate (p.a.)	**Nominal Rate (p.a.)**
$i = 0.06$	$i^{(2)} = 0.059126$
$i = 0.12$	$i^{(12)} = 0.113866$
$i = 0.082432$	$i^{(4)} = 0.08$
$i = 0.092025$	$i^{(2)} = 0.09$

$i^{(m)}$ per annum convertible m^{thly} or m times per year
is identical to
an effective rate of interest $\frac{i^{(m)}}{m}$ per $\frac{1}{m}$ year.

It is usually easier to change to the appropriate time unit and use the effective rate of interest, but you might still come across this (pre-calculator) notation. It can be useful for remembering some of the annuity formula.

The relationship between the effective interest rate i p.a. and the nominal interest rate $i^{(m)}$ p.a. is

$$\boxed{1+i=\left(1+\frac{i^{(m)}}{m}\right)^m.}$$

Exercise 4.1 The effective rate of interest is 3.5% per annum. Express this as the equivalent annual nominal rate of interest:

(i) convertible quarterly.

(ii) convertible 12 times a year.

Exercise 4.2 The nominal rate of interest per annum convertible half-yearly is 5% per annum. Express this as the equivalent annual

(i) effective rate of interest.

(ii) nominal rate of interest convertible monthly.

4.2 Nominal Rates of Discount

Recall that $d=1-v$ is the effective rate of discount per annum. It could be interpreted as the amount of interest payable at the start of the year that is equivalent to i payable at the end of the year.

Example 4.2.1 Generally, bank or savings accounts pay interest at the end of a set time period. However, Santander, a retail bank, launched a 3-year savings account in 2011 that paid interest upfront. No withdrawals were permitted during the 3 year time period.

The *nominal discount rate* $d^{(m)}$ represents interest of $\frac{d^{(m)}}{m}$ paid at the start of each $\frac{1}{m}$ of a year.

$d^{(m)}$ per annum convertible m^{thly} or m times per year
is identical to
an effective rate of discount $\frac{d^{(m)}}{m}$ per $\frac{1}{m}$ year.

The relationship between the effective discount rate d p.a. and the nominal discount rate $d^{(m)}$ p.a. is

$$\boxed{1-d=\left(1-\frac{d^{(m)}}{m}\right)^m.}$$

Example 4.2.2 The nominal discount rate convertible quarterly is 8% per annum. Express this as the equivalent annual:

(i) effective discount rate.

(ii) effective interest rate.
Solution: We are told $d^{(4)} = 0.08$ p.a. Hence

$$1 - d = \left(1 - \frac{0.08}{4}\right)^4 \qquad \Rightarrow \qquad d = 1 - 0.98^4 = 0.077632.$$

Then as $v = 1 - d = 0.98^4$, we find

$$i = \frac{1}{0.98^4} - 1 = 0.084166.$$

Exercise 4.3 The effective rate of discount is 4% per annum. Express this as the equivalent annual nominal rate of discount:

(i) convertible quarterly.

(ii) convertible half-yearly.

Exercise 4.4 The nominal rate of discount per annum convertible monthly is 6% per annum. Express this as the equivalent annual:

(i) effective rate of discount.

(ii) nominal rate of discount convertible quarterly.

Table 4.1: Equivalent Payments

	0	$1/p$	$2/p$	$3/p$	...	$(p-1)/p$	1
(1)	d						
(2)	$d^{(p)}/p$	$d^{(p)}/p$	$d^{(p)}/p$	$d^{(p)}/p$	...	$d^{(p)}/p$	
(3)		$i^{(p)}/p$	$i^{(p)}/p$	$i^{(p)}/p$	...	$i^{(p)}/p$	$i^{(p)}/p$
(4)							i

Example 4.2.3 £2500 is invested in a savings account. The nominal rate of interest per annum convertible monthly for the first 3 months is 24% and the nominal rate of interest convertible quarterly for the next 9 months is 16%. How much is in the account at the end of the year?
Solution:

$$\begin{aligned}
&\Longrightarrow \quad 2500\left(1 + \frac{i^{(12)}}{12}\right)^3\left(1 + \frac{i^{(4)}}{4}\right)^3 \\
&\Longrightarrow \quad 2500\left(1 + \frac{0.24}{12}\right)^3\left(1 + \frac{0.16}{4}\right)^3 \\
&\Longrightarrow \quad 2500(1 + 0.02)^3(1 + 0.04)^3 \\
&\Longrightarrow \quad 2984.29
\end{aligned}$$

Annuities payable $m$thly

- $a_{\overline{n}|}^{(m)}$: The value at time zero of a level annuity payable $m$thly in arrears at the rate of 1 per unit time over the time interval $[0, n]$.
- $a_{\overline{n}|}^{(m)}$: The value at time zero of a level annuity which pays $1/m$ in arrears m times per year for a period of n years.
- $s_{\overline{n}|}^{(m)}$: The value at time n of a level annuity payable $m$thly in arrears at the rate of 1 per unit time over the time interval $[0, n]$.

Useful results

$$\begin{aligned}
a_{\overline{n}|}^{(m)} &= \sum_{t=1}^{nm} \frac{1}{m} v^{\frac{t}{m}} = \frac{1-v^n}{i^{(m)}} \\
\ddot{a}_{\overline{n}|}^{(m)} &= \sum_{t=1}^{nm} \frac{1}{m} v^{\frac{t-1}{m}} = \frac{1-v^n}{d^{(m)}} \\
a_{\overline{\infty}|}^{(m)} &= \frac{1}{i^{(m)}} \\
\ddot{a}_{\overline{\infty}|}^{(m)} &= \frac{1}{d^{(m)}} \\
s_{\overline{n}|}^{(m)} &= \sum_{t=1}^{nm} \frac{1}{m} (1+i)^{n-\frac{t}{m}} = \frac{(1+i)^n - 1}{i^{(m)}} \\
\ddot{s}_{\overline{n}|}^{(m)} &= \sum_{t=1}^{nm} \frac{1}{m} (1+i)^{n-\frac{t-1}{m}} = \frac{(1+i)^n - 1}{d^{(m)}}
\end{aligned}$$

Exercise 4.5 Calculate the present value as at 1 January 2014 of the payments of £100 made on the first day of each quarter during the calendar years 2014 and 2015, if the effective rate of interest is 8% per annum.

Exercise 4.6 An investor who pays income tax at 20% on all their income buys, at issue, a 25-year bond to be redeemed at par bearing semi-annual (twice per year) coupons of 8% per annum. What price will the investor pay per £100 nominal to obtain a redemption yield after income tax of 9% per annum effective?

Annuities payable $m$thly where m < 1

- $m < 1$: An annuity that pays less frequently than annually. For example, $m = 0.5$ means payments are made once in 2 years i.e. biennially.

Exercise 4.7 Assuming that the rate of interest is i per annum, what is the value at time 0 of a series of 7 payments, each of amount 8, made at time 4, 8, 12, . . . , 28?

Example 4.2.4 Suppose that k and r are integers greater than 1. Consider a series of payments, each of amount X, due at times r, $2r$, $3r$, . . . , kr. What is the value of this series at time 0 on the basis of an effective interest rate i per unit time?

Solution: The situation can be illustrated by replacing the payment of X due at time r by a series of r payments, each of amount Y, due at times $1, 2, \ldots, r$, where Y is chosen such that the future value at time r is equivalent to the single payment they replace i.e. X at time r. This means that $Y s_{\overline{r}|} = X$.
Hence the value of the annuity is:

$$\left(\frac{X}{s_{\overline{r}|}}\right) a_{\overline{kr}|}.$$

Exercise 4.8 A company expects to receive £10,000 from a customer. The amount will be split into ten equal instalments. The first instalment is due in one years' time and the remaining payments will be made at three-year intervals thereafter. Calculate, to the nearest £, the present value of these future receipts at an effective rate of 7% per annum.

Chapter 5

Real Interest Rates

5.1 The changing price of sugar

Suppose I buy 1kg of sugar every year on 1 January. How much does it cost me to buy 1kg of sugar? How does the cost change from year to year?
Suppose I buy 1kg of sugar at the start of every month. Suppose the prices I pay in pence each month are given in Table 5.1.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2000	166.6	167.5	168.4	170.1	170.7	171.1	170.5	170.5	171.7	171.6	172.1	172.2
2001	171.1	172.0	172.2	173.1	174.2	174.4	173.3	174.0	174.6	174.3	173.6	173.4
2002	173.3	173.8	174.5	175.7	176.2	176.2	175.9	176.4	177.6	177.9	178.2	178.5
2003	178.4	179.3	179.9	181.2	181.5	181.3	181.3	181.6	182.5	182.6	182.7	183.5
2004	183.1	183.8	184.6	185.7	186.5	186.8	186.8	187.4	188.1	188.6	189.0	189.9
2005	188.9	189.6	190.5	191.6	192.0	192.2	192.2	192.6	193.1	193.3	193.6	194.1

Table 5.1: The changing price in pence of 1kg of sugar.

Example 5.1.1 The change in the price of 1kg of sugar from January 2000 to January 2005 is

$$\frac{\text{Price of 1kg of sugar in January 2005}}{\text{Price of 1kg of sugar in January 2000}} = \frac{188.9}{166.6} = 1.13385,$$

i.e. the price of 1kg of sugar has increased by 13.385% over 5 years.
Expressed as an annual effective rate, the price change is

$$\left(\frac{\text{Price of 1kg of sugar in January 2005}}{\text{Price of 1kg of sugar in January 2000}}\right)^{1/5} = 1.02544,$$

i.e. the annualised average change in the price of 1kg of sugar is +2.544% p.a. over 5 years.

Example 5.1.2 With £16.66 in January 2000, I can buy 10kg of sugar.
With £16.66 in December 2005, how many kg of sugar can I buy?
Solution: From Table 5.1, the cost of 1kg of sugar in December 2005 is £1.941. I can buy

$$\frac{16.66}{1.941} = 8.58\text{kg of sugar in December 2005.}$$

The *purchasing power* of £16.66 has fallen since 2000. It has fallen because the price of sugar has increased.

Who might care about the changes in the price of 1kg of sugar? More widely, who might care about the changes in the prices of goods and services?

In January 2000, I buy £100 nominal of a special 5-year bond, called a "Sugar Bond". The Sugar Bond pays no coupons but its redemption value is increased by the change in the price of sugar over the 5 years from January 2000 to January 2005.

Exercise 5.1 Using Table 5.1, calculate the redemption payment at January 2005.

Exercise 5.2 Calculate how much sugar can be bought with

(i) £100 in January 2000.

(ii) the redemption payment in January 2005.

The *real value* of the redemption payment from the Sugar Bond stays constant. This is because the redemption payment changes proportionally to the change in the price of sugar.
The *money value* of the redemption payment £100 increases to £113.38535. The money value can be thought of as the number written on a banknote or a coin. Sometimes you may see the money value called the *nominal value* or the *monetary value*.

5.2 Price inflation indices

More generally, price inflation is the change of purchasing power of money. It is measured with reference to an index. Two important indices in the U.K. are

- Retail Prices Index (RPI),
- Consumer Prices Index (CPI).

They measure the monthly change in the prices of a specified set of goods and services. The Office for National Statistics (ONS) publish these indices.

Exercise 5.3 *From the ONS website, find out how these two indices differ.*

Exercise 5.4 Find out who makes use of RPI and CPI.

Example 5.2.1 You have started your dream job. At the end of your first year, your employer offers you a pay rise of 1%. You hear in the news that RPI has increased by 4% over the year and is expected to stay high in the next few years. Would you be happy with your pay rise?

Example 5.2.2 In 2012, you retired from your dream job. In 1962, you had deposited £100 000 in a savings account to provide for your retirement. The account has paid interest at an average effective rate of 6.14% per annum. Thus in 2012, the deposit was worth £1 967 679.
The average *money return* on the deposit is 6.14% per annum.
However, the RPI has increased at an average effective rate of 5.85% per annum from 1962 to 2012. Thus, in terms of 1962 purchasing power, the accumulated 2012 deposit has a real value of

$$\frac{1\,967\,679}{1.0585^{2012-1962}} = 114\,660.$$

In other words, £1 967 679 can buy as much in 2012 as £114 660 could buy in 1962.
The average *real return* or *real yield* on the deposit is +0.27% per annum, which can either be calculated from

$$\frac{1.0614}{1.0585} - 1 = 0.0027 \text{ per annum},$$

or from

$$\left(\frac{114\,660}{100\,000}\right)^{1/50} - 1 = 0.0027 \text{ per annum}.$$

5.3 Retail Prices Index (RPI)

- The RPI measures the monthly change in the prices of a specified set of goods and services.
- It is published one month in arrears (usually in the middle of the following month).

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2005	188.9	189.6	190.5	191.6	192.0	192.2	192.2	192.6	193.1	193.3	193.6	194.1
2006	193.4	194.2	195.0	196.5	197.7	198.5	198.5	199.2	200.1	200.4	201.1	202.7
2007	201.6	203.1	204.4	205.4	206.2	207.3	206.1	207.3	208.0	208.9	209.7	210.9
2008	209.8	211.4	212.1	214.0	215.1	216.8	216.5	217.2	218.4	217.7	216.0	212.9
2009	210.1	211.4	211.3	211.5	212.8	213.4	213.4	214.4	215.3	216.0	216.6	218.0
2010	217.9	219.2	220.7	222.8	223.6	224.1	223.6	224.5	225.3	225.8	226.8	228.4
2011	229.0	231.3	232.5	234.4	235.2	235.2	234.7	236.1	237.9	238.0	238.5	239.4
2012	238.0	239.9	240.8	242.5	242.4	241.8	242.1	243.0	244.2	245.6	245.6	246.8
2013	245.8	247.6	248.7	249.5	250.0	249.7	249.7	251.0	251.9	251.9	252.1	253.4
2014	252.6	254.2	254.8	255.7	255.9	256.3	256.0	257.0	257.6	257.7	257.1	257.5

Table 5.2: RPI since 2005. Earlier figures are identical to those in Table 5.1.

Example 5.3.1 The change in RPI from October 2012 to October 2013 is

$$\frac{\text{RPI(Oct 13)}}{\text{RPI(Oct 12)}} - 1 = \frac{251.9}{245.6} - 1 = 0.02565,$$

i.e. the prices of the basket of goods and services on which the RPI is based has increased by +2.565%.

Example 5.3.2 The change in RPI from October 2011 to October 2012 is

$$\frac{\text{RPI(Oct 12)}}{\text{RPI(Oct 11)}} - 1 = \frac{245.6}{238.0} - 1 = 0.03193,$$

i.e. the prices of the basket of goods and services on which the RPI is based has increased by +3.193%.

Example 5.3.3 The annualised average change in RPI from October 2011 to October 2013 is

$$\left(\frac{\text{RPI(Oct 13)}}{\text{RPI(Oct 11)}}\right)^{1/2} - 1 = \left(\frac{251.9}{238.0}\right)^{1/2} - 1 = 0.02879,$$

i.e. the prices of the basket of goods and services on which the RPI is based has increased at an average effective rate of +2.879% per annum.

Exercise 5.5 Calculate the annualised average change in RPI from January 2005 to November 2013.

Exercise 5.6 Does the RPI always increase?

Example 5.3.4 In the 2010 Budget, the Government announced that "The Government will adopt the CPI for the indexation of benefits, tax credits and public service pensions from April 2011" (Source: HM Treasury, *Budget 2010*, page 17, paragraph 1.43). Previously, RPI had been used.

- What reason did the Government give for the change?
- Is there an impact on the recipients of the benefits, tax credits and pensions?

Chapter 6

Force of Interest and Continuously-paid Cashflows

6.1 The Force of Interest

A sum of money $A(0)$ is invested at time 0 in a bank account and it grows continuously. At time t the amount of money in the bank account is $A(t)$.

Definition 6.1 *The force of interest at time t is*

$$\delta(t) = \frac{1}{A(t)} \frac{dA(t)}{dt} \tag{6.1}$$

Remark: From equation (6.1), we see

$$\delta(t) = \frac{d}{dt}(\log(A(t))$$
$$\Rightarrow \int_{t_1}^{t_2} \delta(t)dt = \log A(t_2) - \log A(t_1)$$

Rearranging, we find the important general formula:

$$A(t_2) = A(t_1) \exp\left(\int_{t_1}^{t_2} \delta(s)ds\right). \tag{6.2}$$

For example, set $t_1 = 0$ and $t_2 = t$. Then

$$A(t) = A(0) \exp\left(\int_0^t \delta(s)ds\right).$$

If we know $\delta(s)$ from $s = 0$ to $s = t$, then we can calculate

- the accumulation of $A(0)$ to time t.
- the present value of $A(t)$ at time 0.

For example, suppose the force of interest is constant, $\delta(t) = \delta$ for all $t \geq 0$ then the accumulation of $A(0) = 100$ from time 0 to:

- time 1 is $A(1) = 100 \exp\left(\int_0^1 \delta ds\right) = 100e^{\delta}$.
- time 2 is $A(2) = 100 \exp\left(\int_0^2 \delta ds\right) = 100e^{2\delta}$.
- time t is $A(t) = 100 \exp\left(\int_0^t \delta ds\right) = 100e^{\delta t}$.

Suppose we had been given an annual effective of rate of interest i. Then at time t, the accumulation of $A(0) = 100$ is

$$A(t) = 100(1+i)^t.$$

Therefore

$$A(t) = 100e^{\delta t} = 100(1+i)^t$$

i.e

$$e^{\delta} = 1 + i.$$

Exercise 6.1 Given a constant force of interest $\delta(t) = 5\%$ p.a, for all $t \geq 0$, and assuming time is measured in years, calculate

(i) the PV at time $t = 0$ of £150 paid at time $t = 1$.

(ii) the PV at time $t = 0.5$ of £200 paid at time $t = 3.25$.

(iii) the accumulation at time $t = 4$ of £50 paid at time $t = 1$.

Exercise 6.2 Write down a formula for the annual effective interest rate i in terms of the constant force of interest δ.

More generally,

- a constant force of interest δ p.a.
- an effective rate of interest i p.a.
- a nominal rate of interest $i^{(p)}$ p.a. convertible p times per year
- a nominal rate of discount $d^{(q)}$ p.a. convertible q times per year
- an effective rate of discount d p.a.

are all equivalent if and only if the following holds:

$$e^{\delta} = 1 + i = \left(1 + \frac{i^{(p)}}{p}\right)^p = \left(1 - \frac{d^{(q)}}{q}\right)^{-q} = (1-d)^{-1}.$$

Exercise 6.3 Write down a formula for the annual discount rate v in terms of the constant force of interest δ.

Exercise 6.4 Given that the annual effective rate of interest is 12%, calculate d, $d^{(4)}$, $d^{(12)}$, $i^{(4)}$, $i^{(12)}$ and δ.

Another approach to the constant force of interest

The force of interest δ can also be interpreted as a nominal rate of interest compounded infinitely often, with

$$\delta = \lim_{m\to\infty} i^{(m)}.$$

The reasoning is as follows. Over one year, £1 accumulates to

- $1+i$, given a nominal rate of interest $i^{(1)} = i$ convertible yearly.
- $(1+\frac{i^{(4)}}{4})^4$, given a nominal rate of interest $i^{(4)}$ convertible quarterly.
- $(1+\frac{i^{(365)}}{365})^{365}$, given a nominal rate of interest $i^{(365)}$ convertible daily.
- $(1+\frac{i^{(8760)}}{8760})^{8760}$, given a nominal rate of interest $i^{(8760)}$ convertible hourly.
- $(1+\frac{i^{(m)}}{m})^m$, given a nominal rate of interest $i^{(m)}$ convertible $\frac{1}{m}$-ly.

As $m\to\infty$ (i.e. continuous compounding), $i^{(m)}\to\delta$ and

$$\lim_{m\to\infty}\left(1+\frac{\delta}{m}\right)^m = e^{\delta}$$

Example 6.1.1 Find the accumulated amount at time 20 of a cash flow of £1 000 at time 0, assuming a force of interest $\delta(t)$ p.a. given by

$$\delta(t) = 0.06 + 0.0005t - 0.00005t^2.$$

Solution: Accumulated value at time 20 is

$$\begin{aligned}
&1\,000\exp\left(\int_0^{20}\delta(s)ds\right)\\
=&1\,000\exp\left(\int_0^{20}(0.06+0.0005s-0.00005s^2)ds\right)\\
=&1\,000\exp\left(\left[0.06s+\frac{0.0005}{2}s^2-\frac{0.00005}{3}s^3\right]_{s=0}^{s=20}\right)\\
=&1\,000\times\exp\left(\frac{3.5}{3}\right)\\
=&3\,211.27.
\end{aligned}$$

Example 6.1.2 Find the accumulated amount at time 20 of a cash flow of £1 000 at time 0 and a further £1 000 at time 10, assuming a force of interest $\delta(t)$ p.a. given by

$$\delta(t) = 0.06 + 0.0005t - 0.00005t^2.$$

Solution:

Time	0	10
Cashflow	1 000	1 000
AV of cashflow at time 20	$1\,000 \exp\left(\int_0^{20} \delta(s)ds\right)$ $= 1\,000 \times \exp\left(\frac{3.5}{3}\right)$	$1\,000 \exp\left(\int_{10}^{20} \delta(s)ds\right)$ $= 1\,000 \times \exp\left(\frac{1.675}{3}\right)$

Hence the accumulated value at time 20 is

$$1\,000 \times \exp\left(\frac{3.5}{3}\right) + 1\,000 \times \exp\left(\frac{1.675}{3}\right) = £4\,959.03$$

Exercise 6.5 *Calculate the present value at time 0 of a cashflow of 100 at time 4, and 1000 at time 10 assuming a force of interest per time unit given as follows:*

$$\delta = \begin{cases} 0.04 & 0 \le t < 5 \\ 0.14 - 0.02t & 5 \le t \end{cases}$$

6.2 Continuously-paid cashflows

Let $M(t)$ denote the total payment made between time 0 and time t. Then, the rate of payment is

$$\rho(s) = M'(s), \text{ for all } s \ge 0.$$

For $0 \le t_1 < t_2$, the total payment between time t_1 and time t_2 is

$$M(t_2) - M(t_1) = \int_{t_1}^{t_2} M'(s)ds = \int_{t_1}^{t_2} \rho(s)ds.$$

Example 6.2.1 Start with zero money in the bank, i.e. $M(0) = 0$, and assume no interest. Put money in the bank from time 0 at the constant rate of £500 p.a, i.e $\rho(t) =$ £500 p.a. for $t \ge 0$.

After one year, there is £500 in the bank, i.e. $M(1) = 500$. After 1.5 years, there is £750 in the bank, i.e. $M(1.5) = 750$. In general,

$$M(t) = M(0) + \int_0^t \rho(s)ds = 0 + \int_0^t 500ds = 500t.$$

Present Values and Accumulated Values

For times s and t with $s < t$, denote by

- $V(s,t)$ the present value at time s of 1 unit paid at time t.
- $A(s,t)$ the accumulation at time t of 1 unit paid at time s.

Remark: $V(s,t) = \frac{1}{A(s,t)}$.

Example 6.2.2 For a constant force of interest δ per annum,

- $V(s,t) = e^{-\delta(t-s)}$.
- $A(s,t) = e^{\delta(t-s)}$.

Present Values

Consider a cashflow paid continuously at the rate

$$\rho(t) \quad \text{p.a.}$$

for T years starting at time 0. Partition the time interval $[0,T]$ into tiny disjoint segments, each of length dt. Consider one of these tiny segments $[t, t+dt)$.

If dt is very small, then the total payment between time t and time $t+dt$ is

$$M(t+dt) - M(t) \approx \rho(t)dt.$$

We treat the amount paid between t and $t+dt$ as paid exactly at time t, i.e. we treat the continuous payment made over $[t, t+dt)$ as a discrete cashflow $\rho(t)dt$ paid at time t.

The PV at time 0 of the discrete cashflow is $V(0,t)\rho(t)dt$.

Repeat for all the tiny segments intervals, then sum them to find the total PV at time 0. Letting $dt \to 0$, we find in the limit that the present value at time 0 of the entire cash flow between time 0 and time T is

$$\int_0^T \rho(t)V(0,t)dt = \int_0^T \rho(t) \exp\left(-\int_0^t \delta(u)du\right) dt.$$

Example 6.2.3 Find the present value at time 0 of a cashflow paid continuously at the constant rate £500 p.a. from time 0 to time 10, assuming a constant force of interest δ p.a.

Solution: As $\rho(t) = 500$ and $V(0,t) = e^{-\delta t}$,

$$\begin{aligned}
\text{PV } (0) &= \int_0^{10} \rho(t)V(0,t)dt \\
&= \int_0^{10} 500e^{-\delta t}dt \\
&= 500\left[\frac{e^{-\delta t}}{-\delta}\right]_{t=0}^{t=10} \\
&= 500\frac{1-e^{-10\delta}}{\delta} \\
&= 500\frac{1-v^{10}}{\delta} \qquad \text{with} \quad v = e^{-\delta}.
\end{aligned}$$

Exercise 6.6 Assume time is measured in years.

(i) Given a constant force of interest $\delta = 3\%$ p.a. calculate the PV at time $t = 0$ of a cashflow paid continuously from time $t = 0$ to time $t = 2$ at the rate £10 000 p.a.

(ii) Given a constant effective rate of interest $i = 6\%$ p.a, calculate the PV at time $t = 0$ of a cashflow paid continuously from time $t = 0$ to time $t = 2$ at the rate £10 000 p.a.

(iii) Given a constant force of interest $\delta = 3\%$ p.a, calculate the PV at time $t = 0$ of a cashflow paid continuously from time $t = 0$ to time $t = 2$ at the rate $\rho(t) = e^{-0.05t}$ p.a, for all $t \in [0, 2]$.

More generally, consider a cash flow paid continuously at the rate $\rho(t)$ p.a. for T years starting at time S. Partition the time interval $[S, T]$ into tiny disjoint segments, each of length dt. Consider one of these tiny segments $[t, t + dt)$.

If dt is very small, then the total payment between time t and time $t + dt$ is

$$M(t + dt) - M(t) \approx \rho(t)dt.$$

We treat the amount paid between t and $t + dt$ as paid exactly at time t, i.e. we treat the continuous payment made over $[t, t + dt)$ as a discrete cashflow $\rho(t)dt$ paid at time t.

The PV at time r of the discrete cashflow is $V(r, t)\rho(t)dt$.

Repeat for all the tiny segments intervals, then sum them to find the total PV at time r. Letting $dt \to 0$, we find in the limit that the present value at time r of the entire cash flow between time S and time T is

$$\int_S^T \rho(t)V(r, t)dt = \int_S^T \rho(t) \exp\left(-\int_r^t \delta(u)du\right) dt.$$

Example 6.2.4 Find the present value at time 2 of a cashflow paid continuously at the constant rate £1 000 p.a. from time 5 to time 20, assuming a constant force of interest δ p.a.

Solution: As $\rho(t) = 1\,000$ and $V(2, t) = e^{-\delta(t-2)}$,

$$\begin{aligned}
\text{PV } (2) &= \int_5^{20} \rho(t)V(2, t)dt \\
&= \int_5^{20} 1\,000e^{-\delta(t-2)}dt \\
&= 1\,000\left[\frac{e^{-\delta(t-2)}}{-\delta}\right]_{t=5}^{t=20} \\
&= 1\,000\frac{e^{-3\delta} - e^{-18\delta}}{\delta} \\
&= 1\,000v^3\frac{1 - v^{15}}{\delta} \qquad \text{with} \quad v = e^{-\delta}.
\end{aligned}$$

Exercise 6.7 Assume a constant force of interest $\delta = 3\%$ p.a. Calculate the present value at time $t = 1$ of a cashflow paid continuously at the constant rate £10 000 p.a. from time $t = 3$ to time $t = 14$.

Accumulated Values

For accumulated values, we use exactly the same approach.

Consider a cash flow paid continuously at the rate $\rho(t)$ p.a. for T years starting at time S. Partition the time interval $[S, T]$ into tiny disjoint segments, each of length dt. Consider one of these tiny segments $[t, t + dt)$. The total payment between time t and time $t + dt$ is approximately $\rho(t)dt$.

The AV at time T of the discrete cashflow is $A(t, T)\rho(t)dt$.

Repeat for all the tiny segments intervals, then sum them to find the total PV at time T. Letting $dt \to 0$, we find in the limit that the accumulated value at time T of the entire cash flow between time S and time T is

$$\int_S^T \rho(t)A(t,T)dt = \int_S^T \rho(t) \exp\left(\int_t^T \delta(u)du\right) dt$$

Example 6.2.5 Find the accumulated value at time 10 of a cashflow paid continuously at the rate £1 000 p.a. from time 0 to time 10, assuming a constant force of interest δ p.a.

Solution: As $\rho(t) = 1\,000$ and $A(t, 10) = e^{\delta(10-t)}$,

$$\begin{aligned}
\text{AV } (10) &= \int_0^{10} \rho(t)A(t,10)dt \\
&= \int_0^{10} 1\,000e^{\delta(10-t)}dt \\
&= 1\,000\left[\frac{e^{\delta(10-t)}}{-\delta}\right]_{t=0}^{t=10} \\
&= 1\,000\frac{e^{10\delta}-1}{\delta} \\
&= 1\,000\frac{(1+i)^{10}-1}{\delta} \qquad \text{with} \quad 1+i = e^{\delta}.
\end{aligned}$$

Exercise 6.8 Assume a constant force of interest $\delta = 3\%$ p.a. Calculate the accumulated value at time $t = 14$ of a cashflow paid continuously at the constant rate £10 000 p.a. from time $t = 3$ to time $t = 14$.

Exercise 6.9 Assume a constant force of interest $\delta = 4\%$ p.a. Calculate the accumulated value at time $t = 12$ of a cashflow paid continuously at the constant rate £100 p.a. from time $t = 3$ to time $t = 5$.

Exercise 6.10 The force of interest is

$$\delta(t) = \begin{cases} 0.04 & \text{for } 0 \le t < 5 \\ 0.03 & \text{for } t \ge 5. \end{cases}$$

Calculate the present value at time $t = 2$ of a payment stream paid continuously from $t = 5$ to $t = 10$ at a rate of £50 per annum.

Different types of cashflows The accumulation at time $T > 0$ of

1. one discrete payment $C(t_1)$ made at time t_1 is

$$A(T) = C(t_1) \exp\left(\int_{t_1}^{T} \delta(s)ds\right).$$

2. n discrete payments $C(t_k)$ made at times t_k, for $k = 1, \ldots, n$ is

$$A(T) = \sum_{k=1}^{n} C(t_k) \exp\left(\int_{t_k}^{T} \delta(s)ds\right).$$

3. a continuous payment paid at rate $\rho(t)$ for all $t \in [S, T]$, for some $S < T$, is

$$A(T) = \int_{S}^{T} \rho(t) \exp\left(\int_{t}^{T} \delta(u)du\right) dt.$$

International actuarial notation for continuously-paid cashflows

Assume a constant force of interest δ p.a.

Constant annuity: constant rate of payment

Denote by $\bar{a}_{\overline{n}|}$ the PV at the start of a cashflow which is paid continuously for n years and has a constant rate of payment 1 unit p.a.

Important formula:

$$\bar{a}_{\overline{n}|} = \frac{1 - v^n}{\delta}$$

Example 6.2.6 The PV at time 0 of a cashflow paid continuously at the rate £1 000 p.a. from time 0 to time 10 years, assuming a force of interest of 5% p.a, is

$$1\,000\bar{a}_{\overline{10}|} = 1\,000\frac{1 - e^{-10\times 0.05}}{0.05} = £7\,869.39.$$

Example 6.2.7 A cashflow is paid continuously from time 0 to time 10 years. The initial rate of payment is £1 000 p.a. and increases continuously thereafter at the rate 3% p.a. effective. Calculate the PV at time 0 of the cashflow, assuming a force of interest of 5% p.a.

Solution: The rate of payment is $\rho(t) = 1\,000(1.03)^t$ at time $t \in [0, 10]$. Then

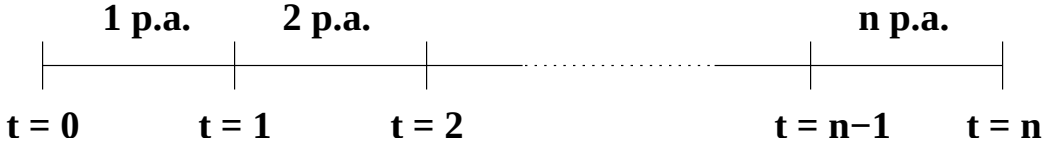
$$\begin{aligned} PV(0) &= \int_0^{10} \rho(t)e^{-\delta t}dt \\ &= \int_0^{10} 1\,000(1.03)^t e^{-0.05t}dt \\ &= 1\,000\int_0^{10} e^{t\ln(1.03)}e^{-0.05t}dt \\ &= 1\,000\int_0^{10} e^{-(0.05-\ln(1.03))t}dt \end{aligned}$$

Set $\hat{\delta} = 0.05 - \ln(1.03)$. Then

$$PV(0) = 1\,000\bar{a}^{\hat{\delta}}_{\overline{10}|} = 1\,000\frac{1 - e^{-10\times\hat{\delta}}}{\hat{\delta}} = £9\,044.16.$$

Increasing annuity: rate of payment k units p.a. in year k

Denote by $(I\bar{a})_{\overline{n}|}$ the PV at the start of a cashflow which is paid continuously for n years and has a rate of payment k units p.a. in year k, i.e.



Important formula:

$$(I\bar{a})_{\overline{n}|} = \frac{\ddot{a}_{\overline{n}|} - nv^n}{\delta}.$$

Proof (one way):

$$\begin{aligned}(I\bar{a})_{\overline{n}|} &= \bar{a}_{\overline{1}|} + 2v\bar{a}_{\overline{1}|} + 3v^2\bar{a}_{\overline{1}|} + \cdots nv^{n-1}\bar{a}_{\overline{1}|} \\ &= \bar{a}_{\overline{1}|}\left(1 + 2v + 3v^2 + \cdots + nv^{n-1}\right) \\ &= \bar{a}_{\overline{1}|}(I\ddot{a})_{\overline{n}|} \\ &= \frac{1-v}{\delta}\frac{\ddot{a}_{\overline{n}|} - nv^n}{d} = \frac{\ddot{a}_{\overline{n}|} - nv^n}{\delta}, \qquad \text{since } d = 1 - v.\end{aligned}$$

Increasing annuity: rate of payment t units p.a. at time t

Denote by $(\bar{I}\bar{a})_{\overline{n}|}$ the present value of a cashflow which is paid continuously for n years and has a rate of payment

$$\rho(t) = t, \qquad \text{at } t \in [0, n].$$

Important formula:

$$(\bar{I}\bar{a})_{\overline{n}|} = \frac{\bar{a}_{\overline{n}|} - nv^n}{\delta}.$$

Proof: For you to do as an exercise (see Tutorial 2).

Actuarial notation for accumulations to time n:

$$\begin{aligned}\bar{s}_{\overline{n}|} &= (1+i)^n\bar{a}_{\overline{n}|} \\ (I\bar{s})_{\overline{n}|} &= (1+i)^n(I\bar{a})_{\overline{n}|} \\ (\bar{I}\bar{s})_{\overline{n}|} &= (1+i)^n(\bar{I}\bar{a})_{\overline{n}|}.\end{aligned}$$

Example 6.2.8 Find the value at time 15 of a cashflow paying continuously at a rate $\rho(t)$ p.a. where

$$\rho(t) = \begin{cases} 10 & \text{for } 0 \le t < 5 \\ 12 & \text{for } 5 \le t < 15 \end{cases}$$

assuming a constant force of interest δ.

Solution: Can you see that

$$\text{Value at time } 15 = e^{15\delta} \times 10\bar{a}_{\overline{5}|} + e^{10\delta} \times 12\bar{a}_{\overline{10}|}$$

Alternatively, we can figure this out the long way:

$$\begin{aligned}
\text{PV}(15) &= \int_0^{15} \rho(t) e^{\int_t^{15} \delta(u)du} dt \\
&= \int_0^{15} \rho(t) e^{(15-t)\delta} dt \\
&= \int_0^{5} \rho(t) e^{(15-t)\delta} dt + \int_5^{15} \rho(t) e^{(15-t)\delta} dt \\
&= \int_0^{5} 10 e^{(15-t)\delta} dt + \int_5^{15} 12 e^{(15-t)\delta} dt \\
&= 10e^{15\delta} \int_0^{5} e^{-\delta t} dt + 12e^{15\delta} \int_5^{15} e^{-\delta t} dt \\
&= 10e^{15\delta} \bar{a}_{\overline{5}|} + 12e^{10\delta} \bar{a}_{\overline{10}|}.
\end{aligned}$$

Exercise 6.11 An investor buys an annuity which pays him cash continuously for 5 years. He receives payment in the first year of £3,000, thereafter his annual receipts increase by £500 per annum. Calculate the purchase price of the annuity, assuming that the annual effective rate of interest is 6%.

Exercise 6.12 An investor agrees to make investments continuously for the next 10 years. She decides that she can afford to invest $£20t$ at time t, $0 \leq t \leq 10$. Calculate the value of the investment at time 0, assuming that the annual effective rate of interest is 4%.

Chapter 7

Loans and Loan Schedules

7.1 Three Ways to Repay a Loan

For illustration, suppose the time unit is the year. You borrow an amount £L from a financial institution, and the loan is to be completely paid back by the end of n years.

Terminology: We say that this loan has *amount* £L, and a *term* of n years.

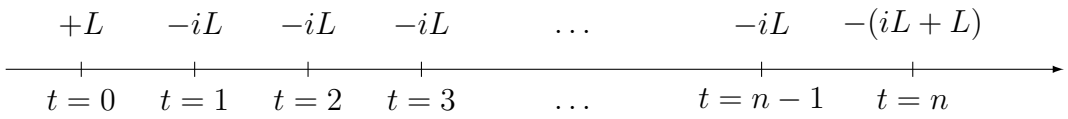
How may you arrange to repay this loan? Three possible ways are shown below. Each is illustrated by a timeline, assuming that the effective rate of interest is a constant i p.a..

1. Repay the loan and all interest on it at the end of the term.



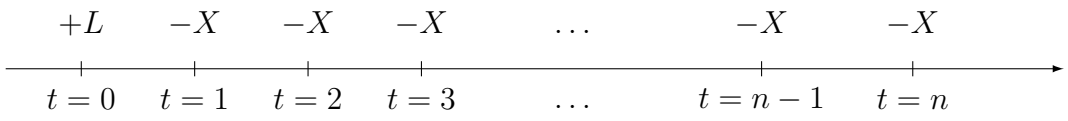
   For the lender, this is the riskiest option, especially if the loan is for a long term. Potentially, they do not find out whether or not they will be repaid, with interest, until the end of n years. Much could have happened to the borrower during that time that the lender may not know about. Hence, this scheme tends to be used only for very short-term loans.

2. Repay interest only during the term of the loan, and then repay the amount of the loan at the end of the term.



This is better for the lender, as at least the interest is being repaid regularly, which means that the lender must be in touch with the borrower. It still leaves the entire amount of the loan to be repaid at the end of the term however, which may put a strain on the borrower's resources at that time. This pattern of repayments is the basis for *fixed interest securities* or *bonds*, where the borrower is usually a government or corporation. It is also the basis of *interest-only mortgages* in the U.K.

3. Repay a level amount each year (or month) calculated so that when the final such payment is made at the end of the term, the loan has been completely paid off. Suppose the level repayment is calculated to be £X.



The idea here is that each repayment of X covers the interest that has accrued since the last repayment, and also repays a slice of the original loan. Hence, clearly, $X > iL$. This is the least risky of the three ways to repay a loan, from the lender's point of view, because it relies only on the borrower having sufficient income to pay X each year, which is something that can be checked in advance, rather than having the resources to repay the whole loan in n years' time, which is harder to know in advance. Loans of this kind are the subject of this chapter.

7.2 An Example

The obvious question is: what should £X be for a loan of £L over n time units (which now need not be a year)? We begin with a concrete example, with the answer supplied, and then consider how we might have obtained that answer.

Suppose the (constant) interest rate is 5% p.a. effective. Suppose you borrow £10,000 for a term of four years. You agree to make level annual repayments of £2,820.12 p.a., in arrears. We show below, in the form of a *loan schedule*, how this will exactly repay the loan and all interest (to within a small rounding error).

Time (Yrs)	Repayment	Interest Content	Capital Content	Capital Outstanding
0				10,000.00
1	2,820.12	500.00	2,320.12	7,679.88
2	2,820.12	383.99	2,436.13	5,243.75
3	2,820.12	262.19	2,557.93	2,685.82
4	2,820.12	134.29	2,685.83	−0.01

We shall explain what happens in the first year.

- At $t = 0$ you receive the loan, so the *capital outstanding* (the amount of the loan you still owe) starts at £10,000.
- Then at $t = 1$ you make the first repayment of £2,820.12.
- By then interest of $£0.05 \times 10,000$ has accrued, so you first use part of the repayment to pay that off. This is the *interest content* of the repayment.
- The remainder of the repayment, equal to £2,320.12, is used to repay part of the loan itself, and is called the *capital content* of the repayment.
- As a result, the capital outstanding falls by the amount of the capital content, so by the end of the year you owe only £7,679.88 of the original loan.

Then in each subsequent year, you repeat the process. Because, each year, you have repaid a further slice of the original loan, the capital outstanding keeps falling, so you owe less interest each year. Thus the split of the repayment between interest content and capital content changes each year, as you pay a little less interest and are able to repay a larger slice of the original loan.

Comment: All the quantities in the loan schedule (including the repayment) are rounded to pence, as they must be since these represent real amounts in actual transactions. As a result, the capital outstanding after the last repayment has been made is usually a few pence (plus or minus). This is of no real importance, but the final repayment would usually be adjusted so that the loan is exactly repaid, as below in this example.

Time (Yrs)	Repayment	Interest Content	Capital Content	Capital Outstanding
0				10,000.00
1	2,820.12	500.00	2,320.12	7,679.88
2	2,820.12	383.99	2,436.13	5,243.75
3	2,820.12	262.19	2,557.93	2,685.82
4	2,820.11	134.29	2,685.82	0.00

Observation: The alert reader may have spotted that, rounded to two decimal places:

$$2,820.12 = \frac{10,000}{a_{\overline{4}|}^{0.05}}.$$

This is not a coincidence.

7.3 The Principle of Equivalence

We often need to compare two or more series of cashflows and decide:

- whether they are worth the same; or

- whether one is worth more than the other.

To do this, we need to formulate a precise definition of the 'worth' of a series of cashflows. We have one to hand, namely their value, at a given point of time, using a given set of interest rates.

Definition 7.1 Consider one series of cashflows $c_1, c_2, \ldots, c_n$ payable at times $t_1, t_2, \ldots, t_n$, and another set of cashflows $d_1, d_2, \ldots, d_m$ payable at times $s_1, s_2, \ldots, s_m$. Suppose we are given a set of accumulation/discount factors $V(t, s)$ for all t and s. Then we say the two sets of cashflows are *equivalent at time* t if we can find some time t at which they have the same value, using the given accumulation/discount factors. That is:

$$\sum_{r=1}^{n} c_r V(t, t_r) = \sum_{r=1}^{m} d_r V(t, s_r).$$

Notes:

- If two sets of cashflows are equivalent at any time t, then they are also equivalent at any other time s, since their values at time s may be obtained by multiplying both sides of the equation above by $V(s, t)$. Therefore we can drop the specification of a time t from the definition, and just say that the two sets of cashflows are equivalent.
- The accumulation/discount factors are an essential part of the definition. Two sets of cashflows that are equivalent under one interest rate regime will generally not be equivalent under a different interest rate regime.
- Where the given accumulation/discount factors come from we leave open. Two commonly used possibilities are:
  - the rates of interest obtained in the financial markets, for example those quoted in the *Financial Times* or online equivalent; or
  - the rate of return on equity that a financial institution sets as a target for its sales and trading activities.
- Often, one of the sets of cashflows is a single cashflow payable now, and the other is a series of one or more cashflows payable in the future. Then we regard the single cashflow payable now as the *price* of the cashflows payable in the future.

We are now in a position to state the idea behind the Principle of Equivalence.

Principle of Equivalence: We assume that any person or entity is indifferent about which of two equivalent series of cashflows they own. That is, they would be willing to exchange one set of cashflows for any other equivalent set of cashflows.

Comments:

- If one set of cashflows has a higher value than another, it is logical to suppose that the person or entity would prefer to own the set of cashflows with the higher value.

- Equivalence in the sense above will be just one of many factors affecting the person or entity's desire to own one or the other of a pair of equivalent series of cashflows. Other factors will include their purpose or intention, and the risks involved. For example a house owner will only be interested in buying house insurance, not selling it.
- The Principle of Equivalence is just the simplest of many principles that can be formulated to model why and when two parties would be willing to make an exchange under a financial contract. It is the simplest, since it ignores risk and does not model the effect of the person or entity's overall financial wealth on the desirability of making the exchange. Most other principles must involve probability theory to model risk and utility theory to model the economic background to the transaction.
- We will most often use the Principle of Equivalence as a way of finding the fair price of a financial asset. That is, one party is paying a single lump sum at outset (the price) in exchange for the right to receive a stream of cashflows generated by the asset in the future. Some examples are:
  - when you buy shares in a company the price you pay is exchanged for the right to receive future dividends and the proceeds of eventually selling the shares; or
  - when a bank gives you a loan, the amount of the loan is the price the bank pays for the right to receive the future repayments from you.

Example 7.3.1 A bank lends you £10,000 for four years. The effective rate of interest is 5% p.a.. Show that the following three ways of repaying the loan are equivalent.

(i) A single repayment of £12,155.06 at the end of four years.

(ii) A payment of £500 at the end of each of the next four years, and repayment of the £10,000 at the end of four years.

(iii) Level annual repayments of £2,820.12, paid yearly in arrears.

Answer: We show that all three series of cashflows have present value £10,000 at outset. This justifies £10,000 as the 'price' to be paid for any of them.

(i) $\text{PV} = 12,155.06(1.05)^{-4} = £10,000.00.$

(ii) $\text{PV} = 500a_{\overline{4}|}^{0.05} + 1,000.00(1.05)^{-4} = £10,000.00.$

(iii) $\text{PV} = 2,820.12a_{\overline{4}|}^{0.05} = £10,000.000.$

Example 7.3.2 You have been offered an investment that will pay you £200 at the end of each of the next 24 months. What is the maximum price you should pay if the effective rate of interest is 0.5% per month?

Answer: You would be willing to pay the price equivalent in value to these cashflows, at the stated interest rate, or less if that were possible. That price is;

$$PV = 200a_{\overline{24}|}^{0.005} = £4,512.57.$$

7.4 The Mathematics of Loan Schedules

Let us generalize the numerical example of a loan schedule given previously. Let L be the amount of a loan made now, with term n time units. The loan is to be repaid by level repayments of X per time unit, payable in arrears. The effective rate of interest is i per time unit. By the principle of equivalence, we have:

$$L = Xa_{\overline{n}|}^{i} \qquad \Rightarrow \qquad X = \frac{L}{a_{\overline{n}|}^{i}}.$$

We will now show, by working through the loan schedule, the following.

(i) The capital outstanding just after the rth repayment is $Xa_{\overline{n-r}|}^{i}$ $(r = 1, 2, \ldots, n-1)$.

(ii) The interest content of the rth repayment is $iXa_{\overline{n+1-r}|}^{i}$.

(iii) The capital content of the rth repayment is Xv^{n+1-r}.

Consider the end of the first time unit, when the first repayment of X is made. The interest then accrued is:

$$iL = iXa_{\overline{n}|}^{i} = iXa_{\overline{n+1-1}|}^{i}$$

so this is the interest content of the first repayment. Then the capital content must be:

$$\begin{aligned} \text{capital content} &= X - iXa_{\overline{n}|}^{i} \\ &= X(1 - ia_{\overline{n}|}^{i}) \\ &= X\left(1 - i\left(\frac{1-v^n}{i}\right)\right) \\ &= Xv^n \end{aligned}$$

and the capital outstanding falls to:

$$\begin{aligned} \text{capital outstanding} &= L - Xv^n \\ &= Xa_{\overline{n}|}^{i} - Xv^n \\ &= X(v + v^2 + \ldots + v^n - v^n) \\ &= X(v + v^2 + \ldots + v^{n-1}) \\ &= Xa_{\overline{n-1}|}^{i}. \end{aligned}$$

Thus (i), (ii) and (iii) are verified at $t = 1$. We continue to work through the repayment at time $t = 2$ (that is, $r = 2$). The interest then accrued is:

$$iXa_{\overline{n-1}|}^{i}$$

so this is the interest content of the second repayment. Then the capital content must be:

$$\begin{aligned}\text{capital content} &= X - iXa^{i}_{\overline{n-1}|} \\ &= X(1 - ia^{i}_{\overline{n-1}|}) \\ &= X\left(1 - i\left(\frac{1-v^{n-1}}{i}\right)\right) \\ &= Xv^{n-1}\end{aligned}$$

and the capital outstanding falls to:

$$\begin{aligned}\text{capital outstanding} &= Xa^{i}_{\overline{n-1}|} - Xv^{n-1} \\ &= X(v + v^2 + \ldots + v^{n-1} - v^{n-1}) \\ &= X(v + v^2 + \ldots + v^{n-2}) \\ &= Xa^{i}_{\overline{n-2}|}.\end{aligned}$$

Thus (i), (ii) and (iii) are verified at $t = 2$. Continuing along these lines, just after the $(n-1)th$ repayment has been made, the capital outstanding is $Xa^{i}_{\overline{n-(n-1)}|} = Xa^{i}_{\overline{1}|} = Xv$. The interest content of the final repayment is then iXv and the capital content is $X - iXv = X(1-d) = Xv$, which exactly pays off the capital outstanding.

Example 7.4.1 For the four-year loan used in the numerical example above, verify the following.

- Interest content of 3rd repayment = £262.19 = $0.05 \times 2,820.12 \times a^{0.05}_{\overline{2}|}$.
- Capital content of 2nd repayment = £2,436.13 = $2,820.12(1.05)^{-3}$.
- Capital outstanding after 1st repayment = £7,679.88 = $2,820.12a^{0.05}_{\overline{3}|}$.

Note that the capital outstanding after the rth repayment is $Xa^{0.05}_{\overline{n-r}|}$ which is the present value at time r of all future repayments that remain to be made. In fact this generalises to any pattern of repayments, level or not, and to any set of interest rates, constant or not.

Principle: The capital outstanding at any time t is the value at time t of all future repayments.

7.5 Changing the Terms of a Loan

It is common for a borrower to ask for a change in the term of a loan, for example:

- to extend or shorten the loan period;
- to miss one or more payments; or

- repay part of the loan early.

The lender will then need to calculate a new repayment amount. The principle always used is to calculate the capital outstanding at the time of the requested change, and then calculate the level (or otherwise) repayment that will repay that capital outstanding under the terms of the amended loan.

Example 7.5.1 A borrower takes out a loan of £70,000 to be repayable by level annual repayments over 20 years, where the repayments are calculated at an effective rate of interest of 7% p.a..

(i) How much is each annual instalment?

Answer: Using the Principle of Equivalence, we know that the loan of £70,000 must equal the present value of the repayments, so, denoting the repayment by X, we have:

$$70,000 = X a_{\overline{20}|}^{0.07} \qquad \Rightarrow \qquad X = \frac{70,000}{a_{\overline{20}|}^{0.07}} = £6,607.50.$$

(ii) What is the new repayment amount, if the borrower asks for the term to be extended by five years, immediately after making the 13th repayment?

Answer: Just after the 13th repayment, there are 7 instalments left to pay, so the outstanding loan is:

$$X a_{\overline{7}|}^{0.07} = £35,609.73.$$

Now, instead of paying this off over 7 years, the investor wants to pay this off over 12 years ($12 = 7 + 5$). To calculate the new instalment, the outstanding loan must be equivalent to the present value of the 12 outstanding instalments, each of X' say. We calculate:

$$35,609.73 = X' a_{\overline{12}|}^{0.07} \qquad \Rightarrow \qquad X' = \frac{35,609.73}{a_{\overline{12}|}^{0.07}} = £4,483.34.$$

(iii) If, instead of extending the term, the borrower had asked to miss the 14th and 15th repayments, what would each of the remaining 5 ($5 = 7 - 2$) repayments be?

Answer: Over the two years when the borrower did not make any repayments, the capital outstanding would have increased by 7% p.a. because the interest accrued and not repaid would be added to the capital outstanding. Hence this will stand at $£35,609.73(1.07)^2 = 40,769.58$. We equate this amount to the five outstanding repayments, whose amount is denoted X''.

$$40,769.58 = X'' a_{\overline{5}|}^{0.07} \qquad \Rightarrow \qquad X'' = \frac{40,769.58}{a_{\overline{5}|}^{0.07}} = £9,943.32.$$

(iv) What is the new repayment amount, denoted X''', if the borrower had decided to repay £5,000 at time $t = 13$, together with the 13th repayment?

Answer: The outstanding loan after the 13th repayment was £35,609.73 (see (ii) above). We reduce this by £5,000 and re-calculate the instalments over the remaining seven years.

$$30,609.73 = X''' a_{\overline{7}|}^{0.07} \qquad \Rightarrow \qquad X''' = \frac{30,609.73}{a_{\overline{7}|}^{0.07}} = £5,679.73.$$

7.6 Changing the Interest Rate

Often more than one interest rate is involved during the term of the loan. This may be because:

- interest rates were planned to change during the term, for example, as a marketing device the lender charged a lower-than-normal rate of interest for the first few years; or
- the lender changes the interest rate for all borrowers to reflect changed market conditions[1].

Example 7.6.1 You borrow £100,000 over 30 years, to be repaid by level annual repayments. The rate of interest will be 6% p.a. effective for the first 15 years and 7% p.a. effective thereafter. Find the repayment amount, denoted X.

Answer: At $t = 0$, £100,000 must be equivalent to the present value of all future repayments, so we can write:

$$100,000 = X(a_{\overline{15}|}^{0.06} + (1.06)^{-15} a_{\overline{15}|}^{0.07}) \qquad \Rightarrow \qquad X = £7,400.47.$$

Example 7.6.2 You borrow £50,000 from a bank, to be repaid over 25 years by level annual repayments initially based on an effective rate of interest of 3% p.a.. Just after the fifth repayment has been paid, the bank raises its interest rate to 5% p.a. effective. What is the new repayment?

Answer: First, we calculate the initial repayment amount, denoted X.

$$50,000 = X a_{\overline{25}|}^{0.03} \qquad \Rightarrow \qquad X = \frac{50,000}{a_{\overline{25}|}^{0.03}} = £2,871.39.$$

Next, we calculate the capital outstanding just after the fifth repayment, which is the present value of the outstanding 20 ($20 = 25 - 5$) repayments

[1]This is normal practice for mortgages in the UK, where mortgage interest rates are usually linked to the Bank of England base rate. Practice in the USA is different, where the interest rate is fixed when the loan is taken out.

$$\text{Capital outstanding} = X a_{\overline{20}|}^{0.03} = £42,719.03.$$

Finally, we equate the outstanding capital to the present value of the new repayment amount, denoted X', on the new terms, namely at 5% p.a. effective interest.

$$42,719.03 = X' a_{\overline{20}|}^{0.05} \qquad \Rightarrow \qquad X' = \frac{42,719.03}{a_{\overline{20}|}^{0.05}} = £3,427.89.$$

7.7 Non-Level Repayments

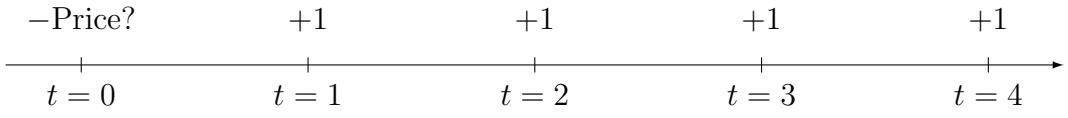
There is no need for payments to always be level.
No new principles are involved.

Chapter 8

Yields

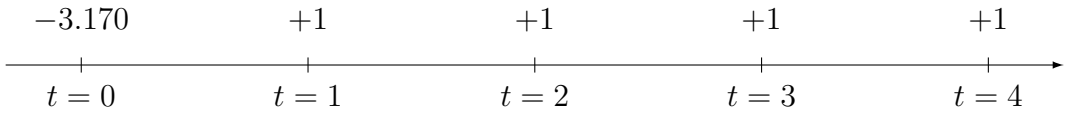
8.1 An Example

Consider an investment that will pay you a cashflow of 1 at the end of each of the next four time units. The question is, what price should you be prepared to pay to make this investment? This is illustrated by the following timeline.



The Principle of Equivalence gives us one answer. Given a suitable interest rate regime, the price (or maximum price) is the present value of the future cashflows. For example, suppose the effective rate of interest is 10% per time unit. Then the price, denoted P, is:

$$P = a_{\overline{4}|}^{0.1} = 3.170.$$



However, suppose that this investment is actually a financial contract that is traded in a financial market. Then the price of the investment will be determined by the mix of buyers and sellers at any given time, who will trade when their various interests and objectives can be achieved at a given price. That is, the price does not emerge as the result of a single present value calculation using some given interest rate or rates; it emerges in a much more chaotic way from a process of barter. What you can actually observe is the market price of the investment.

Suppose, on a given date, that price is $P = 3.000$. Then a different question presents itself: what rate of interest is consistent with a market price of 3.000 for this investment? This rate of interest, if it exists, is called the *yield* on the investment. In other words, we must solve for the unknown i in:

$$3.000 = a^{i}_{\overline{4}|}$$

or, in full:

$$3.000 = v^1 + v^2 + v^3 + v^4$$

or, equivalently:

$$3.000 - v^1 - v^2 - v^3 - v^4 = 0$$

or, also equivalently, but shifting our point of view to the end of the contract instead of the beginning:

$$3.000(1+i)^4 - (1+i)^3 - (1+i)^2 - (1+i)^1 - (1+i)^0 = 0.$$

A solution of these is a candidate for the yield on this investment. We note just now that this raises some potentially tricky questions.

- These equations are all polynomial equations (of order four) in the variable v or $(1+i)$. (In the latter case, we could expand the powers and write it as a polynomial in the variable i, but there is nothing to be gained from doing so.) It is well-known that a polynomial of order n has n roots, or solutions. So, there may be more than one candidate for the yield.

- A real number i only makes sense as a rate of interest, in normal circumstances, if $i \geq -1$. We hope that $i > 0$ so that investment leads to growing wealth, but under some investments there is a risk of losing money, which corresponds to negative interest[1], $i < 0$. However, $i < -1$ would mean losing more than the investor had to begin with. This is not possible under most classes of investment. Therefore, only solutions $i \geq -1$ will be acceptable candidates as a yield.

We will return to these points in Section 8.3.

The practical question is, how can we solve any of the equations above to obtain the yield? We offer two methods.

[1]In exceptional circumstances, even bank interest can be negative. For example when the outlook for most asset classes is dire, investors may be willing to pay a bank interest of, say, 0.5% p.a. just to be sure of keeping 99.5% of the value of their money. From the investor's point of view, $i = -0.005$ p.a. and they are receiving negative interest.

8.1.1 Use Suitable Computer Software

Most software products capable of reasonably sophisticated quantitative work will include a function or module that can be set up to solve simple equations, including polynomial equations like the above.

In particular, *Microsoft*© *Excel* has a functions like 'Solver' and "Goal Seek' for doing this. The vast majority of yield calculations (solving equations like the above for the unknown v or i) will nowadays be done in this way.

Alternatively, if you are adept at programming in a general-purpose language, such as *R*, *Python*, *C#* or *C++*, programmes can be written to carry out this task.

8.1.2 Trial and Error, and Linear Interpolation

Suppose a computer and suitable software is unavailable, which will be the case under examination conditions in many university and professional settings. Then you need a method of finding yields to an acceptable degree of approximation, using pencil, paper and calculator.
The basic idea is simple, if somewhat crude. We will run through it with the example above.

- Choose one of the equations above. In these notes, we will prefer an equation in which $(1+i)$ is the variable, and that has been rearranged so that the right hand side is zero. We therefore choose to illustrate using the last equation above. To emphasise that we are dealing with a function of i, we will denote the left hand side $f(i)$ and write:

$$f(i) = 3(1+i)^4 - (1+i)^3 - (1+i)^2 - (1+i)^1 - (1+i)^0$$

  so the equation we wish to solve is:

$$f(i) = 3(1+i)^4 - (1+i)^3 - (1+i)^2 - (1+i)^1 - (1+i)^0 = 0.$$

- Try different candidate values for i (or v) until you have two, say i_1 and i_2, reasonably close together, for one of which $f(i)$ has a positive sign and for the other of which $f(i)$ has a negative sign. Then as long as f is a continuous function, a solution of $f(i) = 0$ must lie between i_1 and i_2.

- As long as i_1 and i_2 are reasonably close together, and $f(i)$ is not curving too sharply between i_1 and i_2, assuming $f(i)$ to be a straight line between i_1 and i_2 will give a reasonably accurate approximation to where the true curve $f(i)$ crosses the axis, that is, the true solution i^* to $f(i) = 0$. This is called *linear interpolation*.

By definition, when the price was $P = 3.170$ the yield was 0.1 (or 10%) per time unit, so $i = 0.1$ seems a sensible place to start looking. We find successively that:

$$\begin{aligned} f(0.10) &= -0.2487 < 0 \\ f(0.11) &= -0.1555 < 0 \\ f(0.12) &= -0.0588 < 0 \\ f(0.13) &= +0.0416 > 0 \end{aligned}$$

so the yield is between 12% and 13% per time unit. If $f(i)$ was linear between these ordinates, it would cross the $x-$axis at:

$$i^* \approx 0.12 + 0.01 * \frac{0.0588}{0.0588 + 0.0416} = 0.12586 \approx 0.126$$

so the yield is approximately $i^* \approx 0.126$ or 12.6% per time unit.

Comments:

- Just for demonstration, we will show the same process using one of the other equations above, namely $3.000 = a_{\overline{4}|}^{i}$. We find successively that:

$$\begin{aligned} a_{\overline{4}|}^{0.10} &= 3.170 > 3.000 \\ a_{\overline{4}|}^{0.11} &= 3.102 > 3.000 \\ a_{\overline{4}|}^{0.12} &= 3.037 > 3.000 \\ a_{\overline{4}|}^{0.13} &= 2.974 < 3.000 \end{aligned}$$

  so, again, the solution lies between 12% and 13% per time unit. Now linear interpolation gives:

$$i^* \approx 0.12 + 0.01 \frac{3.000 - 3.037}{2.974 - 3.037} = 0.12587 \approx 0.126.$$

  So the answers for i* are extremely close.

- For a small change in the price, $P = 3.000$ instead of $P = 3.170$, a relative change of less than 6%, we get quite a large change in the yield, from 10% to 12.6% per time unit.

- A yield can be thought of as the answer to the question: "if this investment were a bank account, what effective rate of interest would I receive?"

8.2 Equations of Value

The key step in finding the yield on a financial contract is to use its cashflows to construct an *equation of value*, which is the equation involving a function of v or i that we try to solve. Suppose that the financial contract has cashflows $c_1, c_2, \ldots, c_n$ at times $t_1 < t_2 <$

$\ldots < t_n$. Our practice will be to use the value of the cashflows at the last time t_n, so we define the a function $f(i)$ to be:

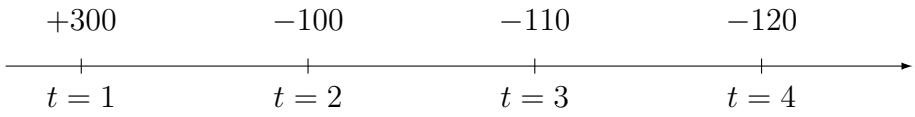
$$f(i) = \sum_{r=1}^{n} c_r(1+i)^{t_n - t_r}.$$

Then the equation $f(i) = 0$ is called the *equation of value*[2]. Note that $f(i)$ by itself is not the equation of value: an equation must contain an equals sign. We define i^* to be the yield on the financial contract if and only if the following holds:

$$i^* \textbf{ is the unique real number } \geq -1 \textbf{ satisfying } f(i) = 0.$$

If no such i^* exists, we say the yield is undefined. It is very important that yields are used *only* when they exist, according to the very careful definition above[3].

Example 8.2.1 What is the yield on the following set of cashflows?



(Note that we have changed the signs of the cashflows from the previous examples: this does not affect the yield.)

Answer: We formulate an equation of value at time $t = 4$:

$$f(i) = 300(1+i)^3 - 100(1+i)^2 - 110(1+i)^1 - 120 = 0.$$

Trial and error solutions lead to:

$$\begin{aligned} f(0.04) &= -5.1008 < 0 \\ f(0.05) &= +1.5375 > 0 \end{aligned}$$

and then linear interpolation gives:

$$i^* \approx 0.04 + 0.01\,\frac{5.1008}{5.1008 + 1.5375} = 0.04768 \approx 0.048.$$

As a rough check that we are not far out, note that $f(0.048) = 0.196 \approx 0$.

[2]Just for example, if we chose to formulate an equation of value at time t_1 in terms of present values, we would define a function:

$$f_v(v) = \sum_{r=1}^{n} c_r v^{t_r - t_1}$$

and the equation of value would be $f_v(v) = 0$.

[3]Again, if we instead formulate present values at time t_1, the yield is the unique real number i^*, ≥ -1, satisfying $f_v((1+i)^{-1})) = 0$.

Example 8.2.2 You are offered an investment which will pay £1 at the end of each of the next ten years, and £2 at the end of each of the succeeding ten years. The price of the investment is £15. What is the yield?

Answer: We formulate an equation of value at time $t = 20$ years:

$$f(i) = 15\,(1+i)^{20} - s^{i}_{\overline{20|}} - s^{i}_{\overline{10|}} = 0.$$

Trial and error solutions lead to:

$$\begin{aligned} f(0.06) &= -1.8594 < 0 \\ f(0.07) &= +3.2333 > 0 \end{aligned}$$

and then linear interpolation gives:

$$i^* \approx 0.06 + 0.01\,\frac{1.8594}{1.8594 + 3.2333} = 0.06365 \approx 0.064.$$

As a rough check that we are not far out, note that $f(0.064) = 0.0327 \approx 0$.

8.3 Existence and Uniqueness of Yields

8.3.1 Yields Do Not Always Exist

When cashflows occur once per time unit (some may be zero) the equation of value:

$$f_v(i) = \sum_{r=1}^{n} c_r (1+i)^{t_n - t_r} = 0$$

is a polynomial equation in the variable i, as already noted. It can fail to provide a valid yield in two circumstances:

1. There are no real solutions greater than or equal to -1. Either all the solutions are complex, or any real solutions are less than -1.

2. There is more than one real solution greater than or equal to -1. Then we have no way of preferring one to the other as a yield.

Example 8.3.1 Consider the following simple cashflows:



We can certainly formulate an equation of value, at time $t = 1$, as follows:

$$f(i) = (1 + i) + 1.1 = 0 \quad \text{equivalently} \quad i = -2.1.$$

Hence these cashflows do not have a yield — it does not exist or is undefined.

Example 8.3.2 Consider the following simple cashflows:

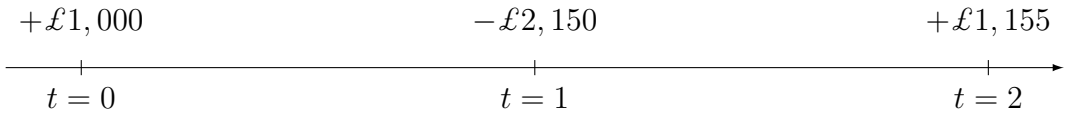


We can certainly formulate an equation of value, at time $t = 2$, as follows:

$$f(i) = (1 + i)^2 + 1.1 = 0 \quad \text{equivalently} \quad (1 + i)^2 = -1.1$$

but this has no real solutions for i: we have $i = \pm\sqrt{-1.1} - 1$. Hence these cashflows do not have a yield — it does not exist or is undefined[4].

Example 8.3.3 Consider the following simple cashflows:



We can certainly formulate an equation of value, at time $t = 2$, as follows:

$$f(i) = 1,000(1 + i)^2 - 2,150(1 + i) + 1,155 = 0$$

or more simply:

$$f(i) = (1 + i)^2 - 2.15(1 + i) + 1.155 = 0.$$

However, this factorises as follows:

$$f(i) = ((1 + i) - 1.1)((1 + i) - 1.05) = 0.$$

Hence 0.05 and 0.1, or 5% per time unit and 10% per time unit are both candidates for the yield. As we have no reason to choose between them we must decree again that the yield is undefined — it does not exist.

[4]Mathematicians should not confuse the conventional use of i to denote an effective interest rate in actuarial notation, with i conventionally used to denote $\sqrt{-1}$ in analysis.

8.3.2 Steffensen's Theorem

Fortunately, we can guarantee that the yield does exist in cases which cover many important financial contracts.

Steffensen's Theorem: Suppose that a financial contract has cashflows $c_1, c_2, \ldots, c_n$ at times $t_1 < t_2 < \ldots < t_n$. Then, if the cashflows change sign exactly once, the yield exists.

Comment: This covers the important case of buying a financial asset or taking out a loan. There is a single payment up front, in return for a stream of payments in future.

Sketch proof: Without loss of generality, suppose all the negative cashflows precede all the positive cashflows. Specifically, suppose $c_1, c_2, \ldots, c_k$ are negative and $c_{k+1}, \ldots, c_n$ are positive. Formulate the equation of value as:

$$\sum_{r=1}^{k} c_r(1+i)^{t_n - t_r} + \sum_{r=k+1}^{n} c_r(1+i)^{t_n - t_r} = 0.$$

Change to time t_k by multiplying through by $(1+i)^{t_k - t_n}$ and rewrite the equation of value as:

$$\sum_{r=1}^{k} c_r(1+i)^{t_k - t_r} - \sum_{r=k+1}^{n} -c_r(1+i)^{t_k - t_r} = 0.$$

Call the first sum $g_1(i)$ and the second sum $g_2(i)$, so the equation of value is:

$$g_1(i) - g_2(i) = 0.$$

Then elementary considerations show that:

- $g_1(i) < 0$, is a decreasing function of i, $\lim_{i \to -1} g_1(i) = c_k$, and $\lim_{i \to \infty} g_1(i) = -\infty$.
- $g_2(i) < 0$, is an increasing function of i, $\lim_{i \to -1} g_2(i) = -\infty$, and $\lim_{i \to \infty} g_2(i) = 0$.

Together, and given that $g_1(i)$ and $g_2(i)$ are continuous functions of i, these imply that the equation of value has exactly one root in $[-1, \infty)$. [Hint: draw a graph.]

8.4 Prices and Yields

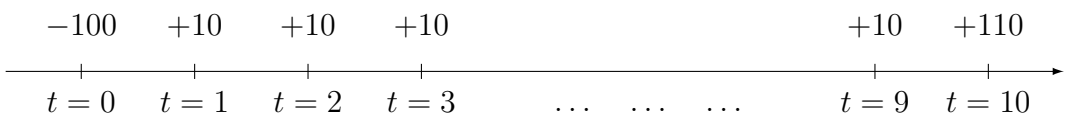
We are often faced with an investment opportunity (for example to buy shares, to invest in a new project) which will produce some future cashflows $c_1, c_2, \ldots, c_n$ at times $t_1 < t_2 < \ldots < t_n$. As always in these notes, we assume that all future cashflows are certain. We then need to decide whether or not to make the investment. In doing so, we are making a choice or decision: do I choose this investment over all the other opportunities that may be available? Then we seek objective criteria to guide this choice. One possible criterion is to choose the investment opportunity with the largest yield.

There are two obvious questions:

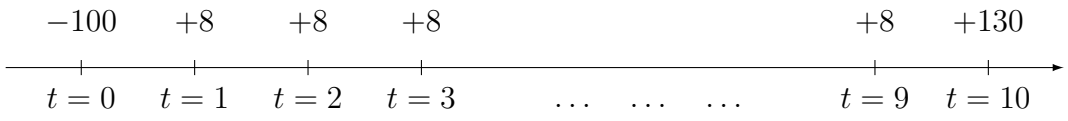
- If we want to earn a given yield on the investment, what price should we pay?
- If we pay a given price for the investment, what yield do we earn?

Example 8.4.1 We are considering two investments, represented by the timelines below.

Investment A:



Investment B:



On the basis of yield, which do we prefer?

bf Answer: It is clear that the yield on Investment A is 10% per time unit.

We formulate the equation of value for Investment B at $t = 10$ as follows:

$$f(i) = 100(1+i)^{10} - 8s_{\overline{10}|}^{i} - 122 = 0.$$

Trial and error solutions lead to:

$$\begin{aligned} f(0.09) &= -6.807 < 0 \\ f(0.10) &= +9.875 > 0 \end{aligned}$$

and then linear interpolation gives:

$$i^* \approx 0.09 + 0.01\,\frac{6.807}{6.807 + 9.875} \approx 0.094.$$

Hence we prefer Investment A, which has the higher yield. Note that this is despite the fact that Investment B returns more cash during the ten years: £202 against Investment A's £200. However, the cash advantage that Investment B delivers is through the large final payment of £130, and the time value of money erodes this.

Chapter 9

Measuring Fund Performance

9.1 Fund Performance

Insurance companies and pension funds hold large funds on behalf of their members. Members pay premiums or contributions into the fund, which are invested by the fund manager on their behalf, and later their benefits or pensions are paid out of the fund. So money is constantly being paid into and out of the fund.

Clearly the members expect their fund manager to obtain as good a rate of return as possible. They want to measure this by comparing the rate of return on their fund with those achieved by other forms of investment, and by other fund managers.

In this chapter, we will look at how we can measure the performance of a fund manager, and consequently, how we can compare a fund with other investments.

The salient features are the following.

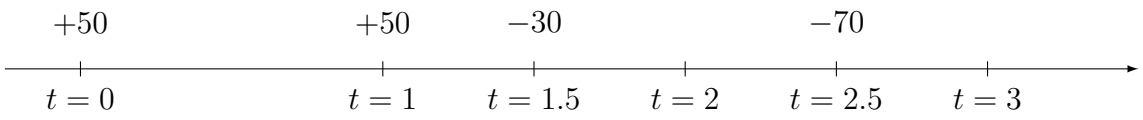
- Although new funds do start up and old funds are wound up, a fund may be an ongoing entity. We may want to measure its performance over some period of time before which it existed and after which it will continue.

- Cashflows are coming into and out of the fund very frequently. A fund may have thousands or even millions of members, generating cashflows both in and out on a weekly, daily or even more frequent basis.

We have already considered the yield as a method of comparing alternative investments. We begin by considering it as a candidate measure of fund performance.

9.2 Money-Weighted Rate of Return (Yield)

Example 9.2.1 Suppose that at time $t = 0$ a fund holds investments valued a t 50 and the the cashflows in and out of the fund during the next three time units are as in the timeline below



and that at time $t = 3$ the fund holds investments valued at 80.

We can formulate an equation of value, choosing to do so at $t = 3$. In doing so, we treat the 50 already in the fund at time $t = 0$ as if it were a cashflow in at that time, because it certainly must be included. Likewise, we treat the 80 in the fund at time $t = 3$ as if it were a cashflow out at that time.

$$f(i) = 100\,(1+i)^3 + 50\,(1+i)^2 - 30\,(1+i)^{1.5} - 70\,(1+i)^{0.5} - 80 = 0.$$

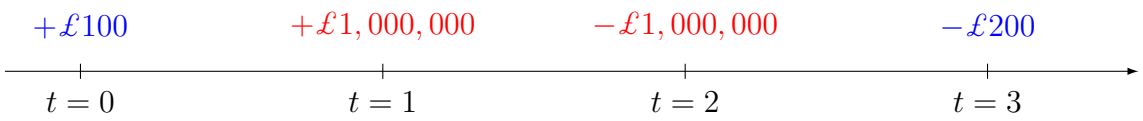
We can solve this for the yield, $i^* = 0.0856$ per time unit.

Comments:

- We set up a simple example above, just to illustrate the ideas which we will repeatedly use throughout this chapter. It has only four cashflows, whereas a real fund may have many thousands. Solving an equation of value in these circumstances may be very challenging and certainly messy.
- We contrived this example so that positive cashflows preceded negative cashflows, and by Steffensen's theorem the yield is sure to exist. With cashflows in and out of a real fund all the time, there is no guarantee that the yield will exist, and it might also be very difficult to try to check this.
- In the context of fund performance measurement, the yield is usually known as the *money-weighted rate of return* (MWRR), for reasons that will become apparent in the next example.

Example 9.2.2 Consider a fund with two members, A and B.

- At $t = 0$ it has nothing in it.
- Member A puts in £100 at time $t = 0$ and takes out £200 at time $t = 3$.
- Member B puts in £1 million at time $t = 1$ and takes out £1 million at time $t = 2$.
- at $t = 3$ the fund has nothing in it.



Investor A obtained a yield of $\left(\frac{200}{100}\right)^{1/3} - 1 \approx 0.26$ or about 26% per time unit and is presumably very happy. Investor B obtained a yield of nil and is presumably not so happy. If we formulate and solve the equation of value for the whole fund we write down:

$$f(i) = 100\,(1+i)^3 + 1,000,000\,(1+i)^2 - 1,000,000\,(1+i)^1 - 200 = 0$$

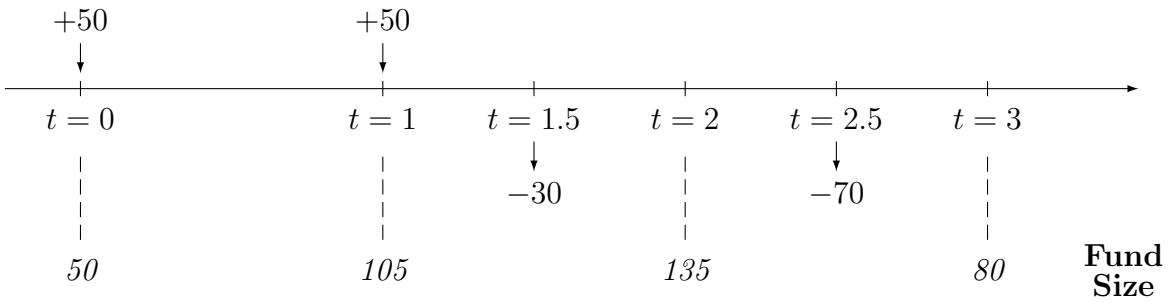
which we solve, getting $i^* \approx 0.0001$, or about 0.01% per time unit. The reason is that the equation of value is dominated by the largest cashflows. This is the origin of the name 'money-weighted' rate of return. It can be seen as a drawback. Here, clearly, the fund manager did very well indeed over the three time units as a whole, but has a poor performance in the middle of the period. Only the latter is revealed by the MWRR because the largest cashflows occurred in the middle of the time period.

9.3 Linked Internal Rates of Return

In order to calculate the yield, or MWRR, achieved on an ongoing fund, we had to know the value of the investments in the fund at the start and end of the period over which we wished to calculate the yield. So, in Example 1, the last cashflow was at time $t = 2.5$, but we did not know the fund size then, we knew the fund size at time $t = 0$ and $t = 3$ only. Therefore, we had to calculate the MWRR between those two times.

If we knew the fund size (the value of the investments held by the fund) at some intermediate times, we could calculate the yield over each of those sub-periods separately, and then link them together multiplicatively, using the corresponding accumulation factors, to find a yield over the whole period. This is called a *linked internal rate of return* (LIRR).

Example 9.3.1 Suppose in Example 9.2.1 we knew the fund size at some intermediate times, as follows. Note that the fund size at any time is usually given *before* any cashflow occurring at that time.



So, for example, the fund size is 105 at time $t = 1$ just before the cashflow of 50 into the fund, so at the start of the second sub-period the fund size will be $105 + 50 = 155$. These points where the fund size is known define three sub-periods, on each of which we can calculate a yield or MWRR. Define:

$$\begin{aligned} i_1 &= \text{the yield or MWRR on the period } t=0 \text{ to } t=1 \\ i_2 &= \text{the yield or MWRR on the period } t=1 \text{ to } t=2 \\ i_3 &= \text{the yield or MWRR on the period } t=2 \text{ to } t=3. \end{aligned}$$

Then on each sub-period, we formulate an equation of value, and solve it, as follows.

$$f(i_1) = 100\,(1+i_1)^1 - 105 = 0 \qquad \Rightarrow \qquad i_1^* = 0.05.$$

$$f(i_2) = 155\,(1+i_2)^1 - 30\,(1+i_2)^{0.5} - 135 = 0 \qquad \Rightarrow \qquad i_2^* = 0.0713.$$

$$f(i_3) = 135\,(1+i_3)^1 - 70\,(1+i_3)^{0.5} - 80 = 0 \qquad \Rightarrow \qquad i_3^* = 0.1482.$$

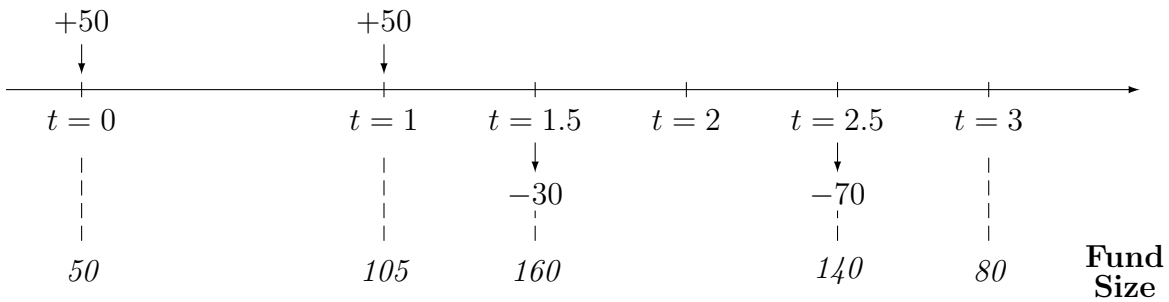
We then define the *linked internal rate of return* (LIRR) as the geometric average of the MWRRs (i.e. the yields), weighted by the lengths of the sub-periods (which in this example are all the same).

$$1+\text{LIRR} = \left((1.05)^1(1.0713)^1(1.1482)^1\right)^{1/3} \qquad \Rightarrow \qquad \text{LIRR} = 0.089.$$

Hence, the LIRR is approximately 8.9% per time unit.

Comments:

- The LIRR is not uniquely defined. It depends on the sub-division of the whole period into sub-periods, and this is an essential part of the statement of a LIRR. For example, we may have chosen a different subdivision as follows.



we now have four subperiods, so define:

$$\begin{aligned} i_1 &= \text{the yield or MWRR on the period } t=0 \text{ to } t=1 \\ i_2 &= \text{the yield or MWRR on the period } t=1 \text{ to } t=1.5 \\ i_3 &= \text{the yield or MWRR on the period } t=1.5 \text{ to } t=2.5 \\ i_4 &= \text{the yield or MWRR on the period } t=2.5 \text{ to } t=3 \end{aligned}$$

and then formulate and solve an equation of value for the yield on each sub-period, as follows:

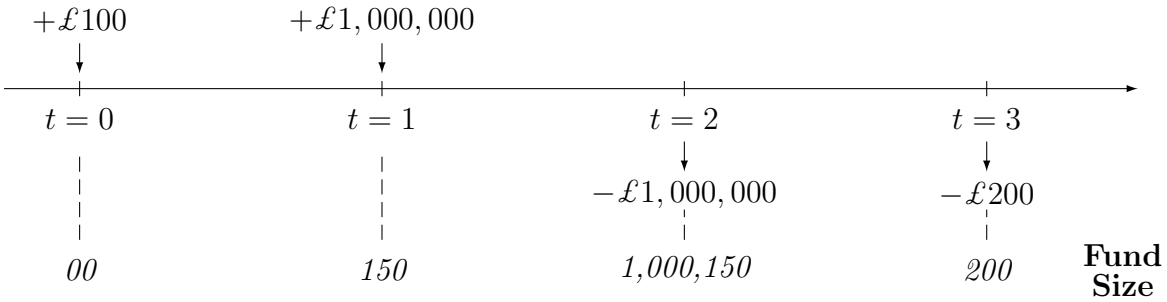
$$\begin{aligned} f(i_1) &= 100\,(1+i_1)^1 - 105 = 0 \quad \Rightarrow \quad i_1^* = 0.05 \\ f(i_2) &= 155\,(1+i_2)^{0.5} - 160 = 0 \quad \Rightarrow \quad i_2^* = 0.0656 \\ f(i_3) &= 130\,(1+i_3)^1 - 140 = 0 \quad \Rightarrow \quad i_3^* = 0.0769 \\ f(i_1) &= 70\,(1+i_4)^{0.5} - 80 = 0 \quad \Rightarrow \quad i_1^* = 0.3061. \end{aligned}$$

The LIRR is then the geometric average of the MWRRs (i.e. the yields), weighted by the lengths of the sub-periods (which in this example are not all the same).

$$1+\text{LIRR} = \left((1.05)^1(1.0656)^{0.5}(1.0769)^1(1.3061)^{0.5}\right)^{1/3} \quad \Rightarrow \quad \text{LIRR} = 0.1008.$$

Hence, the LIRR is approximately 10.1% per time unit. So the choice of sub-periods, in this case, made a difference of just over 1% to the result.

- It is clear, comparing the LIRRs above with the MWRR, that we gain much additional information from the process. We can now see, for example, that the fund performance was much influenced by an exceptionally good period in the last half time unit.
- The LIRR, with properly chosen sub-periods, is much less influenced by very large cashflows. Suppose in Example 9.2.2 we knew the fund sizes as follows.



So the relevant equations of value and yields on each sub-period are:

$$\begin{aligned} f(i_1) &= 100\,(1+i_1)^1 - 150 = 0 \quad \Rightarrow \quad i_1^* = 0.50 \\ f(i_2) &= 1,000,150\,(1+i_2)^{0.5} - 1,000,150 = 0 \quad \Rightarrow \quad i_2^* = 0 \\ f(i_3) &= 150\,(1+i_3)^1 - 200 = 0 \quad \Rightarrow \quad i_3^* = 0.333333. \end{aligned}$$

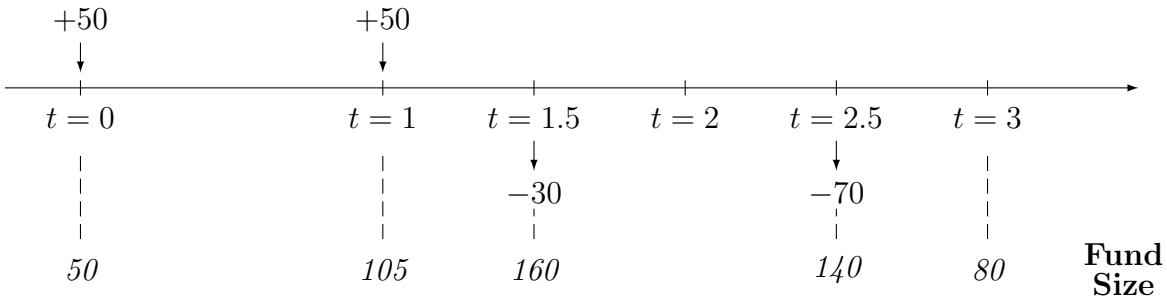
The LIRR is then the geometric average of the MWRRs (i.e. the yields), weighted by the lengths of the sub-periods:

$$1+\text{LIRR} = \left((1.50)^1(1.00)^1(1.333333)^1\right)^{1/3} \qquad \Rightarrow \qquad \text{LIRR} \approx 0.26.$$

Hence, the LIRR gives a more accurate impression of the fund performance over the whole period, eliminating the distortion cause by the large cashflows in the middle.

9.4 Time-Weighted Rate of Return

Suppose we know the fund size at the instant before every cashflow in or out of the fund (and at the beginning and the end, if there is not then a cashflow). We have actually already seen an example; it was the second LIRR calculation above based on Example 1.



Instead of calculating an LIRR, we can simplify the process, because there are no intervening cashflows between the times when the fund size is known. All we need do is calculate a *growth factor* for each period. Define:

$$\begin{aligned} g_1 &= \text{growth factor between } t = 0 \text{ and } t = 1 = \frac{105}{100} \\ g_2 &= \text{growth factor between } t = 1 \text{ and } t = 1.5 = \frac{160}{155} \\ g_3 &= \text{growth factor between } t = 1.5 \text{ and } t = 2.5 = \frac{140}{130} \\ g_4 &= \text{growth factor between } t = 2.5 \text{ and } t = 3 = \frac{80}{70}. \end{aligned}$$

The total growth factor over the whole period, as if we had invested 1 unit for the whole period, is just:

$$\frac{105}{100} \times \frac{160}{155} \times \frac{140}{130} \times \frac{80}{70} = 1.334.$$

Then the *time-weighted rate of return* (TWRR) is defined to be the geometric average rate of return per time unit, based on the total growth factor:

$$(1 + \text{TWRR}) = (\text{total growth factor})^{1/3} = 1.1008$$

so the TWRR is 10.08% per time unit.

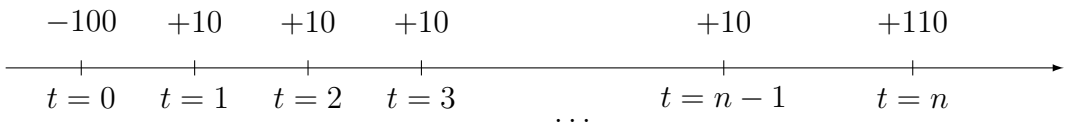
Comment: The TWRR is identical to the LIRR if the times at which the fund size is known that define the sub-periods used to calculate the LIRR include, as a subset, the times at which all the cashflows occur.

Chapter 10

Discounted Cashflows

10.1 Yields Again

Consider the following cashflows representing (for example) profits from investment in a company or venture.



The yield on this transaction is 10% effective per time unit, because $i^* = 0.1$ is the solution of the equation of value:

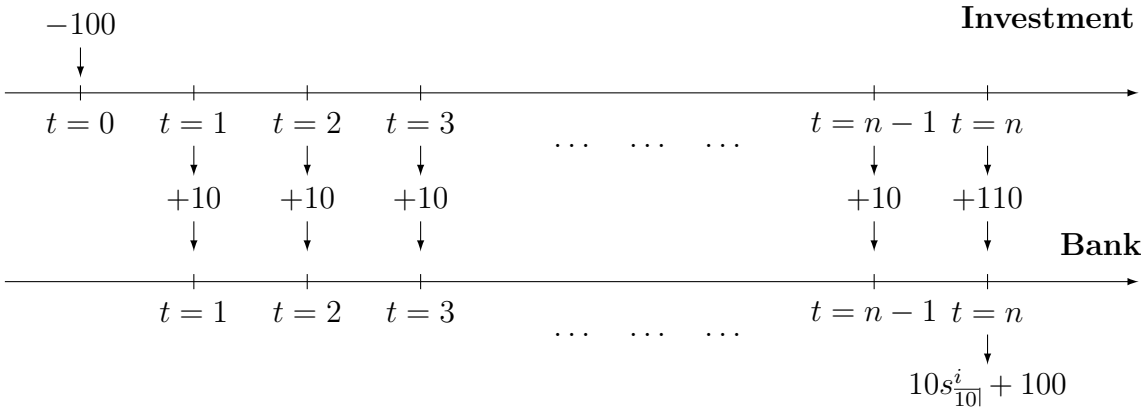
$$f(i) = -100(1+i)^n + 10s_{\overline{n}|}^{i} + 100 = 0.$$

We can rearrange this in a form recognizable as a statement of the Principle of Equivalence:

$$100 = 10\, a_{\overline{n}|}^{0.1} + 100\,(1.1)^{-n}.$$

The yield can be interpreted in terms of the Principle of Equivalence as follows: *if* money can be invested now and in the future (say in a bank account) to earn 10% per time unit effective interest, *then* we would be willing to exchange £100 now for the right to receive the future cashflows. The point here is that it really is necessary to be able to invest money to earn 10% per time unit — and this is not a given.

Looked at another way, if you invest in a project (or, indeed, buy a bond) that delivers cashflows to you at future points in time, you then have to *do something* with that money. Putting it in a bank account to earn more interest is an obvious option. In fact, it would be more realistic to split this investment into two parts: the part that delivers the cashflow to you, and the bank account in which you invest the cashflows:

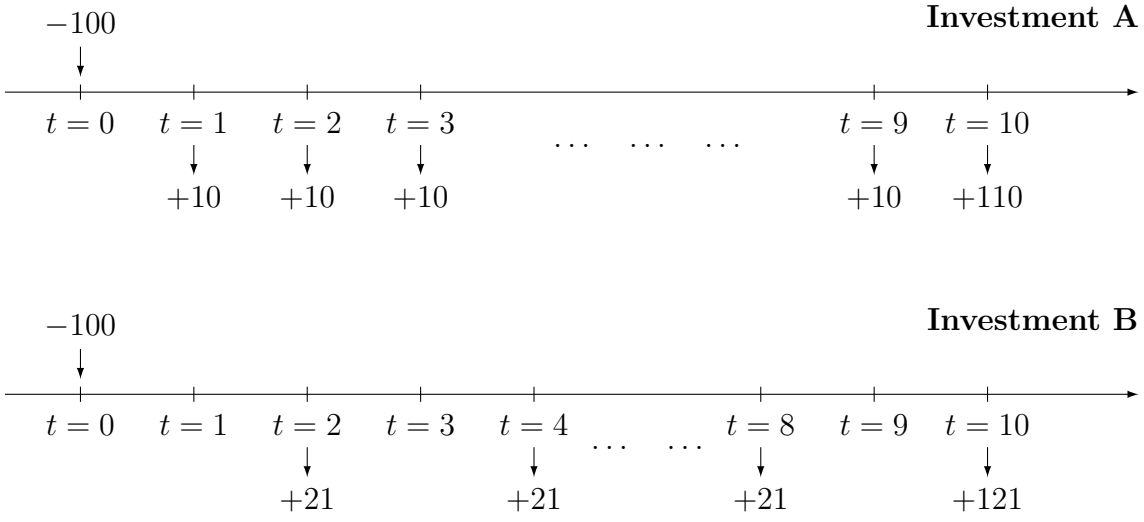


So the statement that the original investment has a yield of 10% per time unit effective is mathematically correct, but in the real world it only leads to an interpretation in terms of the Principle of Equivalence if market interest rates *actually are* 10% per time unit. If they are not, $i \neq 0.1$, then we see from the diagram above that once the intermediate cashflows are washed out, your transaction actually looks like the following:



And, it is the Principle of Equivalence we use to compare investments.

Consider the following two investments.



The yield on both investments is 10% per time unit effective (check this as an exercise). So *if* you could get a bank account that paid 10% interest per time unit effective, *then*:

- we would pay £100 for either investment; and
- we would be indifferent between which of A or B we bought.

But, suppose you could get a bank account that paid 15% per time unit effective. Then note that:

$$\begin{aligned} 100\,(1.15)^{10} &= 404.56 \\ 100 + 10\,s_{\overline{10}|}^{0.15} &= 303.04 \\ 100 + 21\,s_{\overline{5}|}^{0.3225} &= 298.32. \end{aligned}$$

So now:

- we would not pay £100 for either investment; and
- we would rank A above B.

Questions: In view of the above, why are yields important? And, what exactly are 'market rates of interest'?

Answers: 'Market rates of interest' in fact *are* the yields observed in the bond markets, on which fixed interest securities of various terms are traded.

10.2 Net Present Values

Definition 10.1 Given a set of cashflows $c_1, c_2, \ldots, c_n$ at times $t_1, t_2, \ldots, t_n$, their *net present value function* (NPV) is defined as:

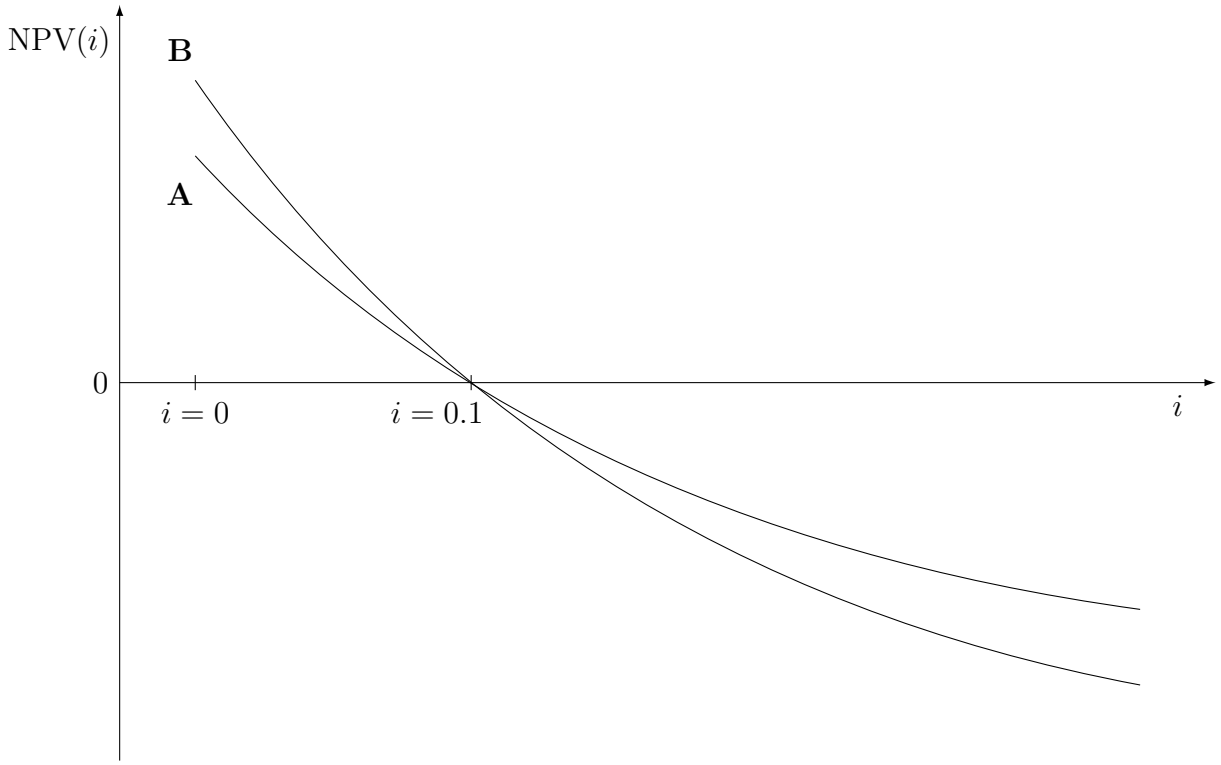
$$\mathrm{NPV}(i) = \sum_{r=1}^{n} c_r\,(1+i)^{-t_r}.$$

That is, the present value of the cashflows, at time $t = 0$, as a function of the (constant) effective rate of interest i. By comparing NPVs of different investments at the market rate of interest we can choose between different investments (this is the Principle of Equivalence again).

Example 10.2.1 Consider Investment A and Investment B from the previous example. Their NPV functions are:

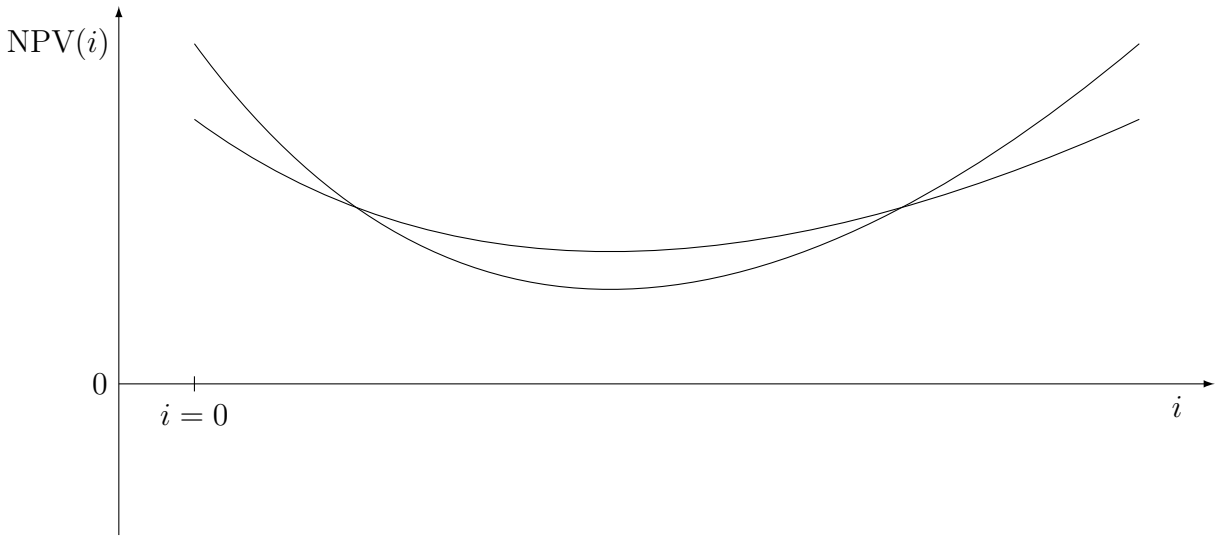
$$\begin{aligned} \mathrm{NPV}_A(i) &= -100 + 10\,a_{\overline{10}|}^{i} + 100(1+i)^{-10} \\ \mathrm{NPV}_B(i) &= -100 + 21\left((1+i)^{-2} + (1+i)^{-4} + (1+i)^{-6} + (1+i)^{-8} + (1+i)^{-10}\right) + 100(1+i)^{-10} \end{aligned}$$

We illustrate these functions in the following diagram.



Comments:

- The yield exists if and only if the equation NPV(i) = 0 has a unique solution not less than -1. However the NPV function always exists and is informative even if the yield does not exist. For example the diagram below shows the NPV functions of two investments that have no yield, but can still be meaningfully compared.



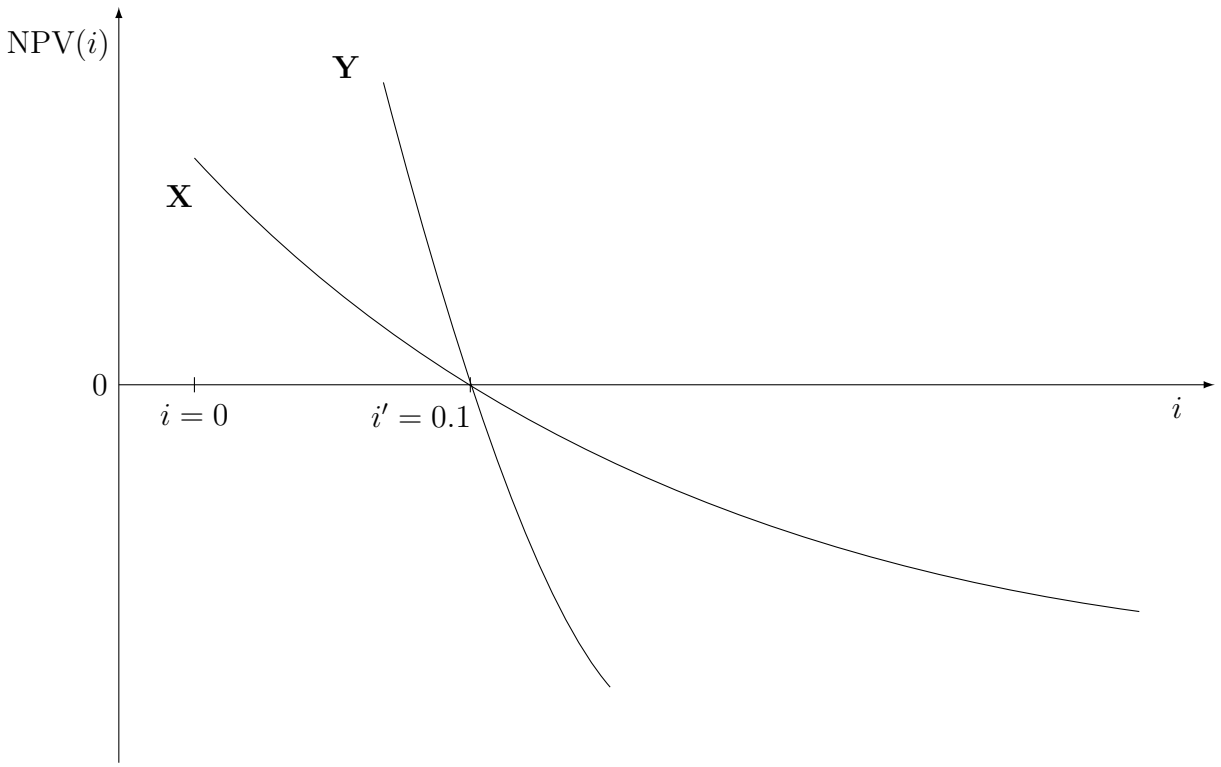
- The NPV functions of two investments may cross more than once, as in the previous examples. This means that we will prefer one to the other in different *regions* of market interest rates.

- NPV functions are additive: If we have two investments A and B and we combine their cashflows to create a new investment which we may call A+B, then:

$$\mathrm{NPV}_{A+B}(i) = \mathrm{NPV}_A(i) + \mathrm{NPV}_B(i).$$

  This is important when we form *investment portfolios* consisting of a collection of assets available in the market.

- Consider the two investments, X and Y, in the figure below.



  They have the same NPV at rate of interest i'. If the market interest rate is close to i', we should be indifferent between X and Y, according to the Principle of Equivalence. But Y is clearly much more sensitive to a *change* in the interest rate. For some purposes we would try to avoid this kind of sensitivity risk. Therefore, ranking of NPVs at a given interest rate (or the Principle of Equivalence) are not the only factors to be considered in practice.

- If the yield exists it is called the *internal rate of return* or IRR. This is the name most often used in corporate finance[1].

[1]The different name is a reflection of the fact that the concept of the yield has been reinvented several times, in different disciplines. We will meet yet another name for the yield in a later chapter.

Example 10.2.2 Which of investments A or B in the previous example would you prefer if the market rate of interest was: (i) 5% p.a. effective?; (ii) 20% p.a. effective?

Answer: Calculate the NPVs of the two investments.

(i) 5% p.a. effective.

$$\begin{aligned} \text{NPV}_A(0.05) &= 10\, a_{\overline{10}|}^{0.05} + 100\,(1.05)^{-10} - 100 = £38.61 \\ \text{NPV}_B(0.05) &= 21\, a_{\overline{5}|}^{0.1025} + 100\,(1.05)^{-10} - 100 = £ = 40.49. \end{aligned}$$

Hence we prefer Investment B.

(i) 20% p.a. effective.

$$\begin{aligned} \text{NPV}_A(0.20) &= 10\, a_{\overline{10}|}^{0.20} + 100\,(1.20)^{-10} - 100 = £ - 41.92 \\ \text{NPV}_B(0.20) &= 21\, a_{\overline{5}|}^{0.44} + 100\,(1.20)^{-10} - 100 = £ = -43.83. \end{aligned}$$

Hence we prefer Investment A, although we might also prefer to look elsewhere entirely.

10.3 The Discounted Payback Period

Most investments involve an initial outlay followed by income or profits. If an investment has a positive NPV (at interest rate i) there must be a first time when the initial outlay has been recovered, allowing for interest. This time is the *discounted payback period* (DPP), sometimes called the *break-even point*, at interest rate i .

Definition 10.2 Suppose we have a set of cashflows $c_1, c_2, \ldots, c_n$ at times $t_1, t_2, \ldots, t_n$, such that all negative cashflows precede all positive cashflows (investments followed by profits), and an effective rate of interest i per time unit such that $\text{NPV}(i) > 0$. The Discounted Payback Period (DPP) at effective rate of interest i per time unit is the smallest k such that:

$$\sum_{r=1}^{k} c_r (1+i)^{-t_r} \geq 0.$$

Example 10.3.1 You invest 100 at time $t = 0$, and receive casflows of 30 at times $t = 1, 2, 3, 4, 5$. The market interest rate of 5% p.a. effective. Find NPV(0.05) and the DPP.

Answer: $\text{NPV}(0.05) = -100 + 30\, a_{\overline{5}|}^{0.05} = 29.88$. This is positive, so the DPP exists and is the smallest k such that:

$$
\begin{aligned}
& 30\, a_{\overline{k}|}^{0.05} - 100 \geq 0 \\
\Rightarrow \quad & \frac{1-(1.05)^{-k}}{0.05} \geq \frac{10}{3} \\
\Rightarrow \quad & (1.05)^{-k} \leq 0.83333\ldots \\
\Rightarrow \quad & (1.05)^{k} \geq 1.2 \\
\Rightarrow \quad & k \geq \frac{\log 1.2}{\log 1.05} \\
\Rightarrow \quad & k = 4 \text{ years.}
\end{aligned}
$$

Comment: You may think of the investment in terms of a loan schedule, as in Chapter 7. The investment of 100 is treated as a 'loan' and each year's profit is applied first to repay 'interest' on the loan and then to repay a slice of the 'loan'. The DPP will automatically appear as the first repayment to reduce the capital content of the 'loan' to zero or less than zero. In the example above, the schedule would look like the following.

Time (Yrs)	'Repayment'	'Interest Content'	'Capital Content'	'Capital Outstanding'
0				100.00
1	30.00	5.00	25.00	75.00
2	30.00	3.75	26.25	48.75
3	30.00	2.44	27.56	21.19
4	30.00	1.06	28.94	−7.75
5	30.00	−0.39	30.39	−38.14

In the last year, the 'interest content' was negative because the 'loan account' was now in credit and earning interest rather then owing interest, and therefore the 'capital content' was larger than the 'repayment' itself. Note that $38.14\,(1.05)^{-5} = 29.88 = \text{NPV}(0.05)$.

10.3.1 Accumulated Profit

Complex investment decisions can be evaluated by working out the net cashflows and calculating the doscounted or accumulated profit. These can include money that is borrowed to finance projects, and the way in which borrowed money has to be repaid. We focus on accumulated profit.

Example 10.3.2 Suppose you can both borrow and invest money at 6% p.a. effective. You borrow £100 million, which is to be repaid by level annual repayments over ten years. The money finances a project that will make a profit of £18 million at the end of each of the *second* to tenth years inclusive.

(i) Find the net cashflows.

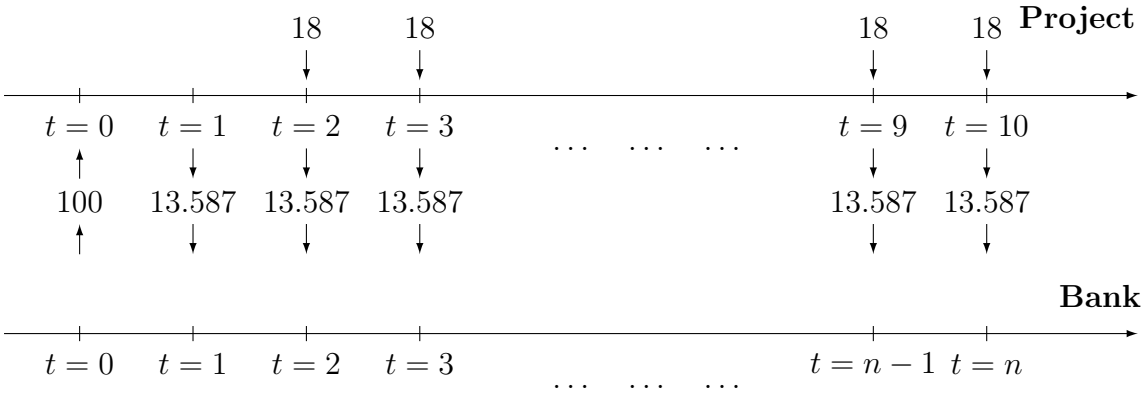
(ii) What is the accumulated profit at time $t = 10$?

(iii) What is NPV(0.06)?

For convenience, you may work in thousands of pounds.

Answers:

(i) The repayment is $100/a_{\overline{10}|}^{0.06}$ million = £13.587 million. The cashflows to and from the bank and the project are illustrated in the following related timelines



from which it is evident that the net cashflows to/from the project are those of the following timeline.



What happened to the initial £100 million? From the point of view of the borrower, it was money that came in (from the bank) and then went straight out again (spent on the project) so these cashflows cancelled out.

(ii) The accumulated profits are:

$$-13.587\,(1.06)^9 + 4.413\,s_{\overline{9}|}^{0.06} = £27.756 \text{ million.}$$

(iii) NPV(0.06) $= 27.756\,(1.06)^{-10} = £15.499$ million.

Methods that make direct use of cashflows such as these, allowing for several transactions at one point between different parties, are known as Disounted Cashflow (DCF) methods. In the example above, the investor had to deal with cashflows to and from the project, and cashflows to and from the bank. There can be many parties involved in a complicated project.

DCF methods can be very useful because each cashflow is considered separately and incorporated into the investor's net position explicitly. This allows for factors that would be very difficult to handle just using the traditional actuarial accumulated and discounted quantities. It does, also, mean that the use of a computer is necessary in any realistic example. We will illustrate its use to deal with one complicating factor, namely the fact that you cannot, in reality, borrow and invest money at the same rate of interest.

Example 10.3.3 You wish to borrow £100 to invest in a project that will return profits of £18 at the end of each of the next ten years. You borrow the money from a bank that charges interest of 8% p.a. effective. However, if you deposit or invest money you can earn only 6% p.a. effective interest. You have two possible ways to repay the loan,

(i) By net level repayments of £14.90 p.a., payable in arrears (check this!).

(ii) By using the entire £18 p.a. (profit from the investment project) to repay the loan as soon as possible, thereafter investing the profits to earn 6% p.a. interest.

Which method would you prefer?

Comment: In this problem, we cannot easily work with discounted quantities, because we do not have a single, unambiguous rate of interest at which to do the discounting[2]. What we can calculate unambiguously is the accumulated profit after ten years.

Answer: As in the last example, the initial £100 is 'money in and money out' and disappears from the net cashflows. The investor has a net profit of (18.00 - 14.90) - £3.10 p.a., and can invest this to earn 6% p.a. effective interest, so the accumulated profit after ten years is $3.1\, s_{\overline{10}|}^{0.06} = £40.86$.

If the investor applies the entire profits to repay the loan as soon as possible we need to find the time when the loan is repaid, in other words, the DPP. This is the smallest time k such that the accumulated profits exceed the accumulated loan, that is such that:

[2]Later, you will encounter a quantity called a *risk discount rate* which could be used to produce a present value in a case like this. It has its roots in the economic theory of risky investments and we will not deal with it here. We will say, though, that the risk discount rate would probably not be either 6% p.a. or 8% p.a. in this problem.

$$\begin{aligned} & 18\, s_{\overline{k}|}^{0.08} \geq 100\,(1.08)^k \\ \Rightarrow \quad & \frac{1-(1.08)^{-k}}{0.08} \geq \frac{100}{18} \\ \Rightarrow \quad & (1.08)^{-k} \leq 0.55555\ldots \\ \Rightarrow \quad & (1.08)^k \geq 1.8 \\ \Rightarrow \quad & k \geq \frac{\log 1.8}{\log 1.08} \\ \Rightarrow \quad & k = 8 \text{ years.} \end{aligned}$$

It would be purely coincidental if the 8th profit paid off the loan exactly, so we need to calculate the outstanding loan at time $t = 8$. It is:

$$18\, s_{\overline{8}|}^{0.08} - 100\,(1.08)^8 = £6.37.$$

We can now accumulate this, and future profits, to time $t = 10$ at 6% p.a. effective:

$$\text{Accumulated profit} = 6.37\,(1.06)^2 + 18\,(1.06) + 18 = £44.24.$$

Compare this with £40.86 when the load was repaid by level repayments. Intuitively this makes sense; if we are having to pay a higher rate of interest on borrowings it is best to pay off the borrowed money as fast as possible.

If you compare the above to the 'loan schedule' shown in Section 10.3, you will see that we are doing just the same as we did there, except that when the loan was paid off the interest rate changed.

Example 10.3.4 An investor wishes to build a factory costing £100,000,000 to make computer games consoles. It is expected to produce profits of £50,000,000 at the end of each of the second, third, fourth and fifth years, after which it will be written off as valueless because the next generation of consoles will have arrived. The project can be financed in either of two ways as follows:

(A) The capital can be borrowed at 7% *per annum* effective interest. The profits will be used to pay off the loan as soon as possible, and once that is done, further profits can be invested to earn 5% p.a. effective interest.

(B) The investor can issue a bond with annual coupons of 6%, payable yearly in arrears, and term five years. It will be issued at par but must be redeemed at 110%. If the investor needs to borrow any more money it will be at 7% p.a. effective, while any profits can be invested to earn 5% p.a. effective.

Which method of funding should the investor choose? For convenience, you may work in thousands of pounds.

Answer: Since either method, potentially, involves more than one interest rate, we cannot compute a simple discounted quantity, but we can compute the accumulated profit after five years.

(A) We note that there is no profit at time $t = 1$, so by that time the loan will have accrued interest and will stand at £107 million. First we must find the DPP, when the loan is completely paid off. This is the smallest number of payments k such that:

$$\begin{aligned} & 50\, s_{\overline{k}|}^{0.07} \geq 107\,(1.08)^k \\ \Rightarrow \quad & \frac{1-(1.07)^{-k}}{0.08} \geq \frac{107}{50} \\ \Rightarrow \quad & (1.07)^{-k} \leq 0.8502\ldots \\ \Rightarrow \quad & (1.07)^{k} \geq 1.1762 \\ \Rightarrow \quad & k \geq \frac{\log 1.1762}{\log 1.07} \\ \Rightarrow \quad & k = 3 \text{ payments.} \end{aligned}$$

The net profit to the investor when the loan is paid off is:

$$50\, s_{\overline{3}|}^{0.07} - 100\,(1.07)^4 = 29.665.$$

Hence the accumulated profit at time $t = 5$ is:

$$29.665\,(1.05) + 50 = £81.15 \text{ million.}$$

(B) The investor must borrow another £6 million at $t = 1$ to pay the coupon then due, as there is not yet any income. As in the last example, it is sensible to pay off as much as possible as soon as possible, so at time $t = 2$ the investor has net cashflow:

$$50 - 6 - 6\,(1.07) = £37.58 \text{ million.}$$

At $t = 3, 4, 5$ the investor's net cashflow is (50 - 6) = £44 million. Hence the accumulated profit at time $t = 5$ is:

$$37.58\,(1.05)^3 + 44\, s_{\overline{3}|}^{0.05} - 110 = £72.21 \text{ million.}$$

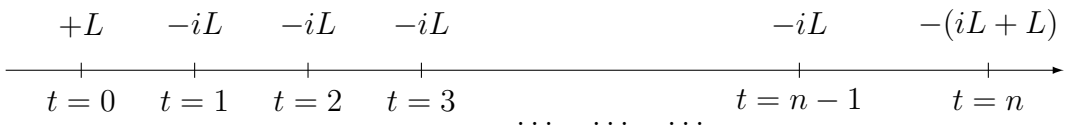
Hence the investor chooses method A.

Chapter 11

Fixed Interest Securities

11.1 Introduction and Terminology

A government or corporation can raise money in the capital markets by issuing *fixed interest securities* (FIS), also called *bonds*. Investors will lend money to the *issuer* (borrower) and in return will receive fixed interest payments at fixed dates plus repayment of the loan at the end of the term. This is the second of the three ways to repay a loan introduced in Section 7.1, represented by the timeline below:



In practice, this is a basic pattern around which many variations are woven.

The loan is usually very large, millions or even billions of currency units. The loan will be split into smaller units that can be traded on an exchange.

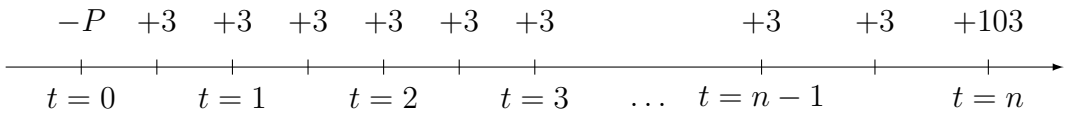
Example 11.1.1 A company raises £100,000,000 by issuing 1,000,000 bonds, each one a loan of face value £100. These can be bought and sold on a stock exchange.

Definition 11.1 The *nominal amount* (also called face value) of a bond is the amount of the loan it represents (£100 in the example above).

Definition 11.2 The interest payments are called *coupons*[1], usually expressed as a percentage per year of the nominal amount. They are always payable in arrears. It is common for coupons to be payable half-yearly or quarterly.

[1]The origin of the name 'coupons' is the fact that when bonds were actually issued as paper certificates, they sometimes had coupons appended, which had to be clipped and sent to the issuer of the bond to claim the interest payment.

Example 11.1.2 A bond of nominal value £100 carries coupons of 6% p.a. payable half yearly[2].



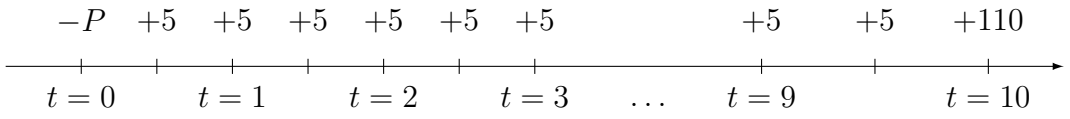
Here, the coupon is 6% p.a. but is payable half-yearly, meaning that for each £100 nominal a coupon of £3 is payable every half-year.

Example 11.1.3 A bond of nominal value £100 carries coupons of 10% p.a. payable quarterly. Then a coupon of £2.50 is payable every three months.

Definition 11.3 The loan is repaid or *redeemed* at the end of the *term*. The date at which it is redeemed is known as the *redemption date* (or maturity date). Often the amount that is repaid is different from the nominal amount. The *redemption amount* per £100 nominal is the *redemption rate*, often expressed as a percentage.

Example 11.1.4 A company issues a 10-year bond to be redeemed at 105%, with coupon of 10% p.a. payable half-yearly. The nominal amount of each bond is £100. What repayments are made?

Answer: The repayments are represented in the following timeline.



The final payment consists of the final coupon (£5) and the redemption amount (£105).

Terminology:

- If a bond is redeemed at less than 100% we say it is redeemed *at a discount*.
- If a bond is redeemed at 100% we say it is redeemed *at par*.
- If a bond is redeemed at more than 100% we say it is redeemed *at a premium*.

In the example above, we might also have said that the bond would be redeemed at a premium of 5%.

[2]Since coupons are always paid in arrears, we omit this from the definition of the coupon.

Terminology: Bonds are issued under many different names.

- Bonds issued by the UK Government are called *gilts* (short for 'gilt-edged securities', reflecting a feature of the old paper certificates).
- Bonds issued by the US Government are called *Treasuries* or *T-bonds*.
- Bonds issued by corporations are sometimes called *debentures*.
- Bonds issued by public sector corporations are sometimes called *loan stocks*.
- High-risk bonds issued by debt-financed companies are called *junk bonds*.

There are many other examples.

11.2 Bond Prices

At any time, *including the issue date*, investors will decide what price to pay for a bond. This is the basis of buying and selling bonds in the exchange. An important criterion is the yield compared to the yield available on other financial assets.

Example 11.2.1 A company issues a 10-year bond to be redeemed at 105%, with coupon of 10% p.a. payable half-yearly (same example as before). An investor wishes to buy this bond on its issue date, to give a yield of 11% p.a. effective (perhaps because she can obtain that yield on another investment). What price should she pay?

Answer: Applying the Principle of Equivalence, the investor will equate the present value of future income at 11% p.a. with the unknown price P:

$$P = 5\, a_{\overline{20}|}^{0.053565} + 105\,(1.11)^{-10} = 97.4492 = £97.45 \quad \text{(rounded)},$$

noting that 11% p.a. effective is equivalent to 5.3565 per half year effective[3]. Thus, if the market's expectation, when the bond is issued, is that 10-year securities should be yielding 11% p.a., the bond will be issued at a price below its face value of £100.

Terminology:

- If a bond is issued for less than its nominal amount we say it is issued *at a discount*.
- If a bond is issued for its nominal amount we say it is issued *at par*.
- If a bond is issued for more than its nominal amount we say it is issued *at a premium*.

[3]The easiest way to think about this bond is to regard it as having term twenty half-years and a coupon of 5% per half year.

Comments:

- The nominal amount of the loan is just a theoretical figure on which the coupon and redemption rates are based (it is usually £100).
- The amount of capital the borrower can actually raise on the issue date is the price that investors are willing to pay for the future income stream.
- As well as the yield on alternative investments, investors will consider the credit risk of the issuer, that is, the risk that they might default on interest or capital payments. The greater the risk, the higher the yield they will require. We do not consider this further in these notes.
- There is an inverse relationship between yields and bond prices.
- Once a bond is issued, it can be traded on an exchange at any time until it is redeemed. The price will depend on:
  - the remaining term to redemption; and
  - market conditions such as the yields obtainable on other investments.

Example 11.2.2 Returning to the bond in Example 11.2.1, just after the 10th coupon has been paid the investor sells the bond. At that time, the market for comparable 5-year bonds is indicating a yield of 7% p.a. effective. What price does the investor obtains for selling the bond?
Answer

$$P = 5\, a_{\overline{10}|}^{0.034408} + 105\,(1.07)^{-5} = 116.5709 = £116.57 \quad \text{(rounded)}$$

noting that 0.034408 effective per half-year is equivalent to 7% effective p.a..

Example 11.2.3 The investor in Example 11.2.2 has now:

- bought the bond for £97.45;
- received ten coupons of £5 each; and
- sold the bond after five years for £116.57.

What yield has she obtained on the completed transaction?

Answer: We formulate an equation of value using the half-year as time unit, because coupons are receivable half-yearly:

$$f(i) = -97.45\,(1+i)^{10} + 5\, s_{\overline{10}|}^{i} + 116.57 = 0.$$

Trial and error solutions lead to:

$$\begin{aligned} f(0.06) &= +7.956 > 0 \\ f(0.07) &= -6.047 < 0 \end{aligned}$$

and then linear interpolation gives:

$$i^* \approx 0.06 + 0.01 \frac{7.956}{7.956 + 6.047} \approx 0.06568 \quad \text{per half year.}$$

Thus the effective yield p.a. is approximately 13.6%.

11.3 Bond Yields

There are two important yields on bond transactions.

- The yield obtained by buying a bond and holding it to redemption is the *redemption yield*. This is what is quoted in financial newspapers and websites. If all taxes are ignored, the yield is called the *gross redemption yield*, abbreviated GRY. In the last example, the investor bought the bond on the issue date would have obtained a GRY of 11% p.a. had she held it to redemption, while the investor who bought the bond from her after five years would obtain a GRY of 7% p.a. if they held it to redemption.
- If a bond is sold before redemption, the investor will obtain (or realise) a yield that depends on both the buying price and the selling price. The first investor above sold the bond after five years to realise a yield of approximately 13.6% p.a.. Clearly the realised and redemption yields are equal if a bond is held to redemption.

The main calculations involving bonds are:

- Given a required yield, or current yield in the financial markets; what is the price?
- Given the buying price, what is the redemption yield?
- Given both buying and selling prices, what is the yield on the completed transaction?

The main complications are *tax*, and *special features* such as optional redemption dates. We will consider taxes next.

11.4 Income Tax

In the UK (and most other countries), coupon payments are regarded as income by the tax authorities. Therefore, if the holder of a bond is liable to pay income tax, their net coupon income will be reduced by the amount of income tax payable. Income tax is *not* levied on sale proceeds or redemption payments. Tax will affect both yields and prices.

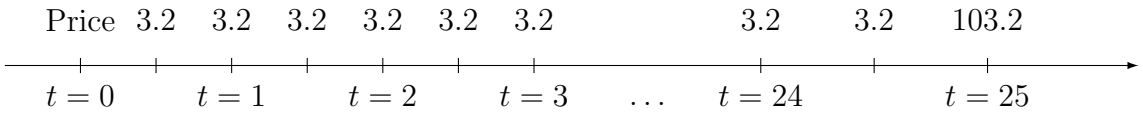
Example 11.4.1 An investor who pays income tax at 20% on all their income buys, at issue, a 25-year bond to be redeemed at par bearing *semi-annual* (twice per year) coupons of 8% p.a.. What price will he pay per £100 nominal to obtain a redemption yield after income tax of 9% p.a. effective?

Comment: The redemption yield after allowing for tax is known as the *net redemption yield* or NRY. Obviously NRY $\leq$ GRY, and the two are equal if there are no taxes.

Answer: First we must calculate the net coupon after income tax per £100 nominal of the bond. This is:

$$(1 - 0.2) \times 8/2 = £3.2 \quad \text{per half year.}$$

The timeline allowing for income tax is shown below.



We have:

$$P = 3.2\, a_{\overline{50}|}^{0.044031} + 100\,(1.09)^{-25} = 75.8449 = £75.84 \quad \text{(rounded)}.$$

11.5 Capital Gains Tax (CGT)

Capital Gains Tax, abbreviated CGT, may be payable on any gain made on the sale or redemption of a bond. The *gain* is defined as the difference between the price received on sale or redemption and the price originally paid. However, there is no relief from CGT on losses, i.e. negative gains.

Example 11.5.1 On 1 January 2025 an investor buys two bonds:

- Bond A: Price = £101
- Bond B: Price = £99.

After one year, the investor sells both bonds for £100 each. How much tax does he pay, if the rate of CGT is 30%? (that is, tax equal to 30% of any capital gain is levied).

Answer: The investor might argue that his net gain is:

$$(100 - 101) + (100 - 99) = 0$$

by netting off the loss on Bond A against the profit on Bond B. Unfortunately, the tax authorities do not work that way. He has to pay CGT on any individual bonds that made a profit, and neither pays nor receives any tax on individual bonds that make a loss. Therefore, he must pay no CGT on Bond A, and:

$$0.3\,(100 - 99) = £0.30$$

CGT on Bond B.

The fact that there is no rebate of CGT on losses means that the equation used to find the price of a bond varies depending on whether CGT is payable. It will be helpful in the remainder of this section to define some notation.

Notation: Define the following, in which the time unit is given by the frequency with which coupons are paid (for example, if coupons are payable half-yearly, the time unit is the half-year).

$$\begin{aligned} n &= \text{the outstanding term of a bond} \\ g &= \text{the amount of each coupon payment per £100 nominal, before income tax} \\ P &= \text{the price paid per £100 nominal} \\ R &= \text{the sale or redemption amount per £100 nominal, before any CGT} \\ t_1 &= \text{the rate of income tax} \\ t_2 &= \text{the rate of capital gains tax} \\ i &= \text{the effective yield per time unit obtained after all taxes.} \end{aligned}$$

We have two cases.

Case 1: $R \leq P$, so no CGT is payable.

$$P = (1 - t_1)\, g\, a_{\overline{n}|}^{i} + R\,(1+i)^{-n}.$$

Case 2: $R > P$, so CGT is payable.

$$P = (1 - t_1)\, g\, a_{\overline{n}|}^{i} + (R - t_2(R - P))\,(1+i)^{-n}.$$

Example 11.5.2 An investor liable to CGT at rate 30% buys a 15-year bond with an annual coupon of 9%, payable annually, to be redeemed at par. The price paid is £95 per £100 nominal. What net redemption yield does the investor obtain?

Answer: We have $R = 100 > 95 = P$ so CGT would be payable if the investor held the bond to redemption. This is an example of Case 2. We formulate the equation of value at the end of the term as usual, so:

$$f(i) = 95\,(1+i)^{15} - 9\, s_{\overline{15}|}^{i} - (100 - 0.3(100 - 95)) = 0.$$

Trial and error solutions lead to:

$$\begin{aligned} f(0.09) &= -16.712 < 0 \\ f(0.10) &= +12.386 > 0 \end{aligned}$$

and then linear interpolation gives:

$$i^* \approx 0.09 + 0.01\,\frac{16.712}{16.712 + 12.386} = 0.0957 \approx 0.096 \quad \text{p.a..}$$

If both buying and selling prices are known, it is straightforward to formulate an equation of value, as above, and to find the yield. It is less straightforward to find the price for a given net redemption yield, because the price itself determines whether or not CGT is payable. In order to find the price, we need to know whether or not CGT will be payable, and to know whether or not CGT will be payable, we need to know the price. Sometimes the answer is obvious from the circumstances.

Example 11.5.3 In Example 11.5.2, what price should the investor have paid to obtain a net redemption yield of 8% p.a. effective?

Answer: We know that a price of £95 gave a NRY of about 9.6%. Therefore a NRY of less than 9.6% will result in a price higher than £95. This is quite a large reduction in yield, so it is a reasonable guess that the price may rise above £100, so no CGT will be payable. Let us proceed on that assumption:

$$P = 9\, a_{\overline{15}|}^{0.08} + 100\,(1.08)^{-15} = 108.5595 = £108.56 \quad \text{(rounded)}.$$

We now *and we must* verify that our guess was correct: $108.56 > 100$ so we were correct; no CGT will be payable.

Example 11.5.4 In Example 11.5.2, what price should the investor have paid to obtain a net redemption yield of 10% p.a. effective?

Answer: We know that a price of £95 gave a NRY of about 9.6%. Therefore a NRY of more than 9.6% will result in a price lower than £95. So CGT will be payable (this is not now a guess, we have deduced it). Therefore:

$$P = 9\, a_{\overline{15}|}^{0.1} + (100 - 0.3(100 - P))\,(1.1)^{-15}.$$

We can solve this for P, to obtain $P = 91.8054 = £91.81$ (rounded).

Fortunately, we do not have to guess whether or not CGT will be payable; there is a test that will always tell us.

Result: With the notation as above, $P < R$, and CGT is payable, if and only if:

$$i > \frac{(1 - t_1)\, g}{R}.$$

Sketch proof: Consider a bond to be redeemed at par, $R = 100$. The net coupons are $(1 - t_1)\, g$ per time unit. The following are intuitively obvious, because of the inverse relationship between prices and yields:

- If the yield was $(1 - t_1)\, g$ per time unit, the price would be $P = 100$ and no CGT will be payable.
- If the yield was less than $(1 - t_1)\, g$, the price would rise, so certainly $P > 100$, and no CGT would be payable.

- If the yield was greater than $(1-t_1)\,g$, the price would fall, so certainly $P < 100$, and CGT would be payable. Note that the CGT would be less than the gain $100 - P$, so the CGT by itself would not negate the gain.

This is just the result we want, in the case that $R = 100$. If $R \neq 100$, we simply normalise the cashflows by considering a bond with a redemption amount of £100 and net coupons of $100\,(1 - t_1)g/R$.

Example 11.5.5 Consider a bond with term 10 years, coupon 10% p.a. payable half-yearly, to be redeemed at 120%. Find the price to give a net redemption yield of 7% p.a. effective, to an investor suject to CGT at rate 25%, and:

(i) subject to income tax at 30%.

(ii) not subject to income tax.

Answer: The time unit is the half year, so the effective yield per half year is 0.034408, and $g = 5, t_1 = 0.3, R = 120$.

(i) We note that $g\,(1 - t_1)/R = 0.029167 < i$ so CGT is payable and:

$$P = 3.5\,a_{\overline{20}|}^{0.034408} + (120 - 0.25(120 - P))\,(1.07)^{-10}$$

which we can solve to get $P = £109.70$.

(ii) We note that $g\,(1 - t_1)/R = 0.041667 > i$ so CGT is not payable and:

$$P = 5\,a_{\overline{20}|}^{0.034408} + 120\,(1.07)^{-10} = £132.45.$$

11.6 Real yields

In previous chapters you calculated the *money yield* yield on assets, projects and bonds. The money yield uses the money payments.

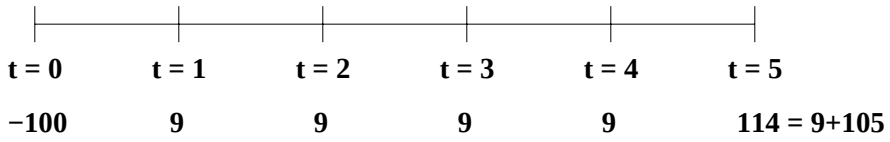
We consider how to calculate the *real yield*. This is the yield which is calculated using the real value of payments. Although the real value of payment is relative to a chosen fixed date, which we call the *base date*, it turns out the real yield doesn't depend on which base date we use.

Here we use i to denote the money yield and use i' to denote the real yield.

Example 11.6.1 (Real yields vs. Money yields)
Consider a 5-year fixed-interest bond issued in January 2005 with a coupon of 9% per annum paid annually and a redemption rate of 105%. The issue price is £100 per £100 nominal.
Question For an investor not subject to tax, calculate the yield:

(i) In money terms.

(ii) In real terms relative to Jan 2005 with reference to the RPI.



Solution: item(i) Solving $100 = 9a_{\overline{5}|} + 105v^5$ for an effective interest rate i p.a, we find $i = 9.822$ % p.a. item(ii) To adjust the bond payments to real values relative to January 2005, we divide each payment by the RPI at the payment date. Then we multiply by the RPI at January 2005. The calculations are shown below.

Time t	0	1	2	3	4	5
Money cashflow	-100	9	9	9	9	114
Payment date	Jan 2005	Jan 2006	Jan 2007	Jan 2008	Jan 2009	Jan 2010
RPI(Payment date)	188.9	193.4	201.6	209.8	210.1	217.9
Real cashflow relative to base date Jan 2005	$\frac{188.9}{188.9} \times -100$ $= -100$	$\frac{188.9}{193.4} \times 9$ $= 8.79059$	$\frac{188.9}{201.6} \times 9$ $= 8.43304$	$\frac{188.9}{209.8} \times 9$ $= 8.10343$	$\frac{188.9}{210.1} \times 9$ $= 8.09186$	$\frac{188.9}{217.9} \times 114$ $= 98.82790$

The real yield i' p.a. effective is the solution to:

$$100 = 8.79059v + 8.43304v^2 + 8.10343v^3 + 8.09186v^4 + 98.82790v^5 \quad \Rightarrow \quad i' = 6.705\%\text{p.a.}$$

Exercise 11.1 The base RPI date at which the real values are calculated (January 2005) is irrelevant for the calculation of i'. By considering the equation of value, explain why.

Exercise 11.2 What is the relationship between the money yield and the real yield?

Example 11.6.2 Today you deposit £1 000 in a bank account paying interest at 0% per annum. What will be the real value of the £1 000 in (i) one year's time? (ii) five years' time? and (iii) ten year's time?
We don't know what RPI will be in the future. All we can do is model the possible values of RPI.

11.7 Index-Linked Gilts (ILGs)

The first *index-linked gilt (ILG)* was issued by the UK Government in March 1981. Similar government bonds in other countries are Treasury Inflation-Protected Securities (TIPS), issued by the US Treasury since 1997, Obligations assimilables du Trésor Indexées sur l'inflation (OATi) issued by the L'Agence Trésor de France since 1998 and Real Return Bonds (RRB) issued by the Bank of Canada since 1991. Colloquially, index-linked bonds are called *linkers*.

As at 30 September 2013, the combined market value of ILGs is nearly £375 billion, compared to a total market value for conventional (fixed-interest) gilts of around £1 140 billion (Source: *DMO Quarterly Review for Jul-Sep 2013*). More information on the UK index-linked gilts, including their history, can be found at the UK Debt Management Office (DMO) website: `http://www.dmo.gov.uk/index.aspx?page=Gilts/Indexlinked`.

Example 11.7.1 The first ILG was the 2% Index-linked Treasury Stock 1996, with an *issue date* of 27 March 1981. For each £100 nominal, it paid:

- a coupon payment of £1 increased with RPI and paid on 16 March and 16 September of each year.
- a redemption payment of £100 increased with RPI and paid on 16 March 1996.

11.7.1 Calculating the money coupon and redemption payments for ILGs

For ILGs, the semi-annual coupon payments and redemption proceeds are adjusted in line with inflation as measured by the RPI. The rough idea is

$$\begin{aligned}\text{Money payment} = {} & \text{Coupon or redemption payment} \\ & \times \frac{\text{Lagged RPI for the month that the payment is made}}{\text{Base RPI figure}}.\end{aligned}$$

For ILGs issued on or after 2005, the *base RPI figure* is the RPI that is 3 months before the issue date. In other words, the *time lag* is 3 months. The reason for introducing a time lag is that it is not possible to index payments to the RPI in the month in which they are paid. A coupon payment in January 2014 cannot be increased using the RPI in January 2014 since the latter figure is not known until after the coupon payment is made.

For ILGs issued before 2005, the time lag is 8 months. The reason was to allow a purchaser of a ILG to know exactly the amount of the next coupon payment. It was changed to a 3-month time lag to be more consistent with other governments' index-linked bonds and increase liquidity and transparency in the ILG market. Additionally, a 3-month time lag provides better inflation protection than an 8-month time lag.

Example 11.7.2 The $4\frac{1}{8}\%$ Index-linked Treasury Gilt 2030 had an issue date of 12 June 1992 and uses an 8-month time lag. Its base RPI figure is 135.1, the RPI for October 1991.

It pays coupons on 22 January and 22 July each year. The latest coupon payments are shown in Table 11.1.

The coupon payment made at 22 July 2011 is calculated as

$$\frac{4\frac{1}{8}}{2} \times \frac{\text{RPI(Jul 2011} - \{8 \text{ months}\})}{\text{Base RPI figure}} = 2.0625 \times \frac{\text{RPI(Nov 2010)}}{\text{RPI(Oct 1991)}} = 2.0625 \times \frac{226.8}{135.1} = 3.4624352.$$

The figure is rounded *down* to 4 decimal places, to give a coupon payment of £3.4624 per £100 nominal.

Coupon payment date	22 Jan 2009	22 Jul 2009	22 Jan 2010	22 Jul 2010	22 Jan 2011
Lagged RPI date	May 2008	Nov 2008	May 2009	Nov 2009	May 2010
Lagged RPI	215.1	216.0	212.8	216.6	223.6
Coupon payment (£ per £100 nominal)	3.2838	3.2975	3.2487	3.3067	3.4135

Table 11.1: Recent coupon payments for the $4\frac{1}{8}\%$ Index-linked Treasury Gilt 2030.

The calculation of the coupons and redemption payments is fixed upon the issuance of each ILG. ILGs with a 3-month time lag have a slightly more complicated calculation that that for ILGs with an 8-month time lag. The rounding convention is also different depending on when the ILG is issued. For example, for the 2% Index-linked Treasury Stock 2035 stock, the figure is rounded *to the nearest* 6 decimal places.

Exercise 11.3 Check the RPI figures and the coupon payments in Table 11.1.

11.7.2 Calculating the money yield on ILGs

Suppose I want to calculate the money yield on an ILG, assuming that I hold it until redemption. To do this I need the future money cashflows. However, I do not know what these will be, as they depend on future inflation.
What can I do to give me an idea of the possible values of the money yield?
Idea: Assume that future inflation is a chosen fixed rate, e.g. 2.5% per annum.
Method: Using the last published RPI figure, project RPI into the future at the chosen fixed rate. Using the projected RPI, I calculate the future coupon and redemption cashflows, which I can then use to calculate a money yield.

Example 11.7.3 Suppose today is 28 November 2013, and the last published RPI is RPI(Oct 2013) = 251.9. Assuming that RPI increases in the future at a constant rate of 2.5% per annum, the projected RPI for November 2013 is calculated as

$$\text{RPI(Nov 2013)} = \text{RPI(Oct 2013)} \times 1.025^{1/12} = 252.418873.$$

Additional projections are shown below.

RPI date	Oct 2013	Nov 2013	Dec 2013	Jan 2014	Feb 2014
Projected RPI for date	215.1	252.418873	252.938814	253.459827	253.981913

Example 11.7.4 Suppose today is 22 January 2013 and the last published RPI figure is RPI(Dec 2012)= 246.8.

Today's market price of £100 nominal of the $4\frac{1}{8}\%$ Index-linked Treasury Stock 2030 is £323.57, just after the payment of the January 2013 coupon. The $4\frac{1}{8}\%$ Index-linked Treasury Stock 2030 was issued in June 1992 and has an 8-month time lag. Its Base RPI figure is 135.1, the October 1991 RPI.

Coupon dates: 22 January, 22 July. Redemption date: 22 July 2030.

Calculate the **money** yield-to-redemption assuming that RPI will increase in the future at a constant effective rate of 2.5% per annum.

Solution To calculate the money yield-to-redemption we need to:

1. Calculate the future money coupon and redemption payments by projecting the last published RPI figure at the assumed inflation rate, and

2. Solve the equation-of-value for the annual money yield i.

Payment date	8-month lagged RPI date	8-month lagged RPI	Money payment
22 Jan 2013	N/A		-323.57
22 Jul 2013	Nov 2012	245.6	$\frac{4.125}{2} \times \frac{\text{RPI(Nov 2012)}}{\text{Base RPI}} = \frac{4.125}{2} \times \frac{245.6}{135.1}$
22 Jan 2014	May 2013	$246.8 \times 1.025^{\frac{5}{12}}$	$\frac{4.125}{2} \times \frac{\text{RPI(May 2013)}}{\text{Base RPI}} = \frac{4.125}{2} \times \frac{246.8 \times 1.025^{\frac{5}{12}}}{135.1}$
22 Jul 2014	Nov 2013	$246.8 \times 1.025^{\frac{11}{12}}$	$\frac{4.125}{2} \times \frac{\text{RPI(Nov 2013)}}{\text{Base RPI}} = \frac{4.125}{2} \times \frac{246.8 \times 1.025^{\frac{11}{12}}}{135.1}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
22 Jan 2030	May 2029	$246.8 \times 1.025^{16\frac{5}{12}}$	$\frac{4.125}{2} \times \frac{\text{RPI(May 2029)}}{\text{Base RPI}} = \frac{4.125}{2} \times \frac{246.8 \times 1.025^{16\frac{5}{12}}}{135.1}$
22 Jul 2030	Nov 2029	$246.8 \times 1.025^{16\frac{11}{12}}$	$\left(100 + \frac{4.125}{2}\right) \times \frac{\text{RPI(Nov 2029)}}{\text{Base RPI}}$ $= \left(100 + \frac{4.125}{2}\right) \times \frac{246.8 \times 1.025^{16\frac{11}{12}}}{135.1}$

The equation of value is obtained from discounting each of the money payments back from the *payment date* to "today's" date of 22 January 2013 at the money yield i, and then summing them up.

$$\begin{aligned} 0 = -323.57 &+ \frac{4.125}{2} \times \frac{245.6}{135.1} \times v^{0.5} \\ &+ \frac{4.125}{2} \times \frac{246.8 \times 1.025^{\frac{5}{12}}}{135.1} \times v^{1} \\ &+ \frac{4.125}{2} \times \frac{246.8 \times 1.025^{\frac{11}{12}}}{135.1} \times v^{1.5} + \cdots + \frac{4.125}{2} \times \frac{246.8 \times 1.025^{16\frac{5}{12}}}{135.1} \times v^{17} \\ &+ \frac{4.125}{2} \times \frac{246.8 \times 1.025^{16\frac{11}{12}}}{135.1} \times v^{17.5} + 100 \times \frac{246.8 \times 1.025^{16\frac{11}{12}}}{135.1} \times v^{17.5}. \end{aligned}$$

To simplify the expression, look for a geometric series, i.e. a series of the form

$1 + r + r^2 + \cdots$.

$$\begin{aligned} 323.57 = &\frac{4.125}{2} \times \frac{245.6}{135.1} \times v^{0.5} \\ &+ \frac{4.125}{2} \times \frac{246.8}{135.1} \times \left(1.025^{\frac{5}{12}} v^1 + 1.025^{\frac{11}{12}} v^{1.5} \cdots 1.025^{16\frac{11}{12}} v^{17.5}\right) \\ &+ 100 \times \frac{246.8 \times 1.025^{16\frac{11}{12}}}{135.1} \times v^{17.5}. \end{aligned}$$

Taking out a factor of $1.025^{-\frac{1}{12}} v^{0.5}$ from the terms in the brackets, we obtain

$$\begin{aligned} 323.57 = &\frac{4.125}{2} \times \frac{245.6}{135.1} \times v^{0.5} \\ &+ \frac{4.125}{2} \times \frac{246.8}{135.1} \times 1.025^{-\frac{1}{12}} \times v^{0.5} \times \left(1.025^{0.5} v^{0.5} + 1.025^1 v^1 \cdots 1.025^{17} v^{17}\right) \\ &+ 100 \times \frac{246.8 \times 1.025^{16\frac{11}{12}}}{135.1} \times v^{17.5}. \end{aligned}$$

Define

$$1 + j := \left(\frac{1}{1.025v}\right)^{1/2} = \left(\frac{1+i}{1.025}\right)^{1/2},$$

in which i is the money yield. Then the equation of value can be written as

$$\begin{aligned} 323.57 = &\frac{4.125}{2} \times \frac{245.6}{135.1} \times \frac{1}{(1+i)^{0.5}} \\ &+ \frac{4.125}{2} \times \frac{246.8}{135.1} \times \frac{1}{1.025^{\frac{1}{12}} \times (1+i)^{0.5}} \times \left(\frac{1}{1+j} + \frac{1}{(1+j)^2} \cdots \frac{1}{(1+j)^{34}}\right) \\ &+ 100 \times \frac{246.8 \times 1.025^{16\frac{11}{12}}}{135.1} \times \frac{1}{(1+i)^{17.5}}, \end{aligned}$$

i.e.

$$\begin{aligned} 323.57 = &\frac{4.125}{2} \times \frac{245.6}{135.1} \times \frac{1}{(1+i)^{0.5}} \\ &+ \frac{4.125}{2} \times \frac{246.8}{135.1} \times \frac{1}{1.025^{\frac{1}{12}} \times (1+i)^{0.5}} \times a^{j}_{\overline{34}|} \\ &+ 100 \times \frac{246.8 \times 1.025^{16\frac{11}{12}}}{135.1} \times \frac{1}{(1+i)^{17.5}}. \end{aligned}$$

This is as far as we can reasonably simplify the equation of value. We can solve it using a computer or by hand using trial-and-error. We find $i = 2.187\%$ p.a.

Remark We can use different inflation assumptions.

- Assuming a constant future inflation rate of 2.0% p.a. results in a money yield of $i = 1.710\%$ p.a.

- Assuming a constant future inflation rate of 3.0% p.a. results in a money yield of $i = 2.665\%$ p.a.

- We could use different inflation rates for different periods, such as

  - 3.0% p.a. up to 22 January 2018, followed by
  - 2.5% p.a. from 23 January 2018 to the maturity date.

11.7.3 Calculating the real yield on ILGs

Now we turn to calculating the real yield. For this, I need to the real values of the future coupon and redemption proceeds.

To calculate the real yield-to-redemption,

(i) Calculate the future money coupon and redemption payments by projecting the last published RPI figure at the assumed inflation rate, to calculate

$$\text{Money payment} = \text{Coupon or redemption payment} \times \frac{\text{RPI(Payment date - Time lag)}}{\text{Base RPI figure}}.$$

(ii) Adjust the money values to the real values at a chosen fixed date, e.g. adjusting them to today's purchasing power, we calculate

$$\text{Real payment} = \text{Money payment} \times \frac{\text{RPI(Today's date)}}{\text{RPI(Payment date)}}.$$

(iii) Solve the equation of value for the annual real yield i'.

Example 11.7.5 Suppose today is 22 January 2013 and the last published RPI figure is RPI(Dec 2012)= 246.8. The market price of £100 nominal of the $4\frac{1}{8}\%$ Index-linked Treasury Stock 2030 is £323.57, just after the payment of the January 2013 coupon.

Calculate the **real** yield-to-redemption assuming that RPI will increase in the future at a constant effective rate of 2.5% per annum.

Solution: We have already calculated the money payments. The last published RPI figure is RPI(Dec 2012)= 246.8.

To calculate the real payments at "today's" date of 22 January 2013, we multiple each money payment by

$$\frac{\text{RPI(Jan 2013)}}{\text{RPI(Payment date)}} = \frac{\text{RPI(Dec 2012)} \times 1.025^{\frac{1}{12}}}{\text{RPI(Payment date)}} = \frac{246.8 \times 1.025^{\frac{1}{12}}}{\text{RPI(Payment date)}}.$$

Payment date	RPI(Payment date)	RPI(Jan 2013) / RPI(Payment date)	Real payment at Jan 2013 = Money payment $\times \frac{\text{RPI(Jan 2013)}}{\text{RPI(Payment date)}}$
22 Jan 2013	$246.8 \times 1.025^{\frac{1}{12}}$	1	-323.57×1
22 Jul 2013	$246.8 \times 1.025^{\frac{7}{12}}$	$\frac{1}{1.025^{0.5}}$	$\frac{4.125}{2} \times \frac{245.6}{135.1} \times \frac{1}{1.025^{0.5}}$
22 Jan 2014	$246.8 \times 1.025^{1\frac{1}{12}}$	$\frac{1}{1.025^{1}}$	$\frac{4.125}{2} \times \frac{246.8 \times 1.025^{\frac{5}{12}}}{135.1} \times \frac{1}{1.025^{1}}$
22 Jul 2014	$246.8 \times 1.025^{1\frac{7}{12}}$	$\frac{1}{1.025^{1.5}}$	$\frac{4.125}{2} \times \frac{246.8 \times 1.025^{\frac{11}{12}}}{135.1} \times \frac{1}{1.025^{1.5}}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
22 Jan 2030	$246.8 \times 1.025^{17\frac{1}{12}}$	$\frac{1}{1.025^{17}}$	$\frac{4.125}{2} \times \frac{246.8 \times 1.025^{16\frac{5}{12}}}{135.1} \times \frac{1}{1.025^{17}}$
22 Jul 2030	$246.8 \times 1.025^{17\frac{7}{12}}$	$\frac{1}{1.025^{17.5}}$	$\left(100 + \frac{4.125}{2}\right) \times \frac{246.8 \times 1.025^{16\frac{11}{12}}}{135.1} \times \frac{1}{1.025^{17.5}}$

The equation of value is obtained from discounting each of the real payments back from the *payment date* to "today's" date of 22 January 2013 at the real yield i', and then summing them up. Setting $v := 1/(1+i')$ and cancelling out some of the 1.025 indices gives

$$\begin{aligned} 0 = -323.57 &+ \frac{4.125}{2} \times \frac{245.6}{135.1} \times \frac{1}{1.025^{0.5}} \times v^{0.5} \\ &+ \frac{4.125}{2} \times \frac{246.8}{135.1} \times \frac{1}{1.025^{\frac{7}{12}}} \times v^{1} \\ &+ \frac{4.125}{2} \times \frac{246.8}{135.1} \times \frac{1}{1.025^{\frac{7}{12}}} \times v^{1.5} \\ &\cdots + \frac{4.125}{2} \times \frac{246.8}{135.1} \times \frac{1}{1.025^{\frac{7}{12}}} \times v^{17.5} \\ &+ 100 \times \frac{246.8}{135.1} \times \frac{1}{1.025^{\frac{7}{12}}} \times v^{17.5}. \end{aligned}$$

i.e.

$$\begin{aligned} 323.57 = &\frac{4.125}{2} \times \frac{245.6}{135.1} \times \frac{1}{1.025^{0.5}} \times v^{0.5} \\ &+ \frac{4.125}{2} \times \frac{246.8}{135.1} \times \frac{1}{1.025^{\frac{7}{12}}} \times v^{0.5} \times \left(v^{0.5} + v^{1} + \cdots + v^{17}\right) \\ &+ 100 \times \frac{246.8}{135.1} \times \frac{1}{1.025^{\frac{7}{12}}} \times v^{17.5}. \end{aligned}$$

The equation of value simplifies to

$$323.57 = \frac{4.125}{2} \times \frac{245.6}{135.1} \times \frac{1}{1.025^{0.5}} \times v^{0.5} + \frac{246.8}{135.1} \times \frac{1}{1.025^{\frac{7}{12}}} \times \left(\frac{4.125}{2} \times v^{0.5} \times a^{(2)i'}_{\overline{17}|} + 100 \times v^{17.5} \right).$$

We solve to find $i' = -0.305\%$ p.a.

Remark: The main selling point of an index-linked bond is that it protects the investor from price inflation. However, if we re-calculate the real yield using

- a constant inflation rate 2.0% p.a. then $i' = -0.285\%$ p.a.
- a constant inflation rate 3.0% p.a. then $i' = -0.325\%$ p.a.

The reason that the real yield is not invariant to the inflation assumption is due to the indexation time lag. The impact of the indexation time lag is seen in the last equation through the terms $\frac{1}{1.025^{0.5}}$ and $\frac{1}{1.025^{\frac{7}{12}}}$.

If there was no indexation time lag, then

- the real yield would be independent of future inflation. It would depend only on the market price, and
- the real yield would be known as soon as the investor buys the bond.

In practice, as there is an indexation time lag,

- the real yield, actual or anticipated, depends on future inflation as well as the market price, and
- the investor is exposed to some inflation risk. The longer the time lag, the greater the inflation risk to which the investor is exposed.

Exercise 11.4 Explain the meaning of a negative real yield.

Exercise 11.5 Would you invest in an ILG if it had a negative real yield? Explain your reasoning.

11.7.4 Break-even inflation rate (BEIR)

Index-linked bonds can also give an indication of the market consensus on future inflation. Suppose we have two bonds which are identical in characteristics, except that one is an index-linked bond and the other is a fixed interest bond, i.e.

- FI bond with annual coupon C p.a. payable m times a year and redeemed at $R\%$ in N years.

- IL bond with annual coupon C p.a. payable m times a year and redeemed at $R\%$ in N years.

Define

$$\{\text{Break-even inflation rate}\} := \{\text{Money yield on FI Bond}\} - \{\text{Real yield on IL Bond}\}.$$

The break-even inflation rate gives the market consensus on future inflation (ignoring issues such as the indexation time lag). It is the inflation rate at which the investor is indifferent between holding the FI bond and the IL bond.

Example 11.7.6 Example 11.7.4 continued)
Suppose a FI bond with coupon $4\frac{1}{8}\%$ p.a. paid twice a year on 22 January and 22 July and is redeemed at par on 22 July 2030 gives a money yield of 4.2%. The real yield on the $4\frac{1}{8}\%$ Index-Linked Treasury Stock 2030 is 1.5% p.a. The break-even inflation rate is $4.2 - 1.5 = 2.7\%$. If an investor believes:

- future rate of inflation will be lower (say 2.5% p.a), then the investor should buy the FI bond. Why? Because the investor believes the money yield from the IL bond will be lower than from the FI bond.

- future rate of inflation will be higher (say 3.0% p.a), then the investor should buy the IL bond. Why? Because the investor believes the money yield from the IL bond is higher.

Chapter 12

Duration, Convexity and Immunisation

12.1 Duration

Consider a set of discrete future cash flows C_{t_k} to be paid at time t_k, $0 < t_1 < \ldots < t_n < T$ and a continuous cashflow at rate of payment $\rho(t)$ for $t \in [0, T]$.

The net present value of these cash flows at constant force of interest δ is

$$NPV(\delta) = \sum_{k=1}^{n} C_{t_k} e^{-\delta t_k} + \int_0^T \rho(t) e^{-\delta t} dt.$$

What happens if the force of interest suddenly changes - how will the NPV change? Will it increase or decrease it value? By how much will it increase or decrease? We introduce the concept of *duration* to estimate the latter. It measures the sensitivity of the NPV of a set of cashflows to changes in the interest rate.

Definition 12.1 The *(Macaulay) duration* $\tau(\delta)$ of these cashflows at the force of interest δ is

$$\begin{aligned}\tau(\delta) &= -\frac{1}{NPV(\delta)} \times \frac{d}{d\delta} NPV(\delta) \\ &= \frac{\sum_{k=1}^{n} t_k C_{t_k} e^{-\delta t_k} + \int_0^T t\rho(t) e^{-\delta t} dt}{\sum_{k=1}^{n} C_{t_k} e^{-\delta t_k} + \int_0^T \rho(t) e^{-\delta t} dt}.\end{aligned}$$

Remarks:

(i) The duration is also called the *discounted mean term (DMT)*.

(ii) It is the weighted average time until payment of a set of cashflows, with the weights being the present value of each cashflow.

12.1.1 Use of duration

$\tau(\delta)$ measures the proportionate change in $NPV(\delta)$ due to a small change in the force of interest δ.

$$\tau(\delta) \approx -\frac{1}{NPV(\delta)}\frac{\Delta NPV(\delta)}{\Delta\delta} \qquad \Rightarrow \qquad \frac{\Delta NPV(\delta)}{NPV(\delta)} \approx -\Delta\delta \cdot \tau(\delta).$$

Thus, if the force of interest changes from δ_0 to δ_1 then:

$$\frac{NPV(\delta_1) - NPV(\delta_0)}{NPV(\delta_0)} \approx -(\delta_1 - \delta_0) \cdot \tau(\delta_0)$$

This can be a good approximation for *immediate small* changes in δ.
Aside: Duration calculated as

$$-\frac{1}{NPV(i)} \times \frac{d}{di} NPV(i)$$

is called the *modified duration*. It is not equal to the Macaulay duration; can you find the relationship between them? We consider only the Macaulay duration in this course.

Definition 12.2 An *n-year zero-coupon bond* is a bond which pays only a redemption amount in n years time and no coupons. If we buy £1 nominal of a n-year zero-coupon bond, this means that we receive £1 in n years' time.

Example 12.1.1 If £1 nominal of an 3-year zero-coupon bond costs £0.95, then an investor can pay £0.95 today to receive a payment of £1 in exactly 3 years' time.

Example 12.1.2 Suppose the force of interest is $\delta = 0.05$ p.a. How much should an investor pay today to buy £100 nominal of a 10-year zero-coupon bond?

Solution: $100e^{-10\delta} = 100e^{-10\times 0.05} = 100 \times 0.6065 = £60.65$.

Example 12.1.3 Show that the duration of an n-year zero-coupon bond (ZCB) is n years, i.e. it does not depend on the chosen force of interest.

Solution: Duration at the force of interest δ, i.e. $\tau(\delta)$ is given by the formula:

$$\tau(\delta) = -\frac{1}{NPV(\delta)} \times \frac{d}{d\delta} NPV(\delta)$$

where the net present value of the n-year ZCB is, $NPV(\delta) = 1 \times e^{-\delta n}$.
The derivative of NPV(δ) with respect to δ is

$$\frac{d}{d\delta} NPV(\delta) = -ne^{-\delta n},$$

hence,

$$\tau(\delta) = -\frac{1}{e^{-\delta n}} \times (-ne^{-\delta n}) = n.$$

Exercise 12.1 Show that the duration of an n-year annuity of £1 p.a. paid in arrears is

$$\frac{(Ia)_{\overline{n}|}}{a_{\overline{n}|}}.$$

Solution: The net present value of an n-year annuity of £1 p.a. paid in arrears at the effective interest rate i p.a. can be written as

$$\text{NPV}(i) = \sum_{t=1}^{n}(1+i)^{-t}. \tag{12.1}$$

Note that $(1+i) = e^{\delta}$ and so Equation 12.1 can be re-written as

$$\text{NPV}(\delta) = \sum_{t=1}^{n} e^{-\delta t},$$

where

$$= \frac{d}{d\delta}\text{NPV}(\delta) = \sum_{t=1}^{n} -te^{-\delta t},$$

so the corresponding duration is

$$\tau(\delta) = \frac{-1}{\sum_{t=1}^{n} e^{-\delta t}} \sum_{t=1}^{n} -te^{-\delta t},$$

and since $e^{\delta} = (1+i)$ or $e^{-\delta} = v$, we get,

$$\tau(\delta) = \frac{\sum_{t=1}^{n} tv^t}{\sum_{t=1}^{n} v^t}$$

$$\frac{(Ia)_{\overline{n}|}}{a_{\overline{n}|}}.$$

Exercise 12.2 Show that the duration of an n-year annuity of £1 p.a. paid continuously is

$$\frac{(\bar{I}\bar{a})_{\overline{n}|}}{\bar{a}_{\overline{n}|}} = \frac{(\overline{Ia})_{\overline{n}|}}{\bar{a}_{\overline{n}|}}.$$

Solution: The net present value of an n-year annuity of £1 p.a. paid continuously is

$$\text{NPV}(\delta) = \int_{t=0}^{n} e^{-\delta s} ds. \tag{12.2}$$

where

$$\frac{d}{d\delta}\text{NPV}(\delta) = \int_{t=0}^{n} (-s)e^{-\delta s} ds,$$

so the corresponding duration is

$$\begin{aligned}\tau(\delta) &= \frac{-1}{\int_{t=0}^{n} e^{-\delta s} ds} \int_{t=0}^{n} (-s) e^{-\delta s} ds \\ &= \frac{\int_{t=0}^{n} s e^{-\delta s} ds}{\int_{t=0}^{n} e^{-\delta s} ds} \\ &= \frac{(\bar{I}\bar{a})_{\overline{n}|}}{a_{\overline{n}|}}.\end{aligned}$$

Example 12.1.4 Two investments providing the following two sets of cashflows

Time (years)	Cashflow 1	Cashflow 2
1	50	6.7
2	25	13.4
3	12	26.8
4	6	53.6

Calculate the NPV and the duration of the two sets of cashflows at $\delta = 0.05$.

Solution: Since the duration is the weighted mean term of a cashflow we already know without any calculations that the durations for both cashflows must be between 1 and 4 since all payments are between time 1 and 4. Furthermore, we know without calculations that the duration of cashflow 1 is smaller than the duration of cash flow 2 since cashflow 1 provides higher payments at the beginning of the period and cashflow 2 provides higher payments at the end of the period. This means that (a) the average mean term of cashflow 1 must be smaller than the one of cashflow 2 (we calculate a weighted average mean term) and (b) cashflow 2 is much more sensitive to a change in the force of interest because we receive money relatively late as compared to cashflow 1.

$$\delta = 0.05 \implies v = e^{-0.05}.$$

$$NPV_1(\delta) = 85.4 = NPV_2(\delta)$$

$$\tau_1(\delta) = \frac{\{1 \times 50 \times v + 2 \times 25 \times v^2 + 3 \times 12 \times v^3 + 4 \times 6 \times v^4\}}{NPV_1(\delta)} = 1.68$$

$$\tau_2(\delta) = \frac{\{1 \times 6.7 \times v + 2 \times 13.4 \times v^2 + 3 \times 26.8 \times v^3 + 4 \times 53.6 \times v^4\}}{NPV_2(\delta)} = 3.22$$

Figure 12.1.1 and Figure 12.1.1 show the graphs of $\tau(\delta)$ for the two cashflows that we have considered. It is obvious from these graphs that the sensitivity of the NPV of both cashflows decreases with an increasing force of interest.

Using again the interpretation of the duration as the mean term it is clear that the mean term is decreasing with an increasing force of interest since the importance of payments in the far future becomes less important when the market force of interest is high. This can be compared to the discounted payback period for the two cashflows which is increasing if the market force of interest is increasing.

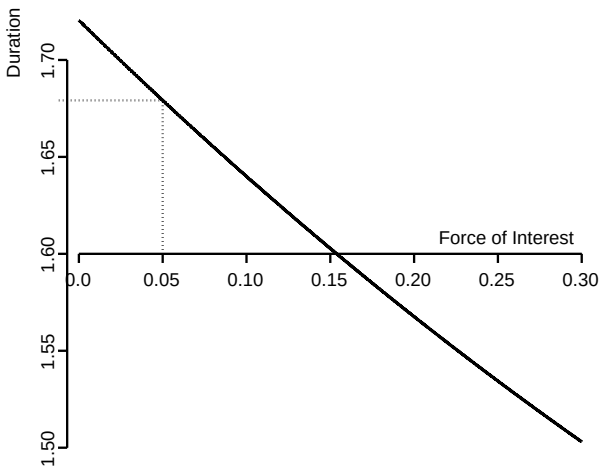


Figure 12.1: Duration of cashflow 1 in Example 12.1.4

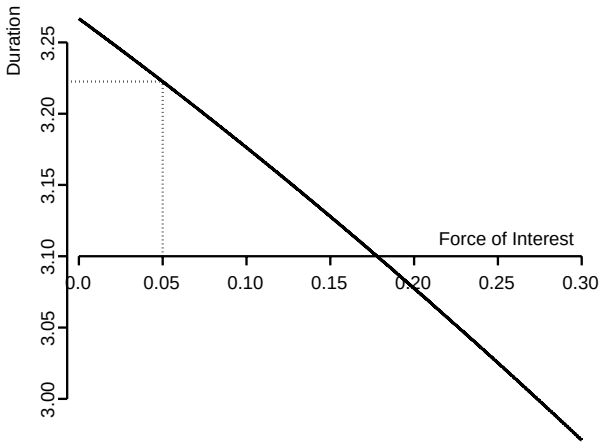


Figure 12.2: Duration of cashflow 2 in Example 12.1.4

Example 12.1.5 Consider the follong fixed-income securities.

A: term 10 years, coupon 12% p.a. paid annually in arrears, redemption at par.

B: term 25 years, coupon 3% p.a. paid annually in arrears, redemption 120%.

$$\begin{aligned}\tau_A(\delta) &= \frac{1\times 12\times v + 2\times 12\times v^2 + \cdots + 10\times 12\times v^{10} + 10\times 100\times v^{10}}{12a_{\overline{10}|} + 100v^{10}} \\ &= \frac{12\left(v + 2v^2 + \cdots + 10v^{10}\right) + 1000v^{10}}{12a_{\overline{10}|} + 100v^{10}} \\ &= \frac{12(Ia)_{\overline{10}|} + 1000v^{10}}{12a_{\overline{10}|} + 100v^{10}}\end{aligned}$$

Similarly,

$$\tau_B(\delta) = \frac{3(Ia)_{\overline{25}|} + 120\times 25\times v^{25}}{3a_{\overline{25}|} + 120v^{25}}.$$

Assume the current force of interest is 4% p.a.

δ	$NPV_A(\delta)$	$NPV_B(\delta)$	$\tau_A(\delta)$	$\tau_B(\delta)$
4%	163.97	90.61	7.1451	17.7975

In the following table, we compare the actual change in the NPV with the approximation of the change in the NPV using the duration, if the force of interest suddenly shifted from 4%.

Sudden shift in force of interest	Actual price change A	Actual price change B	Estimated price change $\approx -(\delta_1-\delta_0)\tau(0.04)$ A	Estimated price change $\approx -(\delta_1-\delta_0)\tau(0.04)$ B
From 4% to 3%	+7.465%	+19.92%	+7.1451%	+17.7975%
From 4% to 5%	-6.843%	-15.98%	-7.1451%	-17.7975%

For the actual price change, we used the following figures:

δ	$NPV_A(\delta)$	$NPV_B(\delta)$
3%	176.21	108.66
5%	152.75	76.13

The values in the second and in the fourth column of the above table should be approximately the same, and so should the values of the third and last column. The difference in this values comes from the fact that $-(\delta_1-\delta_0)\tau(\delta_0)$ provides only an approximation of the correct proportionate change.

12.2 Convexity

Definition 12.3 For any set of discrete cashflows $\{(C_{t_k}, t_k); k = 1, 2, \ldots, n\}$ and a continuous cashflow at rate of payment $\rho(t)$ for all $t \in [0, T]$, we define the *convexity* of the combined cashflows as

$$\begin{aligned} CONV(\delta) &= \frac{1}{NPV(\delta)} \frac{d}{d\delta^2} NPV(\delta) \\ &= \frac{\sum_{k=1}^{n} t_k^2 C_{t_k} e^{-\delta t_k} + \int_0^T t^2 \rho(t) e^{-\delta t} dt}{\sum_{k=1}^{n} C_{t_k} e^{-\delta t_k} + \int_0^T \rho(t) e^{-\delta t} dt}. \end{aligned}$$

Exercise 12.3 Show that the convexity of an n-year zero-coupon bond (ZCB) is n^2 years, i.e. it does not depend on the chosen force of interest.

Solution:

$$\begin{aligned} NPV(\delta) &= e^{-\delta n} \\ \frac{d}{d\delta} NPV(\delta) &= -n e^{-\delta n} \\ \frac{d}{d\delta^2} NPV(\delta) &= n^2 e^{-\delta n} \\ \implies CONV(\delta) &= \frac{1}{NPV(\delta)} \times \frac{d}{d\delta^2} NPV(\delta) \\ &= \frac{n^2 e^{-\delta n}}{e^{-\delta n}} \\ &= n^2 \end{aligned}$$

Exercise 12.4 Show that the convexity of an n-year annuity of £1 p.a. paid in arrears is

$$\frac{\sum_{k=1}^{n} k^2 v^k}{a_{\overline{n}|}}.$$

Solution:

$$\begin{aligned} NPV(\delta) &= \sum_{k=1}^{n} e^{-\delta k} \\ \frac{d}{d\delta^2} NPV(\delta) &= \sum_{k=1}^{n} k^2 e^{-\delta k} \\ \implies CONV(\delta) &= \frac{1}{NPV(\delta)} \times \frac{d}{d\delta^2} NPV(\delta) \\ &= \frac{\sum_{k=1}^{n} k^2 e^{-\delta k}}{\sum_{k=1}^{n} e^{-\delta k}} \\ &= \frac{\sum_{k=1}^{n} k^2 e^{-\delta k}}{\sum_{k=1}^{n} v^k} \\ &= \frac{\sum_{k=1}^{n} k^2 e^{-\delta k}}{a_{\overline{n}|}} \end{aligned}$$

Exercise 12.5 Show that the convexity of an n-year annuity of £1 p.a. paid continuously is

$$\frac{2(\bar{I}\bar{a})_{\overline{n}|} - n^2 v^n}{\delta \bar{a}_{\overline{n}|}}.$$

Solution:

$$\begin{aligned}
NPV(\delta) &= \int_0^n e^{-\delta t}\, dt \\
\frac{d}{d\delta^2} NPV(\delta) &= \int_0^n t^2 e^{-\delta t}\, dt \\
\implies CONV(\delta) &= \frac{1}{NPV(\delta)} \times \frac{d}{d\delta^2} NPV(\delta) \\
&= \frac{1}{\int_0^n e^{-\delta t}\, dt} \int_0^n t^2 e^{-\delta t}\, dt \\
&= \frac{1}{\bar{a}_{\overline{n}|}} \int_0^n t^2 e^{-\delta t}\, dt.
\end{aligned}$$

where (using integration by parts),

$$\begin{aligned}
\int_0^n t^2 e^{-\delta t}\, dt &= -\frac{n^2}{\delta} e^{-\delta n} + \frac{2}{\delta} \int_0^n t e^{-\delta t}\, dt \\
&= -\frac{n^2}{\delta} e^{-\delta n} + \frac{2}{\delta} (\overline{Ia})_{\overline{n}|} \\
\implies CONV(\delta) &= \frac{1}{\bar{a}_{\overline{n}|}} \left(\frac{2}{\delta} (\overline{Ia})_{\overline{n}|} - \frac{n^2}{\delta} e^{-\delta n}) \right) \\
&= \frac{2(\overline{Ia})_{\overline{n}|} - n^2 v^2}{\delta \bar{a}_{\overline{n}|}}.
\end{aligned}$$

Exercise 12.6 By using a Taylor series expansion about $\delta = \delta_0$, show that

$$NPV(\delta) \approx NPV(\delta_0) \left(1 - (\delta - \delta_0)\tau(\delta_0) + \frac{(\delta - \delta_0)^2}{2} CONV(\delta_0) \right).$$

Solution: Taylor's expansion for a function $f(\delta)$ about $\delta = \delta_0$ is

$$f(\delta) = f(\delta_0) + f'(\delta_0)(\delta - \delta_0) + \frac{f''(\delta_0)}{2}(\delta - \delta_0)^2 + \ldots \tag{12.3}$$

If we apply Equation 12.3 to the net present value function, i.e. NPV(δ), we may write the following:

$$NPV(\delta) \approx NPV(\delta_0) + \frac{dNPV(\delta_0)}{d\delta}(\delta - \delta_0) + \frac{dNPV(\delta_0)}{d\delta^2} \frac{(\delta - \delta_0)^2}{2}$$

where

$$\frac{dNPV(\delta = \delta_0)}{d\delta} = -\tau(\delta_0) NPV(\delta_0)$$

and

$$\frac{dNPV(\delta=\delta_0)}{d\delta^2} = CONV(\delta_0)NPV(\delta_0)$$

so we have,

$$NPV(\delta) \approx NPV(\delta_0) - \tau(\delta_0)NPV(\delta_0)(\delta-\delta_0) + CONV(\delta_0)NPV(\delta_0)\frac{(\delta-\delta_0)^2}{2}$$

$$NPV(\delta) \approx NPV(\delta_0)\left(1-(\delta-\delta_0)+\frac{(\delta-\delta_0)^2}{2}CONV(\delta_0)\right).$$

Exercise 12.7 Convexity can be used to provide a better approximation of the change in NPV. In a spreadsheet, use the approximation in Exercise 12.6 to obtain better approximations for the change in NPV for the fixed-income securities A and B of Example 12.1.5. Compare the duration-convexity approximations to the duration-only approximations and the actual price changes.

12.3 immunisation

Set-up: financial institution (e.g. life insurance company or pension fund). The institution has future net liability cashflows of L_{t_k} at time t_k, for $k = 1, 2, \ldots$. It holds assets which provide cashflows A_{s_n} at time s_n, for $n = 1, 2, \ldots$. Let the current force of interest be δ_0 p.a. We assume that

$$NPV_A(\delta_0) = \sum_{n=1}^{\infty} A_{s_n} e^{-\delta_0 s_n} = NPV_{\mathrm{L}}(\delta_0) = \sum_{k=1}^{\infty} L_{t_k} e^{-\delta_0 t_k}. \tag{12.4}$$

Since the NPV is a function of the force of interest, equation (12.4) may no longer hold if the force of interest changes. Hence, the institution faces the following problem.

Problem: How to choose assets so that an instantaneous change in the force of interest from δ_0 to δ_1 will not result in $NPV_A(\delta_1) < NPV_L(\delta_1)$?

"Perfect Matching": Choose $s_1 = t_1$, $s_2 = t_2$, ..., and then $A_{t_k} = L_{t_k}$ for all k (not usually a practical solution).

In cases in which "Perfect Matching" is not possible it might still be possible for the institution to choose their assets such that a small immediate change in the force of interest will not result in insufficient NPV of the assets. This provides at least some protection against the risk associated with a change in the interest rate.

The following theorem shows how a financial institution can protect itself against immediate small changes in the force of interest. This protection is called immunisation.

Redington's theory of immunisation

Suppose that the market force of interest is δ_0 p.a. If the following conditions hold:

$$NPV_A(\delta_0) = NPV_L(\delta_0), \qquad \text{"NPV matching"} \tag{12.5}$$

$$\tau_A(\delta_0) = \tau_L(\delta_0), \qquad \text{"Duration matching"} \tag{12.6}$$

$$CONV_A(\delta_0) > CONV_L(\delta_0), \qquad \text{"Convexity condition"} \tag{12.7}$$

then there exists $\epsilon > 0$ such that for all δ with $|\delta - \delta_0| < \epsilon$ the following holds

$$NPV_A(\delta) \geq NPV_L(\delta),$$

with equality holding only at $\delta = \delta_0$.

Interpretation: If the conditions (12.5), (12.6) and (12.7) are fulfilled, then an *immediate, small* change in the force of interest from δ_0 to δ_1 will give:

$$NPV_A(\delta_1) > NPV_L(\delta_1).$$

Define $f(\delta) = NPV_A(\delta) - NPV_L(\delta)$.

$$\text{Equation (12.5)} \quad \Rightarrow \quad f(\delta_0) = 0.$$

$$\begin{aligned} \text{Equations (12.5) and (12.6)} \quad &\Rightarrow \quad NPV'_A(\delta_0) - NPV'_L(\delta_0) = 0 \\ &\Rightarrow \quad f'(\delta_0) = 0 \\ &\Rightarrow \quad f \text{ has a turning point at } \delta_0. \end{aligned}$$

$$\begin{aligned} \text{Equations (12.5) and (12.7)} \quad &\Rightarrow \quad NPV''_A(\delta_0) - NPV''_L(\delta_0) > 0 \\ &\Rightarrow \quad f''(\delta_0) > 0 \\ &\Rightarrow \quad f \text{ has a minimum point at } \delta_0. \end{aligned}$$

It follows that (i) $f(\delta)$ has a local minimum at $\delta = \delta_0$, and (ii) $f(\delta_0) = 0$.
Hence for all δ close to δ_0,

$$f(\delta) > f(\delta_0) = 0.$$

More precisely, there exists $\epsilon > 0$ such that for all δ with $|\delta - \delta_0| < \epsilon$, we have $NPV_A(\delta) \geq NPV_L(\delta)$,with equality holding only at $\delta = \delta_0$.

Remarks:

1. If conditions (12.5),(12.6) and (12.7) hold, the financial institution is said be immunized (in the sense of Redington) against immediate small changes in the interest rate.

2. The financial institution is immunized against immediate small changes in the interest rate assuming a flat yield curve. (Generalizes to immediate parallel shifts of a yield curve.)

3. Suppose (12.5) and (12.6) hold and there are only discrete cashflows. Then (12.7) is equivalent to

$$\sum_{n=1}^{\infty}(s_n - \tau_A(\delta_0))^2 A_{s_n} \exp(-\delta_0 s_n) > \sum_{k=1}^{\infty}(t_k - \tau_L(\delta_0))^2 L_{t_k} \exp(-\delta_0 t_k) \qquad (12.8)$$

i.e. the "spread of the assets" around the common τ is greater than the "spread of the liabilities". A similar result holds for continuous cashflows.

The next example shows how an institution can apply Redington's theory of immunisation to protect itself against immediate small changes of the force of interest.

Example 12.3.1 A life office has to pay £1 million in 10 years time. The effective rate of interest is $i = 4\%$ p.a.. The available securities are a 5-year and a 15-year zero-coupon bond (ZCB).
Calculate how much of each ZCB the life office should buy so that it is immunised in the sense of Redington.

Solution: Suppose £X is invested in the 5-year ZCB at $t = 0$ and £Y is invested in the 15-year ZCB at $t = 0$.

The value of the liability at time 0 is $NPV_L = £1\,000\,000 \times 1.04^{-10} = £675\,564$.

(i) Equate NPVs: $NPV_A = NPV_L \Rightarrow X + Y = 675\,564$.

(ii) Equate durations:

$$\tau_A = \tau_L \qquad \Rightarrow \qquad \frac{5X + 15Y}{X + Y} = \frac{10\,(1\,000\,000)v^{10}}{1\,000\,000v^{10}}.$$

Hence we have two linear equations in two unknowns:

$$X + Y = 675\,564$$
$$5X + 15Y = 10(1\,000\,000)v^{10}.$$

Solve them for X and Y to find $X = £337\,782$ and $Y = £337\,782$, i.e. half the assets are invested in the 5-year ZCB and half are in the 15-year ZCB at $t = 0$.

(iii) Check if the convexity condition (12.7) holds.

$$CONV_A = \frac{5^2X + 15^2Y}{X + Y} = 125 > 100 = \frac{10^2\,(1\,000\,000)v^{10}}{1\,000\,000v^{10}} = CONV_L.$$

Since it holds, the financial institution is immunized.

Remarks:

(i) Price of the 5-year ZCB per £1 nominal is $P_5 = v^5 = 1.04^{-5} = 0.82193$. Investing $X = £337\,782$ at time 0 in the 5-year ZCB means that the insurer purchases

$$\frac{X}{P_5} = \frac{337\,782}{0.82193} = £410\,962 \text{ nominal of the 5-year ZCB.}$$

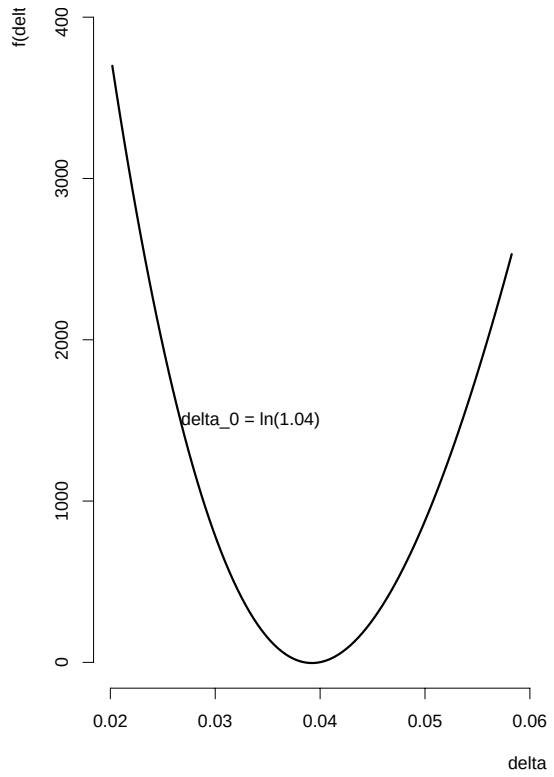


Figure 12.3: $NPV_A - NPV_L$ for i between 2% and 6% (i.e. $\delta \in [\ln 1.02, \ln 1.06]$), for Example 12.3.1.

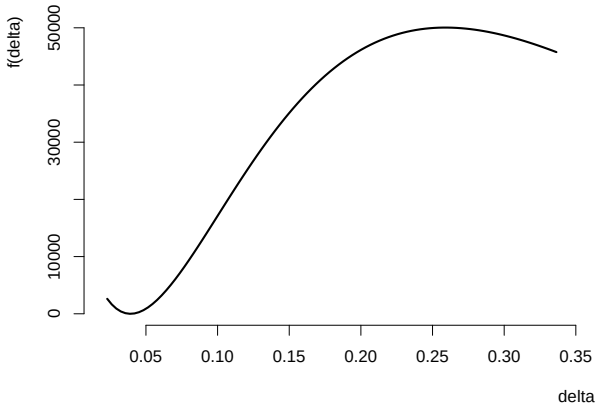


Figure 12.4: $NPV_A - NPV_L$ for i between 0% and 40% (i.e. $\delta \in [\ln 1.02, \ln 1.06]$), for Example 12.3.1.

(ii) Price of the 15-year ZCB per £1 nominal is $P_{15} = v^{15} = 1.04^{-15} = 0.55526$. Investing $Y = £337\,782$ at time 0 in the 15-year ZCB means that the insurer purchases

$$\frac{X}{P_{15}} = \frac{337\,782}{0.55526} = £608\,331 \text{ nominal of the 15-year ZCB.}$$

Exercise 12.8 Show that with the solution above we have:

(i) If interest rates move to 3% p.a. effective, NPV(3%) of (assets-liabilities) $\approx$ +£870.

(ii) If interest rates move to 5% p.a. effective, NPV(5%) of (assets-liabilities) $\approx$ +£703.

Hint: Use the nominal amounts of the ZCBs to calculate their prices under the new interest rate.

Solution: According to Example 12.3.1 in order to be immunised for a liability at time 10 that is equal to £1 million, we need to invest £337 782 in a 5-year ZCB and £337 782 in a 15-year ZCB today. Given the 4% p.a. effective rate, this would mean to collect £410 962 nominal of the 5-year ZCB and £608 331 of the 15-year ZCB.

(i) If interest rates move to 3% p.a., the NPV of assests and liabilities are now

$$\begin{aligned} NPV_A &= 410962 \times (1.03)^{-5} + 608331 \times (1.03)^{-5} \\ &= 744963.9 \\ NPV_L &= 100000 \times (1.03)^{-10} \\ &= 744093.9 \end{aligned}$$

Hence at 3%, $NPV_A - NPV_L = £870$.

(ii) If interest rates move to 5% p.a., the NPV of assests and liabilities are now

$$\begin{aligned} NPV_A &= 410962 \times (1.05)^{-5} + 608331 \times (1.05)^{-5} \\ &= 614617.1 \\ NPV_L &= 100000 \times (1.05)^{-10} \\ &= 613913.3 \end{aligned}$$

Hence at 5%$NPV_A - NPV_L = £703.8$.

Exercise 12.9 Show that if all money had been invested in the 5-year ZCB:

(i) If interest rates move to 3% p.a. effective, NPV(3%) of (assets-liabilities) $\approx$ -£35 093.

(ii) If interest rates move to 5% p.a. effective, NPV(5%) of (assets-liabilities) $\approx$ +£30 088.

Hence we see that immunisation works against profits as well as losses.

Solution: If all money had been invested in the 5-year ZCB, this would be £2 $\times$ 337782 invested in the 5-year ZCB at time zero. We again use the nominal amount of the 5-year ZCB to calculate the new price under the relevant interest rate.

(i) If interest rates move to 3% p.a., the NPV of assests and liabilities are now

$$\begin{aligned} NPV_A &= (2 \times 337782)(1.04)^{-5}(1.03)^{-5} \\ &= 709\,001.4 \\ NPV_L &= 100000 \times (1.03)^{-10} \\ &= 744\,093.9 \end{aligned}$$

Hence at 3%, $NPV_A - NPV_L$ = -£35 092.5.

(ii) If interest rates move to 5% p.a., the NPV of assests and liabilities are now

$$\begin{aligned} NPV_A &= (2 \times 337782)(1.04)^{-5}(1.05)^{-5} \\ &= 614617.1 \\ NPV_L &= 100000 \times (1.05)^{-10} \\ &= 613913.3 \end{aligned}$$

Hence at 5%$NPV_A - NPV_L$ = -£30 087.9.

Exercise 12.10 An insurance company has issued a number of policies for which a total liability of £1 000 000 is payable in exactly 10 years' time. The insurance company holds sufficient funds to cover the liability on the basis of a force of interest of δ per annum and intends to invest this money solely in the purchase of a 5-year and a 20-year zero-coupon bond (ZCB).

What proportion of the company's funds should be invested in the 5-year ZCB to ensure that the discounted mean term of the assets equals the discounted mean term of the liabilities at a constant force of interest of δ per annum given that this is also the market basis for pricing all stocks?

(*Answer: 2/3*)

Solution: Note that *discounted mean term* indicates **duration**. Let us suppose that £X is invested in the 5-year ZCB and £Y is invested in the 20-year ZCB at time zero. We are looking for the proportion of available assets that guarantee

$$\tau_A(\delta) = \tau_L(\delta)$$

which is equal to

$$\frac{5X + 20Y}{X + Y} = 10 \tag{12.9}$$

for

$$\begin{aligned} NPV_A &= X + Y \\ NPV_L &= 1000000e^{-10\delta}. \end{aligned}$$

From Equation 12.9,

$$\begin{aligned} 5X + 20Y &= 10X + 10Y \\ X &= 2Y \end{aligned}$$

This tells us that the protfolio contains $2Y$ units of 5-year ZCB and Y units of 20-year ZCB i.e. a total of $3Y$ units in assets. Meaning the company should invest $2/3$ of its funds into the 5-year ZCB.

Chapter 13

Term Structure of Interest Rates

13.1 Yield Curves

- The *term structure of interest rates* is the variation of bond yields with the terms to redemptions of the bonds.
- Graphically, this is also called a *yield curve*.
- Yields are calculated from market prices. They are determined in the market by demand and supply.
- Yield curves vary from day to day. They can be
  - increasing (upward-sloping), sometimes called "normal",
  - decreasing (downward-sloping), sometimes called "inverted",
  - flat (all yields are equal), or
  - inconsistent term to redemption/yield relationship, sometimes called "humped" (e.g. short-term and long-term yields are equal, but medium-term yields vary).
- Flat and humped yield curves indicate uncertainty. As certainty in the market increases, the yield curve moves towards a normal or an inverted shape.
- There are 3 observed characteristics of the term structure of interest rates:
  - The change in yields of different term bonds tends to move in the same direction.
  - The yields on short-term bonds are more volatile than long-term bonds.
  - The yields on long-term bonds are often higher than short-term bonds.
- Why does the yield curve have the shape it does?

13.1.1 Explanations for the shape of the yield curve

The main explanation is the Expectations Theory. It explains

- The change in yields of different term bonds tends to move in the same direction.
- The yields on short-term bonds are more volatile than long-term bonds.

The third observation:

- The yields on long-term bonds are often higher than short-term bonds.

can be explained by the Liquidity Preference Theory.

An additional explanation is the Market Segmentation Theory/Preferred Habitat Theory, which can be used in conjunction with the Expectations Theory and the Liquidity Preference Theory to explain the shape of the yield curve.

13.1.2 Expectations Theory

- Long-term interest rates reflect investors' expectations of short-term interest rates in the future.
- This means that the long-term rate is composed of all the short term rates up to that long term.
- i.e. the long-term rate up to time T is the geometric mean of all the intervening short-term rates up to time T.

13.1.3 Liquidity Preference Theory

- The more risky the investment, the higher the *expected* return is required to be in order to entice investors.
- Long-dated bonds are more risky than short-dated bonds: when interest rates move, the price of a long-dated bond changes more than the price of a short-dated bond.
- Therefore, investors expect a higher interest rate on long-dated bonds because of the extra risk.

Together, the Expectations Theory and Liquidity Preference Theory suggest that an upward-sloping yield curve is more common than a downward-sloping one.

This is observed in practice.

13.1.4 Market Segmentation Theory

- Different investors are attracted to different types of bonds.
- This can be due to investment requirements,
- e.g. a pension fund is attracted to long-dated, index-linked bonds to match long-term, inflation-linked liabilities,
- or it can be due to regulatory reasons, taxation benefits, etc, e.g. an investor not liable to income tax would be more interested in high-coupon bonds.
- The yield for each "segment" of the market are determined by supply and demand in that segment, e.g. yield on short-term dated bonds is determined by supply and demand among short-term investors.)

For example, the Market Segmentation Theory can explain why yields at the long-end of the yield curve may decrease. However, for the overall shape of the yield curve, we need the Expectations Theory and Liquidity Preference Theory.

time	0	1	2	3	4
Random interest rate		i_1	i_2	i_3	i_4
2-yr ZCB	$-P_2$		100		
4-yr ZCB	$-P_4$				100
FI bond	$-P$	C	C	C	$C+R$

13.2 Zero-Coupon Bonds (ZCBs) and Spot Rates

Definition 13.1 £1 nominal of a n-year **zero-coupon bond (ZCB)** pays £1 after n years and nothing else, i.e. a ZCB is a fixed-interest security which is redeemed at par, with term n years and has **no** coupon payments.

In the UK market, a ZCB is sometimes called a "gilt strip", a gilt which has been stripped of its interest payments, or indeed just one of the coupon payments.

Definition 13.2 The **n-year spot rate of interest** y_n is the annual redemption yield of an n-year ZCB.

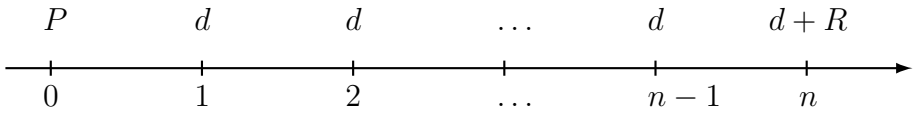
Let P_n be the price today of an n-year ZCB. Then, for $n > 0$,

$$\text{EoV:} \qquad P_n = 1 \times (1+y_n)^{-n} \Rightarrow y_n = P_n^{-\frac{1}{n}} - 1$$

Example 13.2.1 Calculate the spot rate of interest y_5 for a 5-year ZCB currently priced at 0.78.

Solution: $y_5 = 0.78^{\frac{1}{5}} - 1 = 5.095\%$ p.a.

Consider a fixed-interest security which pays redemption proceeds of R at time n, and dividends (coupons) d at times $1, 2, \ldots, n-1, n$. With no-arbitrage, the price P of this security must be equal to the price of a series of ZCBs.



$$\begin{aligned} P &= d(P_1 + P_2 + \cdots + P_n) + RP_n \\ &= d((1+y_1)^{-1} + (1+y_2)^{-2} + \ldots + (1+y_n)^{-n}) + R(1+y_n)^{-n} \end{aligned}$$

If this were not true then we could make money by buying a series of ZCBs and selling the fixed-interest security (or vice versa) to make a profit ($\Rightarrow$ arbitrage). This relationship allows us to consider only ZCBs.

Exercise 13.1 Suppose $P_1 = 0.95$ and $P_2 = 0.90$. Calculate

(i) the 1-year spot rate.

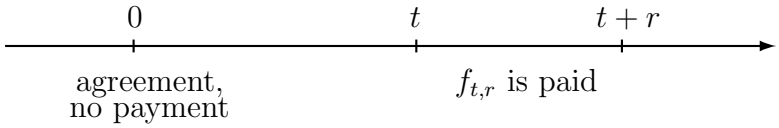
(ii) the 2-year spot rate.

(iii) the price of £100 nominal of a 2-year fixed-interest bond with coupon rate 7% p.a. paid annually in arrears and redemption proceeds 100%.

(iv) the gross redemption yield on the above fixed-interest bond.

13.3 Forward Rates

Definition 13.3 The *f*orward rate $f_{t,r}$ is the effective annual rate of interest agreed now (i.e. at time 0) for an investment made at time t for a period of r years.



What this means is that the present value **at time** t of a unit amount paid at time $t + r$ is

$$\frac{1}{(1+f_{t,r})^r}.$$

13.3.1 The relationship between spot rates and forward rates

Exercise 13.2 Express $f_{0,r}$ in terms of a spot rate.

Consider £1 invested at time 0 for $t+r$ years at the $(t+r)$-spot rate. It will accumulate to $(1+y_{t+r})^{t+r}$.

Alternatively, if we invest the £1 for t years at the t-year spot rate y_t followed by investment from time t to $t+r$ at the forward rate $f_{t,r}$, £1 accumulates to

$$(1+y_t)^t(1+f_{t,r})^r.$$

Under no-arbitrage, these accumulations must be equal, i.e.

$$(1+y_t)^t(1+f_{t,r})^r = (1+y_{t+r})^{t+r}.$$

Rearranging, we get

$$(1+f_{t,r})^r = \frac{(1+y_{t+r})^{t+r}}{(1+y_t)^t}.$$

Definition 13.4 The *n-year par yield* is the coupon rate c_n payable annually in arrears per £1 nominal which would give an n-year bond with coupon rate c_n a price of £1 assuming the bond is redeemed at par.

That is, for a given zero-coupon yield curve, the n-year par yield c_n is given by

$$\begin{aligned} 1 &= c_n[(1+y_1)^{-1} + (1+y_2)^{-2} + \ldots + (1+y_n)^{-n}] + (1+y_n)^{-n} \\ &= c_n[P_1 + P_2 + \ldots + P_n] + P_n \end{aligned}$$

Exercise 13.3 Show that, in terms of zero-coupon bond prices, we have

$$c_n = \frac{1-P_n}{\sum_{k=1}^n P_k}.$$

Example 13.3.1 Consider the following spot interest rates

Term (in years)	Spot rate (in % p.a.)
1	5
2	5.5
3	6

The 3-year par yield c_3 satisfies

$$1 = c_3(1.05^{-1} + 1.055^{-2} + 1.06^{-3}) + 1.06^{-3} \qquad \Rightarrow \qquad c_3 = 5.96\% \text{ p.a.}$$

13.4 Continuous-time rates

Definition 13.5 The *t-year spot force of interest* is the force of interest equivalent to the annual spot rate y_t, i.e.

$$Y_t = \ln(1+y_t).$$

Note that $P_t = e^{-tY_t}$.
Y_t is also called

- the continuously-compounded spot rate, or
- the continuous-time spot rate.

Example 13.4.1 $y_t = 5\% \Rightarrow Y_t = 0.04879 = 4.879\%$.

Definition 13.6 *The* continuous-time forward rate $F_{t,r}$ *is the force of interest equivalent to the annual rate of interest* $f_{t,r}$*, i.e.*

$$F_{t,r} = \ln(1 + f_{t,r}).$$